iCONSULT: A BUSINESS CONSULTANCY INTEGRATED WITH PROJECT MANAGEMENT SYSTEM AND CLIENT MANAGEMENT SYSTEM

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ABSTRACT

Business consultancy continues to evolve with the integration of digital solutions designed to optimize operations and client relations. This study introduces iConsult, a web-based application that integrates project management and client management into a single platform for business consultancies. iConsult addresses common consultancy challenges such as fragmented communication, task inefficiencies, and poor client engagement through features including real-time project tracking, online appointment scheduling, document storage, online payments via PayPal Sandbox, and reports with visualization tools. Developed using modern technologies such as React.js, Node.js, Express.js, MySQL, and APIs like SendGrid, Google Drive, and PayPal, the system was designed and deployed with tools such as Visual Studio Code, Figma, GitHub, and Heroku. The system functionality and reliability were evaluated using ISO/IEC 25010 standards, and the results showed that it was "Highly Acceptable," indicating that it was effective, consistent, and fault-tolerant under a range of circumstances. Tests also showed that the system performs as expected, with key features like file sharing, real-time scheduling, client information management, and efficient consultant workflows. In order to increase its usefulness, it is suggested that future development include enabling in-system document editing, incorporating live chat support, adding multiple payment options, improving backup and recovery systems, and supporting video conferencing platforms like Zoom or Google Meet.

Keywords: Project management system, client management system, business consultancy, appointment scheduling

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

In today's rapidly evolving business landscape, entrepreneurs and companies worldwide constantly seek strategies to stay competitive and achieve sustainable growth. A career in business consulting relies on planning, organization, creativity, and strategic thinking to help businesses troubleshoot and problem-solve effectively (Forage, 2024). "Business consulting involves providing expert advice and guidance to organizations to help them improve their operations, achieve their goals, and solve complex business problems," says Michael Tschudy. A knowledgeable with a broad range of abilities who helps individuals with their undertakings is known as a business consultant. Because of their training and prior work experience, consultants possess expertise.

Additionally, business consultants provide management consulting to help organizations improve their performance and efficiency (Peek, 2024).

In short, consulting is an advising service; hence, in theory, consultants are not accustomed to managing businesses or making decisions for managers. They are unable to make decisions or carry out changes because they lack direct authority. Consulting firms in the Philippines and other nations are only accountable for the caliber and honesty of their recommendations. Clients are responsible for carrying out any obligations arising from doing so.

According to Afsaneh and Chesteen (2019), a mail survey of 126 businesses that had sought assistance from the SBDC at the University of Utah was conducted. According to data gathered from 106 respondents (84% participation rate), the company owners and

managers expressed great satisfaction with the support that they received and thought they were helpful. Additionally, they stated that their firm benefited from the consulting. The main justifications for not putting some recommendations into practice are appropriateness and an absence of consultant experience.

Some organizations employ consulting companies to provide valuable ideas and professional services (O'Mahoney & Markham, 2013). Fincham and Clark (2002) describe the management consultancy industry as one of the fastest-growing industries. In the face of these difficulties, this project suggests integrating a thorough project management system with a client management system within iConsult. This will help consultants overcome these obstacles and improve their capacity to provide clients with excellent results. iConsult wants to maximize its consulting operations, increase client satisfaction, and keep a competitive advantage in the Philippine business consulting market by utilizing technology and implementing effective methods.

Background of the Study

The companies can accomplish their objectives with the help of the business consulting sector. In the past, consultants gave professional advice via in-person meetings, manual project management, and client communication techniques. A study of Azumah, 2024, highlights significant changes in this field. This study examines the limitations of traditional consulting methods and proposes the "iConsult" a business consultancy integrated with a Project Management Sytem and Client Management System that makes managing clients and projects easier for a consultancy firm.

Business consultancies provide basic advisory services, but in recent years, their role

has expanded beyond mere consultations. Modern businesses now rely heavily on essential services from partner organizations. But as companies expand and needs become more specialized, many smaller firms can't keep pace with growing demands. The core issue lies in their outdated consulting approaches. Azumah's 2024 research demonstrates how many still depend on clunky spreadsheets, endless email chains, and disorganized paper trails - systems that can't satisfy today's clients. Even as digital transformation accelerates, this disconnect grows, creating operational headaches. These antiquated methods resulted to wasted time and resources, poor information organization communication breakdowns. The consequences may lead to lost of documents, missed deadlines, client misunderstandings, and damaged business relationships.

The business consultancy field has grown significantly, but the current approaches to project management and client communication still have shortcomings that iConsult can fill. This integrated system combines project and client management because of this shortcomings that traditional methods can’t do. According to Azumah (2024), modern consultancies faces three major challenges which are digital transformation, cultural shifts and ethical concerns and rather than simply improving operational efficiency, consultants are encouraged to adopt to a more thorough solutions that focus on these issues head-on. This shows the importance of streamlined operations, transparent communication channels, and efficient project management systems in which iConsult’s integrated approach directly responds to these needs. By combining project tracking with client management functions, the system shows Azumah’s (2024) findings about modern consultancy need, particularly the growing importance of organizational efficiency together with client satisfaction.

A comprehensive study of Tahri (2021) identified five essential components for

effective project management these are planning tools, portfolio management, content management, collaboration platforms, and satisfaction metrics but somehow, few studies has examined how these systems actually perform in real consultancy environments. Moreover, Yan et al. (2020) successfully implemented a cloud-based system for the university research management, which streamline project application, review, progress management, and display of research results. While demonstrating significance improvements in project tracking and result dissemination. However, this cloud-based systems lacks the integration of client management systems that is needed for business consultancies. This gap shows the need for consultancies to require a specialized solutions that integrates both project management with client relations. Thus, the study iConsult was developed specifically to address this challenges faced by consultancies proposing them this integrated system that enhances both operational and client relationship that directly impact business growth (Tahri, 2021; Yan et al., 2020).

Business consultancies often struggle with unorganized workflows, inefficient communication, and lack of transparency for that the iConsult platform was designed to address this challenges. This web-based application combines project management system with client management tools to organize operations in a consultancy firms. By integrating these functions into a single platform, iConsult enhances efficiency in a project and tasks while improving collaboration between consultants and clients. Key features include consultation appointments, calendar overview, real-time progress updates, files sharing, client and employee management which contributes to a smoother user experience.

In conclusion, the consulting industry clearly needs better tools to manage projects and communicate with clients effectively. Thus, the answer to this problem is iConsult - a web

platform that combines both functions in one place. This system makes consultants' jobs easier in several important ways: projects stay organized, everyone can see what's happening, and teams work together more smoothly. Key features include up-to-the-minute status reports, safe document sharing, and easy appointment booking all of which save time and keep clients satisfied. Beyond solving current workflow issues, iConsult prepares consulting businesses for the challenges of tomorrow's digital workplace.In conclusion, the consulting industry clearly needs better tools to manage projects and communicate with clients effectively. Thus, the answer to this problem is iConsult - a web platform that combines both functions in one place. This system makes consultants' jobs easier in several important ways: projects stay organized, everyone can see what's happening, and teams work together more smoothly. Key features include up-to-the-minute status reports, safe document sharing, and easy appointment booking all of which save time and keep clients satisfied. Beyond solving current workflow issues, iConsult prepares consulting businesses for the challenges of tomorrow's digital workplace.

Objectives of the Study

General Objective

The study focused on developing a user-friendly web-based platform that integrates project and client management system into one platform to enhance efficiency and client satisfaction within business consultancies.

Specifically, the study aimed to:

1. Create and design the system with the following characteristics:
    a) Real-time online consultation scheduling, enabling clients to book appointments instantly.
    b) A project management module for tracking projects, tasks, deadlines, and deliverables.
    c) A centralized client portal handling client's information, appointments, feedback, projects, and payments.
    d) Google Drive API for secure document management for file sharing and storage.
    e) Paypal Sandbox integration for secure payment processing.
    f) The system utilizes data visualization reports to analyze and identify areas for improvement.
2. Develop the system using the following software development tools:
    a) **Front-End Development**: HTML, CSS, and JS, React.js
    b) **Back-End Development**: Node.js

c) **Framework:** Express.js

d) **Database Management**: MySQL

e) **Additional Tools**: PayPal Sandbox, SendGrid Email API, Google Drive API

f) **Development and Hosting**: Jaws DB MYSQL, Heroku

g) **Version control**: GitHub

3. Evaluate the system in terms of functional suitability and efficiency.

4. To evaluate the system's acceptability using the ISO 25010 standard focusing on functional suitability and operational efficiency.

Scope and Limitations of the Study

This study introduces "iConsult", a web platform designed to address common challenges in consultancy business. The platform was designed to improve operational efficiency and client engagement. Its main features is the project management module that allows consultant handle project more efficient and easily track the project process. Though this consultants can create new projects with its details, break down into multiple tasks, assign tasks to employees and monitor urgent and due tasks. This module not only simplify internal operations but clients also get transparent access to view the development of their project.

Furthermore, the iConsult's client portal facilitates two-way collaboration, allowing clients to monitor real-time project development, secure access to all related documents including contracts, reports, transactions, and business-related papers and lastly, share their feedback on the completed project. This feedback module serves as a significant feature, helping consultancies to strengthen client partnerships and consistently meet their needs.

Moreover, the iConsult platform also makes scheduling appointment easier through its integrated system. Clients can easily book consultation from consultants' pre-set availability and receive emails to remind them of their appointment. On the consultant side, they can set their availability both date and time, and manage all appointments schedule in one place. To keep them updated, client and consultants receive notification for new appointments and payment of client. Additionally, the iConsult's reporting feature shows data visualization that provides insights and metrics of the different modules in the system. The platform generates interactive charts, graphs, and other visual formats for easy understanding, enabling consultants to quickly identify trends, see opportunities, and track projects across all operational areas.

The development of the iConsult is built with front-end and backend technologies. For the user interface, HTML and CSS were used for structure and styling while JavaScript and React.js for interactive and responsive interface. For backend technologies runs on Node.js with Express.js for server-side, together with MYSQL for a database management. To enhance the overall functionality, external APIs are also used such as SendGrid API for automated email communications, Paypal API Sandbox for payment processing, and Google Drive API for document sharing and storage. For the deployment and hosting of the system, JawsDB MySQL is used for database hosting and Heroku for application deployment. Throughout development, GitHub version control is used for collaboration with teams to easily track changes and contribution. Lastly, ISO 25010 standards is used in evaluating the system particularly on functional suitability and operational efficiency.

While iConsult offers comprehensive functionalities, it does have several limitations. First, although the system sends emails alerts for consultation reminder, it lacks an

integrated chat feature for real-time messaging, meaning users must switch to external platforms for direct conversations. Second, clients can't independently create their own account and must wait for the consultant to create their accounts first and wait for an email indicating their username and password to prevent duplication of accounts. While this ensure security, it may delay initial access like booking of appointments. Finally, dependencies on third-party APIs means that performance could be affected if services experience downtime. These limitations reflect the application's focus on prioritizing ease of use and main features of the system over advanced functionality.

Chapter 2

CONCEPTUAL FRAMEWORK

This chapter encapsulates the related literature and studies that support the project. It also introduces the conceptual model of the study and the operational definitions of terms. The relationships between the various sources were explored, highlighting their importance and defining the research gap the study aims to bridge.

Review of Related Literature

This section presents the key concepts and ideas that are relevant to this study. It will include discussions on Business Consultancy, Business Consultancy Services, Hypertext Markup Language (HTML), Cascading Style Sheet (CSS), Tailwind, JavaScript, MySql, ReactJS, NodeJS, PhP, Express.js, Github, Visual Studio Code, Netlify, AWS, Paypal Sandbox, Google Calendar API, Twilio, SendGrid Email, and Figma.

Business Consultancy

In today's dynamic business environment, companies increasingly seek external expertise to navigate complex challenges and achieve their goals (Indeed Editorial Team, 2024). Business consultancies, staffed by skilled consultants with industry-specific knowledge, act as strategic partners to their clients (Peek, 2024). Through in-depth analysis of internal operations and external market forces, consultants develop insightful solutions that propel businesses toward success.

At the heart of a successful consultancy lies the strength of its relationships with clients. Research by Da Costa et al. (2020) underscores this established concept, highlighting the

significance of consultant competence, client understanding, transparency, professionalism, and alignment of values. Building trust is paramount, ensuring clients feel confident in the expertise guiding their journey. Effective consultants go beyond generic solutions, tailoring strategies to address each client's specific needs (Da Costa et al., 2020).

The consulting business remain constantly. According to Cui and Santos (2023), the Chinese consulting industry is growing and requires constant adaptation to satisfy changing requirements of clients. The growing significance of computerization businesses is one important aspect of this development. A comprehensive amendment to the way a company operates must occur enable a digital shift to prove effective as stated in “Digital Transformation of Incumbent Firms: A Business Model Innovative Perspective” (2023).

Developing strong connection was only one aspect of a successful consultancy. Consulting firms must place a high priority on innovation and ongoing development in order to stay competitive in changing environment (Cui & Santos, 2023). This includes embracing innovation in all every aspect of the company, including the range of services to operational procedures, using efficient project management techniques, effectively allocation resources, and adjusting to changing client wants and demands (Mevoli, 2023). Consultancies can strengthen their client connections by using technology to improve communications, client service, and teamwork.

Sole Proprietorship

There are many different types of structures in the business world, each serving a particular requirement. Among the most popular and simple constructions, especially for emerging. The lone proprietorship is known as a venture (Twin, 2024). A sole

proprietorship has one owner who runs the company, keeps all earning and is entirely for any obligations or losses sustained. This simplicity in setup and tax filing makes it an attractive option for many small businesses and independent contractors, including consultants, in the initial stages of their practice.

While sole proprietorship offers ease of establishment, it's important to acknowledge their limitations. They lack the liability protection that separates the owner's personal assets from business debts. This can be a significant drawback as a consultancy grows and takes on larger projects.

Business Consultancy Types and Services

In the Philippines, business consultancies offer various services to assist companies in various aspects of their operations. De Castro (2024) highlighted that consulting services can provide valuable perspectives on various business challenges. Here's a look at some of the key consultancy types offered in the Philippines:

Marketing Consulting. Marketing consultant help businesses develop their brand identity and expand their reach (De Castro, 2024). They can assist with creating a compelling brand image, developing marketing campaigns, and crafting effective social media strategies.

Human Resource (HR) Consulting. HR consultants offer guidance on various HR functions, including recruitment, payroll processing, employee performance management, and training (De Castro, 2024). Companies can outsource specific HR tasks or seek broader strategic advice on building a strong workforce.

Strategy Consulting. Strategy consultants collaborate with executives to develop long-term business plans and navigate complex decisions (Peak, 2024). Strategy consultants

offer businesses a comprehensive perspective on critical decisions, including strategic alignment, strategic planning and execution, strategy optimization, market entry guidance, and integration and effectiveness. They work with leadership teams to define objectives, develop a comprehensive plan, and analyze existing strategies for improvement. Their expertise leads to well-informed decisions for long-term success (Nagy, 2024).

Operations Consulting. Operations consultants focus on optimizing day-to-day business processes for efficiency and cost-effectiveness (Peek, 2024). Operation consultants are specialists in streamlining a company's daily operations for maximum efficiency and cost-effectiveness. They analyze business processes, focusing on key metrics like cost, efficiency, and quality. They offer core services such as process management, outsourcing optimization, procurement improvement, supply chain management, and risk management integration. They also offer specialized services like ROI consulting, risk and compliance consulting, e-commerce consulting, and cross-functional collaboration (Nagy, 2024). By leveraging their expertise, businesses can improve efficiency, cost savings, and competitiveness.

Management Consulting. Management consulting is a broad field encompassing strategy, HR, and operations expertise (Indeed Editorial Team, 2024). Management consultants are expert advisors who help companies optimize resource allocation and achieve objectives. They bridge the gap between strategy and execution, analyze existing management practices, and offer specialized services like leadership development, project management expertise, and management framework implementation. By leveraging their knowledge, companies can gain insights, improve practices, and achieve their strategic objectives more effectively (Nagy, 2024).

Information Technology (IT) Consulting. IT consultants offer specialized knowledge of information technology systems (Indeed Editorial Team, 2024). They advise on software selection, system implementation, and data security. IT consultants can also train employees on new systems and ensure smooth integration with existing technology infrastructure.

Legal Consulting. Legal consultants provide guidance on legal matters that could impact a business (Indeed Editorial Team, 2024). They help companies navigate regulations, manage legal risks, and ensure compliance with relevant laws.

Financial Consulting - Financial consultants offer expertise in managing a company's financial health (De Castro, 2024). They assist with investment decisions, debt management, and overall financial planning. Some consultants specialize in areas like taxes, daily expenses, and retirement plans. Below are some key accounting services offered by consultancies (Tabar, 2023):

Bookkeeping. According to De Castro (2024), business consultancies offer valuable assistance to companies in the Philippines in managing their bookkeeping and registering their books of accounts with the Bureau of Internal Revenue (BIR). They handle daily transaction recording and categorization, manage invoicing and customer payments, perform bank reconciliations, generate essential monthly financial statements, and provide payroll processing services. They also help navigate BIR requirements, guide companies through registration, and meet legal requirements.

Tax Services. Business consultancies offer valuable support in managing a company's taxes. According to De Castro (2024), consulting firms assure companies in meeting their tax responsibilities on time, reduce the risk of errors, and improving tax strategies. The company represent by the consultancy during audit, engage in discussions with tax agencies, and defend the

legal rights. By staying current with tax legislation in the Philippines, they support the development of simplified tax management systems. Their expertise gives businesses greater confidence in fulfilling their obligations and optimizing their tax positions.

Forensic Accounting. Forensic accounting consultants can find and examine the misrepresentation within a business, with expert opinions for legal actions.

Accounting Auditing. Accounting Auditing consultants can obtaining an external audits of final statements of the company, certifying accuracy and maintaining compliance with accounting principles, thereby enhancing confidence among investors and stakeholders.

Payable Accounts Management. Consultancies can enhance accounts payable processes by reviewing invoices, approving payments, and ensuring prompt settlements, thus optimizing cash flow management and preventing late payment fees.

Bank Reconciliation. Bank reconciliation is the process of matching the transactions in a company's (or individual's) accounting records with those listed on the bank statement to ensure consistency and accuracy.

Account Receivable Management. Is the process of overseeing and controlling the money owed to a business by its customers (i.e., outstanding invoices or credit sales) to ensure timely collection, maintain healthy cash flow, and minimize bad debts.

Payroll Processing. Consultants manage payroll processing and ensures proper payments, remit taxes, and keeping detailed records which allows businesses to focus on their main task while staying compliant with employment regulations. According to De Castro (2023), consulting firms in the Philippines assist companies in creating payroll system by helping them with registration, payroll process establishment, labor law comprehension, employee data gathering and management.

A web-based system that combines project and client management gives advantages for financial accounting consultancies. This approach helps staff work more efficiently by keeping projects well organized and running smoothly, which allows consultants to work at their best. While a secure place to store documents makes collaboration with clients easier and saves time spent searching for files, it also allows clients to check project updates and important documents instantly through their respective portals. The system's reporting and analytic tool lets consultants monitor project progress, resource allocation, and client satisfaction, allowing them to improve workflows and show clients their worth. For accounting firms, being able to break down big projects into smaller tasks with deadlines and assigned staff becomes an advantage for consultancies.

Project Management System

Many study has shown multiple factors influence project management across different studies like Lalic et al. (2022) study examines how project management methods impacts project success, while Buchynska and Homotiuk (2023) concentrate on communication strategies in global companies. Their findings highlights the importance of formal and informal approaches in achieving successful project outcomes.

Shikov's (2023) study presents research that established an interest in the creation of web applications for managing company projects and ideas. He discusses designing, implementing, and testing a web service designed to help users manage and evaluate projects. The application allows users to develop their own business projects while evaluating others' solutions. The primary goal is to take into consideration the cost categories linked with the project's feasibility research track development versions in an

automated fashion. The study area was looked into, and the system's design and data model required for functioning are explained. The data gathered is used to determine the system's requirements, enabling a thorough design. The chosen technologies for the project's execution are examined, the server and client components are built, and the results of the implementation are displayed.

One study by Purwandari and Raharjo (2020) discusses the challenges faced in Agile projects across PMBOK knowledge areas including project resources, integration, scope, schedule, and stakeholder management. Stakeholder management, which includes problems during Agile adoption and transition, is the most significant hurdle. There was also a study on the difficulties in managing resources, integration, maintaining project scope, and timetable. To address these challenges, use resources like the Agile Practice Guide, PRINCE2 Agile, and the PMBOK Guide. The findings of the study provide insightful information for both researchers and professionals who want to understand and address the challenges with Agile methodologies. To understand this academic perspective, research by Webthesis (2021) provides real world insights by analyzing a consultancy firm's application of Waterfall and Agile approaches for an actual project. This can demonstrate how consultancies address the Agile's complexities in practice.

Ottaviani et al. (2023) recent research shows how the trend of technologies is increasingly built into Project Management Information Systems (PMIS). Weinert and Banaś (2024) examine how these systems handle traditional project management approaches, like storing, organizing, and tracking project data following Project Management Body of Knowledge (PMBOK) guidelines. However, Ottaviani et al. (2023) found that without standard data formats, PMIS limits its utilization of information with

other business tools, wasting data. To fix this, a standard PMIS model was proposed that defines PM elements and describes how they work and connect. Task and resource management, scheduling projects, risk management, and tracking progress are the model that supports essential functions. The study shows the advantages and effects of applying this standard model while recognizing its limitations and suggesting practical ways to implement it.

Additionally, the study highlights the importance of proper project management tools for construction projects, especially when dealing with budget allocation, timelines, and successful outcomes. As managers compare the capabilities and outputs of different Project Management Software (PMS) can do, they can also make informed decisions regarding their project's specific demands. The study also points out the significance of familiarity with how to use their chosen software well and the potential challenges with accessibility and usability. These findings (Franz et al., 2023) help provide valuable insights for managers to choose and use PMS effectively to improve their planning and work processes.

Previous research has explored different PMS functionalities and methods, but real-world industry implementation matters most. Santos et al. (2022) tested four tools, such as Primavera P6, MS Project, ProjectLibre, and MS Excel, on an actual commercial project in the Philippines. Their findings show that choosing the right software must consider factors like the project's budget, scheduling requirements, and how easily staff can access and use the tool.

Client Management System

As researcher widely explored how Customer Relationship Management (CRM)

systems have influenced different sections including its possibility to transform business approaches and strengthen customer relationships. Guerola-Navarro et al. (2022) semi-systematic review of 86 studies to explore CRM effects on entrepreneurial marketing, highlighting its role in identifying and leveraging new opportunities while adapting to meet customer demands. Their review also demonstrate CRM with relationship marketing and customer-driven models, illustrating its broader impact on entrepreneurial innovation. In different sector, Ngelyaratan and Soediantono (2022) focus on the application of CRM in the defense industry, by applying the Kaizen method to improve business processes and customer relationships and reviewing 30 journal articles (2014-2021).

According to Specchia (2022), implementing a CRM system for small and medium organizations will greatly improve customer interactions, sales processes, and business success. This book attempts to equip business owners to use CRM technology efficiently so they can fully profit from this potent tool by emphasizing it is useful, business-oriented features.

According to Yang and Babapour (2023), as technology changes quickly, businesses worldwide are increasingly using Information Technology (IT) tools like E-CRM systems. The goal of this study is to examine how customer satisfaction and E-CRM performance in e-commerce platforms are influenced by information availability, security measures, and system quality. As digital technology has advanced, companies are implementing E-CRM to improve customer relationship, boost loyalty, and increase profits.

In addition to Frangos (2022), email communication analysis has great deal of potential to improve CRM in the fund administration sector. By using a deep learning architecture that combines natural language processing (NLP) with graph-based techniques, the study

offers a scalable and objective way to assess service quality and client interaction.

This study examines how a Customer Relationship Management (CRM) system could solve operational challenges and strengthen the business approach for Umrah division at PT XYZ Tours & Travel Indonesia. Following Situngkir's (2021) framework, it aims to streamline workflows, enhance customer relationships, and help company expand its reach to target customers.

According to Sardjono (2023), Customer Relationship Management (CRM) is integral to business continuity and development, facilitating the establishment and maintenance of sales and customer relationships. However, a retail company found their actual sales falling short of targets even after adopting CRM technology. This study aims to analyze why this gap exists and develops an appropriate statistical model, and suggests practical solutions for improvement.

This study provides insights to the challenges companies face when implementing Social CRM systems. By mapping out these challenges, industry professionals can create specific solutions to address them. According to Raphael (2023), overcoming these challenges helps build more stronger and sustainable relationships among construction projects.

According to Lamrhari (2021), Social CRM has become essential for companies aiming to enhance customer experience and refine their marketing approaches. Companies can better understand what customers want and how they behave, by incorporating social media data into CRM processes. However, managing this vast amount of social media data isn't easy. To address this problem, our research introduces a social CRM analysis method designed to help businesses keep existing customers, attract new ones, together with their

past transactions.

A challenge of integrating social media data into CRM procedures are successfully addressed by the suggested social CRM analytical methodology. The framework uses cutting-edge analytical tools to improve client acquisition, retention, and conversion rates. The insights this study offers can help practitioners and scholar alike make better decisions and build stronger relationships with customers.

The study of Sunding (2023) shows Customer Relationship Management (CRM) systems have become vital tools for businesses, offering a range of functionality to organize and automate organizational processes. While CRM systems are sometimes work as a component of larger ERP platform, their main job is improving how companies interact with their customers. This project focuses on creating a new CRM tool designed specifically for customer service, addressing shortcomings in existing CRM solutions. Customer retention is important for growth and sustainability in the competitive business environment. Many existing CRM tools concentrate too much on getting new customers, often forget about supporting them later which affects satisfaction and loyalty. When customers leave, it damages both growth and profits. Thus, building a CRM system is necessary with a strong focus on customer service.

However, the functionality offered by CRM systems keep growing and changing in what they can do. Data Analytics Capabilities (DAC) is one of the most important upgrades for CRM effectiveness. Jabado's 2024 study of retail industry shows these DAC are essential for making CRM systems work well. His research proves that integrating DAC leads to smarter marketing choices and, see better profits. These findings change how we think about technology's role and offer useful insights for businesses wanting to improve

their CRM systems.

According to Rane (2023), the study analyzes multiple customer satisfaction metrics, including survey formats and how employee satisfaction affects consumer satisfaction. The study also emphasizes how technology affects customer experiences and loyalty. In addition to discussing the ways to track and preserve service quality in online platforms, this also shows how digital tools and social media influence customer interactions. Maintaining positive customer relationships also requires efficient complaint handling and user feedback thus, CRM is one of the most effective for building client loyalty and helping businesses keep their customers engaged over time.

The article discusses why understanding client worth matters and the different approaches to measuring it. It offers practical methods for improving client loyalty through high-quality support. Understanding customer expectations, empowering and training staff, personalizing the client experience, ensuring consistency across touch-points, communicating promptly and effectively, concentrating on continual improvement, rewarding loyal clients, fostering emotional bonds, effectively resolving complaints, measuring and tracking satisfaction among clients, anticipating customer needs, promoting and reacting to client feedback, and investing in technology are some of these tactics.

Hypertext Markup Language (HTML)

According to an article by Sufiyan (2023), *"HTML is primarily used to develop web pages and web applications that run on web browsers like Chrome, Firefox, and Internet Explorer."* Developers use HTML (Hypertext Markup Language) because it is the foundation language for creating and structuring content on the web. Every developer

undoubtedly uses HTML when creating a project. The researchers used HTML as the primary technology since the system is a web-based application.

Additionally, HTML is compatible with other technologies that the researchers used for the system's development, like CSS (Cascading Style Sheets) and JavaScript. Its simplicity and easy-to-understand syntax make it quick learn and implementation, which is why it remains a cornerstone technology for web developers across the globe. Furthermore, HTML's regularly updates the standards set by the World Wide Web Consortium (W3C), ensuring to keep pace to meet the needs of modern web applications, by adding new features and enhancing web accessibility.

Cascading Style Sheet (CSS)

According to an article by Juviler (2024), HTML forms the backbone of website development, enabling developers to structure the elements, from headings, paragraphs, images, tables, forms, lists, and so on. However, while HTML handles content organization, you can't control how these elements look on the page. CSS (Cascading Style Sheets) is a potent tool for managing the visual display of web pages and allowing the separation of content from design, which is why we have it. CSS enables developers to use uniform layouts, fonts, colors, and spacing across several web pages, guaranteeing a unified and expert look. This division improves maintainability because design modifications can be done in a single stylesheet without affecting the HTML content.

As Kumar (2022) states, *"A Cascading Style Sheet, commonly known as CSS, is a layer of design over HTML elements. It is a general idea that HTML is like a skeleton and CSS is the skin over it."* CSS supports responsive design, making the system functional on

various devices, particularly those used by the client. CSS is a tool that enables researchers to develop a user-friendly website, especially for the client's side. One of the ultimate goals of this study is to build a system that is easy to understand and user-friendly. Thus, incorporating CSS is a primary technology that can be used alongside HTML.

JavaScript

According to an article by Ozanich (2022), "JavaScript is the flagship of interactive web development, and as a result, it is all but universal in this industry." This widespread usage and the large range of potent features it provides demonstrate this, making it a crucial tool in contemporary web development. JavaScript is a flexible programming language that is mostly used to create dynamic and interactive web content. It will improve the user experience by enabling researchers to incorporate intricate elements like animations, interactive forms, and real-time updates.

As O'Grady states (2023), *"The JavaScript programming language is text-based and can be used on both client and server sides."* JavaScript is crucial in building an engaging and user-friendly interface. It can power the dynamic elements of your platform, such as real-time project updates, interactive dashboards, and client communication features, which were implemented on the system. Additionally, JavaScript frameworks were a big help throughout the development process.

ReactJS

According to an article by Modan (2023), "*Ever since its launch, it has taken the front-end development space by storm."* Industry giants like Apple, Netflix, and PayPal are using

React. With so many companies, including some of the top brands in the world, adopting React, it must have something significant to offer. Using ReactJS significantly aids researchers in developing the system, as ReactJS offers a component-based architecture that aligns well with the modular nature of such a complex system. This allows developers to break down the user interface into reusable components, promoting code reusability, maintainability, and scalability. Given iConsult's need for user-friendly functionality for project management and client management systems, ReactJS provides a solution that meets the demands of modern business consultancy.

According to an article by a Consummate Custom Software Development company, TatvaSoft (2024), “ReactJS comes with a variety of features including the virtual DOM, JSX, states, props, one-way data binding, React Hooks and more.” ReactJS comes with many features that help the development process. Additionally, researchers used JavaScript as part of front-end development, and ReactJS as its framework is an excellent choice since it is an open-source JavaScript library maintained by Facebook and a community of developers. You can find a vast collection of React tutorials on YouTube and explanatory articles on the internet from professionals and experts, which means it will be easy to learn ReactJS.

MySQL

According to an article by Quach (2024), “*If you’re venturing into the world of databases, chances are you’ve already heard of MySQL and its reputation as a powerhouse of data management.”* A large number of websites, applications, and businesses are built on top of MySQL, an open-source Relational Database Management System. Its popularity

can be attribute to it is reputation for data management and the wealth of developer-created web tutorials that make learning it easy.

MySQL is perfect Client-related data, including contact information and project specifications, can be effectively stored and managed by MySQL. This guarantees that client communications are easily captured and made available for project management. MySQL offers a strong foundation for administrators to manage user accounts, permissions, project data, and system configurations. Its support for complex queries and transactions enables admins to efficiently organize and track projects, assign tasks, and oversee client interactions. MySQL is commonly used in systems with client and admin sides because of its reliability, scalability, and flexibility.

Additionally, as Domantas (2024) states, "*With its Access Privilege System and User Account Management, MySQL sets the security bar high. Host-based verification and password encryption are both available."* The system that will be implemented will store sensitive information about clients and business services. While SQL (Structured Query Language) is not a security feature, databases like MySQL implement security measures to protect data and prevent unauthorized access.

NodeJS

According to an article by Capan (2023), *"Node.js uses non-blocking, event-driven I/O to remain lightweight and efficient in the face of data-intensive real-time applications that run across distributed devices."* The desired system to develop will involve a significant amount of I/O operations, such as reading/writing to databases, handling client requests, and managing project documents. Node.js is highly efficient in handling such I/O

operations due to its asynchronous nature, ensuring fast and responsive interactions.

Karen (2022) states, *"With Node.js, JavaScript is used for both front-end and back-end development, making the language more consistent across the entire application."* With Node.js, researchers would not have to use different languages to develop the desired project and client management systems. For instance, the majority of applications employ distinct languages for the back-end (such as PHP, Ruby on Rails, or Java) and front-end (such as HTML, CSS, and JavaScript). The system is kept lightweight and effective by Node.js's non-blocking, event-driven architecture, which can manage several connections simultaneously without sacrificing performance. This is essential for project management and client management systems, which involve real-time project updates, client communications, and data synchronization.

Vite

According to an article by Gurung (2024), the data collected from the survey and research found most participants favored utilizing Vite over CRA for designing React applications. Because Vite is lighter and faster, the participants said they liked it. Vite was more popular than CRA, according to the popularity index derived from GitHub, npm, and Google trends. More developers are moving from CRA to Vite because of quicker development and building times. Given that the React team has deprecated CRA, Vite may become a viable alternative to CRA, although lacking the same degree of ecosystem as CRA in terms of information and resource availability.

XAMPP

According to an article by Naik (2023b), XAMPP is an open-source, free cross-platform web server. The acronym for Cross-Platform, Apache, MySQL, PHP, and Perl is XAMPP. Programmers may develop and test their code on a local server for the web using XAMPP, a well-known cross-platform web server. Apache Friends developed it, and its native source code is openly editable. Among other computer languages, it has MariaDB, Apache HTTP Server, and PHP and Perl interpreters. A developer may quickly and simply install a WAMP or LAMP stack on the operating system thanks to XAMPP's deployment ease. Additionally, typical add-in software like WordPress and Joomla can be loaded.

Express.js

According to Sharma (2024), Node.js is a fast JavaScript runtime environment used to build server-side applications; however, it lacks the capability to serve files, handle requests, and manage HTTP methods, which is exactly why Express.js comes in handy. Developed on top of Node.js, Express.js is a framework that speeds up the process of creating web apps and APIs for both mobile and online platforms. Its features enable developers to efficiently create single-page, multi-page, and hybrid applications. Moreover, Express.js enhances server management and routing capabilities, thereby saving substantial coding time often cutting it by nearly half while maintaining the efficiency of web and mobile applications (Sharma, 2024). The framework is particularly appealing for new developers because it is written in JavaScript, an accessible language for those without prior programming experience. Additionally, Express.js is designed to be time-efficient, fast, economical, and easy to learn, with an asynchronous nature that further improves

performance. Overall, Express.js not only simplifies the development process but also empowers a broader range of developers to engage in web development, making it a vital tool in the modern web ecosystem.

Github

According to an article by Juviler (2024P), *"Today, GitHub is one of the most popular resources for developers to share code and work on projects together."* Using GitHub for the iConsult project provides numerous advantages that significantly enhance development and project management. GitHub is built on Git's version control foundation that enables collaborative coding seamlessly, track changes, and maintain a history of code changes that is essential for complex system development. Its collaboration features including pull requests, code reviews, and issue tracking, ensure transparent communication and code standards. Additionally, for researchers, Github enhances project coordination with each other, promotes peer feedback, and simplifies communication.

According to an article by Kinsta (2023), GitHub lets anyone host public code repositories at no cost, which explains its strong appeal for open-source work. Its accessibility has made GitHub significant and popular in open-source communities by removing barriers to participation while encouraging transparent teamwork and advancing open-source innovation forward. Therefore, GitHub is a good choice for researchers, as it creates an ideal space for shared coding projects and exchanging technical knowledge.

Visual Studio Code

According to an article by Uspenski (2023), Microsoft's Visual Studio Code (VS Code)

is a free and open-source editor that is compatible across most platforms like Windows, Linux, and macOS operating systems. There are certainly a number of strong benefits to developing the client management and project management system with Visual Studio Code. First of all, developers of all skill levels can use Visual Studio Code because it is a free and robust Integrated Development Environment (IDE). It guarantees compatibility with the technologies necessary for developing the iConsult system through its broad support for a variety of programming languages, including JavaScript, as well as other tools that researchers will need, such as HTML, CSS, and JSX.

As Pedamkar (2023) states, "In normal terms, it facilitates the user's ability to write the code easily. Many say it is half an IDE and half an editor, but the decision is up to the coders." This IDE also supports Windows, which is what the researcher will use. Additionally, VS Code doesn't occupy much drive space; you will need less than 200MB of free space to download it. Furthermore, it has a simple user interface, making coding and troubleshooting easier for researchers.

Heroku

According to Postema (2021), Heroku is a cloud platform that allows users to build, monitor, and scale the app. Heroku features various features, including practicality, dependability, and accessibility. It worked really well to deploy a static website to Heroku. However, there were several difficulties in the process of deploying the website, including using the programming language that Heroku supports, using the Git bash fatal using commands, and converting a static website into a web application.

SendGrid Email

According to Mittal (2024), SendGrid is a comprehensive email service provider that offers a robust Email API for developers, along with a suite of marketing tools designed to streamline business communication. Beyond its core email-sending capabilities, SendGrid offers advanced email analytics to optimize outreach strategies, including open rates, click-through rates, and geographical engagement. Its wide integration network supports over 320 applications, allowing businesses to connect SendGrid with their existing CRM and marketing platforms, enhancing communication processes and overall business operations. The Email API leverages SMTP (Simple Mail Transfer Protocol) to facilitate the sending of large volumes of transactional and triggered emails, such as password resets, account confirmations, and shipping notifications (SendGrid, 2024).

Hypertext Preprocessor(PHP)

According to Apiag (2023), PHP (Hypertext Preprocessor) is explored for its advantages and applications in business consultancy, particularly when integrated into project management and client management systems. PHP is widely acknowledged as a powerful tool for web development, particularly for building dynamic and interactive websites. Originally developed by Rasmus Lerdorf in 1994-1995, PHP has developed into a capable server-side scripting language that is ideal for handling server-side operations and database communications. PHP's key strength lies in how it easily interacts with HTML that enables developers to directly insert PHP code within HTML pages, simplifying web development.

Hypertext Preprocessor (PHP) enhances the development of web applications and

makes it easier to create dynamic content. It supports a wide range of different databases, including PostgreSQL, MySQL, and Oracle. PHP adaptability helps developers build web applications that interact with multiple data sources and handle key functions like storing, accessing, and modifying data. In business consultancies dealing with project and client data, PHP's database features make it easier to create tailored solutions for specific business needs. Since PHP is free and open-source, it is popular among developers and companies. Its accessibility, community support, and regular updates make it an attractive option for businesses of all sizes.

Paypal

In 2023, Shadlou study states that the PayPal API simplifies payments by integrating straight into the platform of the organization. This makes transactions easier for clients while keeping them secure through PayPal’s trusted system. Since many customers already know and trust PayPal for online purchases (B2C), they’re more likely to feel comfortable using the consultancy’s services. The API also accepts multiple payment options including credit cards, debit cards, and PayPal balances giving clients flexibility. This versatility accommodates diverse client preferences, improving the overall user experience and reducing friction in completing transactions. The PayPal API automates payment scheduling for recurring payments such as retainer fees or subscription-based services. This automation not only enhances operational efficiency but also ensures timely and consistent cash flow for the consultancy, contributing to financial stability.

Google Drive

According to Dinatha (2021), it is done through the Internet. This service uses Google Drive cloud storage as a middleman and data storage medium. The capacity of the system to benefit from the information being shared is Google Drive's solution to the interoperability issue. This service's objective is to support an organization's business operations and offer a backup communication channel in the event that the primary system malfunctions. One of the benefits of using Google Drive cloud storage for data interchange is that it is not restricted by IP addresses, allowing for operation anywhere there is an Internet connection. This service requires a Google account, which serves as the sender's identifying information and for authentication.

Figma

According to an article by Miller (2021), *"The user interface is uncluttered and well organized, making it extremely easy to use."* For several reasons, Figma is an excellent choice for creating UI/UX designs for this project. Firstly, Figma is a collaborative design tool promoting effective teamwork and collaboration, it allows team members to work on designs simultaneously. Figma's collaborative capabilities made smooth communication and design idea iteration because the proposed undertaking involves multiple designers. Second, creators of all skill levels may utilize Figma thanks to its intuitive design tools and user-friendly interface. Furthermore, Figma's cloud-based system lets team members access and work on design files from anywhere with an internet connection. This remote collaboration feature proves especially valuable for distributed teams. As Ahwan (2021; 2020) points out, Figma offers free Professional plans for students and educators. Among

various design tools available, Figma stands out for being more budget-friendly than many alternatives. While other high-end options exist, Figma delivers all the essential features needed for the iConsult project at a lower cost

JawsDB MySQL

A relational database that is fully functional and managed for usage with your application is provided by JawsDB, a MySQL/MariaDB Database-as-a-Service (DBaaS) provider. JawsDB offers one-click delivery and cloud-based relational database management, saving you the trouble of hosting, configuring, patching, and maintaining a database. JawsDB is simple to use, and interacting with it requires no special scripting or libraries. All you have to do is pick a plan that suits your requirements, and we'll provide you with the hostname and login information you need to get started.

ISO 25010

Software quality standards include ISO 25010, *"Systems and software engineering – Systems and software Quality Requirements and Evaluation (SQuaRE) – System and software quality models."* Together with helpful advice on how to apply the quality models, it explains the models, which comprise features and sub-characteristics for software product quality as well as software quality in use. According to ISO 25010's characteristics, software quality is determined by how well it complies with design specifications as well as non-functional requirements like security or maintainability. Software quality measurement measures how well the program performs in relation to each of the attributes.

According to ISO/IEC 25010 comprises eight quality characteristics, and these are:

Functional Suitability

Functional suitability is the degree to which a system or product can fulfill explicit and implicit needs.

- **Functional Completeness:** The collection of functions that address all of the designated tasks and user goals.
- **Functional Correctness:** The ability of a system to produce accurate results that meet specifications.

Reliability

The ability of a system to consistently perform required tasks under defined conditions.

- **Maturity:** The ability of a system to know how reliable a system or product meets your dependability needs.
- **Availability**: shows if the system is operational and accessible when needed.
- **Fault Tolerance:** The ability of a system to keep working correctly even when hardware or software problems occur.

Performance Efficiency

The performance efficiency measures how well a system uses available resources to deliver results.

- **Time Behavior.** Describes a system responds to inputs and processing data during operation.
- **Resource Utilization.** This refers to what kind and the quantity of resources a product requires and the amount needed during operation.
- **Capacity.** Refers to the maximum limit of what a system or product can handle.

Usability

Usability is the extent to how effective, efficient, and satisfying a system facilitates users in achieving their goals.

- **Appropriateness Recognizability.** Refers to how easily you can tell if a system fits your requirements.
- **Learnability.** Is the level of ease of use a system or product is.
- **Operability.** Refers to a system or product ease of control and operation.
- **User Error Protection.** The ability of a system to help in error prevention by guiding users and protecting against accidental actions.
- **User Interface Aesthetics.** Refers to how users find the visual of a systems to be visually appealing and attractive.
- **Accessibility.** The ability of a systems that is accessible to all users, with all of its features, regardless of ability.

Security

Security measures a system's ability to protect data from vulnerabilities and threats.

- **Confidentiality.** The ability of a system to ensure that sensitive data can only accessed by authorized users.
- **Integrity.** Its ability to protect programs and data from unauthorized access and changes.
- **Non-repudiation.** Providing evidence that specific actions or events actually happened, makes it impossible for anyone to deny that certain act.
- **Accountability.** To be able to track down and connect actions back to

unauthorized users.

- **Authenticity.** To how reliable we can verify the true origin or identity.

Compatibility

A system, component, or product to function properly and communicate with others when using the same hardware or software environment.

.**Co-existence:** The ability of a product to perform its required functions efficiently while sharing resources and operating environment with other products, without causing negative impacts.

- **Interoperability.** The ability of different systems, products, or components to communicate and utilized information effectively.

Maintainability

The ability of a system's maintainability measures how direct it is to update for improvements, corrections, or adaption to new conditions.

- **Modularity.** Components of a system can be changed separately without having a major impact on other parts.
- **Reusability.** Refers to how easily an asset can be modified and applied to different systems.
- **Analysability.** Refers to how effective an impact assessment can guide planned changes. It also involves identifying flaws, figuring the causes of failures, and determining which components need to be changed.
- **Modifiability.** Refers to how easily a system can be modified without causing errors or lowering its overall quality.

- **Testability.** Refers to how well a system, item, or part can be tested. It also explains how to conduct tests to confirm whether the requirements have been met.

Portability

Portability measures whether software or hardware can work properly when transferred between systems.

- **Adaptability.** Shows how easily it can be modified to work with changing technologies or conditions.
- **Installability.** Refers to how straightforward it is to successfully deploy and, if necessary, completely remove a software product.
- **Replaceability.** Measures whether a product can effectively take the place of a comparable alternative.

The ISO 25010 standard is used in this study to assess system quality. ISO 25010 framework, which was created as a component of the ISO/IEC 25000 series, this framework identifies nine key characteristics that is necessary for evaluating software performance and meets user needs (Santos 2021). These consist of operational suitability, dependability efficiency, maintainability, compatibility, security, usability, suitability, performance efficiency, and functional suitability.

Synthesis of Review of Related Literature

Previous research highlights the challenges that business consultancies encounter when using manual approach. Recent studies of Forage and Peek (2024), indicates that these outdated approached usually lead to missed deadlines, ineffective workflows,

miscommunication issues, and client dissatisfaction. Furthermore, traditional methods of project management and client communication, often involving spreadsheets and email exchange fails to meet clients' expectations for responsiveness (Azumah, 2024).

Recent findings highlight how important project management is for consultancy firms. Scholars like Tahri (2021) says that project success rates are greatly raised by improve strategic planning, workflow coordination, and budget distribution. Despite these insights, the market still lacks specialized tools that address to consultancies' dual needs. Although there are many online project platform available, Yan et al., (2020) shows that the majority of them do not integrate project with essential client relationship features into a single system.

As a result, this research gap presents an opportunity to develop a comprehensive system that addresses the challenges faced by consultants. By integrating project and client management can increase client satisfaction, speed up procedures, and improve communication. This study "iConsult," a web-based application designed to address the consultancy challenges identified in existing studies. The platform combines project management for managing task, timeline, progress tracking, document storage and reports with client features for online booking systems, and manage client information and their history. By focusing on these core functionality, iConsult aims to optimize business consultancy services, improve efficiency, and ultimately drive successful outcomes for both consultants and their clients.

Review of Related Studies

This study presents the related studies and works that have already been conducted that are relevant to the fields of project management and client management.

In 2021, Tahri conducted a study, "Designing a Project Management Integrated System", which successfully attempted to develop a project management system incorporating twelve management functions. These include workflow management, electronic document management, project portfolio management, content management, collaboration platforms, problems management, deliverables management, resource management, change management, configuration management, and satisfaction surveys. In order to facilitate multiple criteria analysis projects and project management maturity evaluation, the finished system incorporates essential elements of the current project management software together with extra capabilities and a dedicated portfolio management module.

A similar work was conducted by IEEE (2020) entitled "University Research Project Management System Based on Cloud Platform." This study designed a cloud-based university research project management system. It comprises the following modules: project application and review, project opening management, project progress management, project completion management, and display of research findings. The solution improves efficiency and saves time for teachers, review experts, and project management staff by enabling distributed submission and network evaluation of project declarations, opening reports, progress reports, and conclusion reports.

A study by Porntrakoon and Moemeng (2022) was titled "Student Project Management

System (SPMS)." This web-based program assists academic administrators, project advisors, and students in managing projects from beginning to end. Project title recommendation, team and project prioritizing, team and project matching, scoring and project management, and report production are some of its primary functions. There are now 228 individuals, 205 projects, 201 teams, 25 project advisors, and two administrators on the system.

In addition, several other works have developed systems with similar features aimed at improving project management processes. These include works by IEEE (2022), Baashar et al. (2020), Hamizah (2021), UTAR (2023), and Li et al. (2022), who describe a project management system built with the Vue3.0 framework. These studies focused on creating platforms that integrate various project management functionality, improve efficiency, and provide comprehensive project information and management tools.

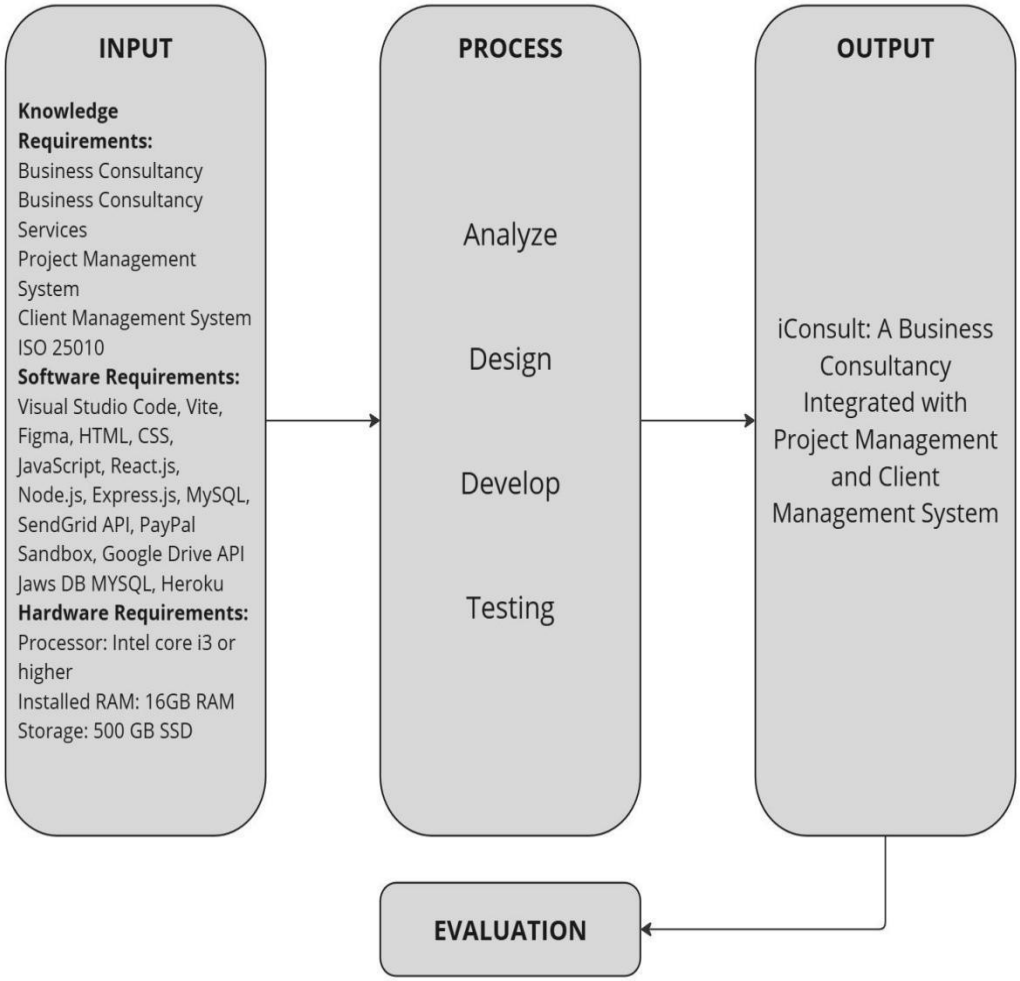
These studies collectively highlight the evolution of project and client management systems, introducing innovative features to streamline management processes and enhance project and client satisfaction. The current study will develop a platform that caters to both project and client management systems. Integrating features from previous research will offer a comprehensive solution for business consultancies, aiming to streamline the project management process and enhance client relationships. Leveraging insights from the authors, the current study aspires to address existing gaps in the market and provide an all-encompassing, user-friendly platform for modern business consultancy needs.

Conceptual Model of the Study

Figure 1

Conceptual Model of the Study

Figure 1 shows the conceptual model of the study, which illustrates the process of conducting a study.



Input

The input phase of the system consists of three main requirements. First, this project requires a consultant with knowledge of business practices, project management systems, client management, and ISO 25010.

Second, the researchers used several Software requirements. These include HTML, CSS, JavaScript, ReactJS, NodeJS, Express.js, and MySQL for handling databases. Additionally, Paypal sandbox, SendGrid API, Google Drive API, Jaws DB MSQL, and Heroku. The researchers used tools like Visual Studio Code, Figma, and GitHub throughout the development process.

Lastly, the researchers used Hardware Requirements that can be used to access the system, an Intel Core i3 processor (or better), 16GB of RAM, and a 500GB SSD.

Process

The four primary stages of this process are analysis, design, development, and testing. In order to comprehend the needs, objectives, and difficulties of the project, extensive investigation and evaluation will be carried out during the analysis phase. The collected data will be converted into a thorough plan throughout the design phase. This entails establishing the integration between the project management and client management systems, designing user interfaces, and developing the system architecture.

After the design phase, a rich package of technologies was used to build the iConsult system. The main IDE was Visual Studio Code, with GitHub for version control and Figma for UI/UX design. To guarantee a quick development process, the front end makes use of HTML, CSS, JavaScript, PHP, Vite, and ReactJs. Node.js and MySQL handled server-

side functions and data storage on the back end, while XAMPP was used to replicate server environments while the project is being developed. Important connectors include SendGrid for notifications, Google Drive API for file storage, and Paypal Sandbox for payments. Finally, the system was deployed using Heroku and Jaws DB MYSQL.

The system was put through extensive testing after development was finished. Finding any flaws, mistakes, or problems with the system's performance, security, or functioning was the goal of this step.

Output

The project's output was web-based, integrating a project management system and a client management system for business consultancies. This application allowed business consultancies to effectively manage projects, clients, appointments, communication, and other relevant aspects of their operation.

Operational Definition of Terms

The following terms are defined to better understand the study.

Clients refers to a customer who seeks professional services from the business consultancy.

Consultants refers to a business professional who provides expert advice and guidance to clients.

Track progress refers to the process of monitoring and evaluating the advancement of a project.

Task refers to the process of creating, assigning, monitoring, and updating tasks within projects.

Project refers to the process of overseeing and controlling all aspects of a project.

Employee refers to a staff member employed by the consultancy firm who handles assigned project tasks or client services.

Payment Status shows whether project fees have been settled, with these three classifications: Paid, Partially Paid, or Unpaid.

Planned Duration refers to the expected timeline for finishing the project, with both the scheduled beginning and conclusion dates.

Status refers to the project's current progress, which can be either Completed or Ongoing.

Professional Fee the total amount paid to the consultant for their services and the time needed to finish the task successfully.

Chapter 3

METHODOLOGY

This chapter includes the project design, project development, operating and testing procedure, and evaluation procedure of the study.

Project Design

The project design of the study is explained below using Unified Modelling Language (UML), System Flow Chart, UI Design, and Wireframe.

Figure 2

Use Case Diagram

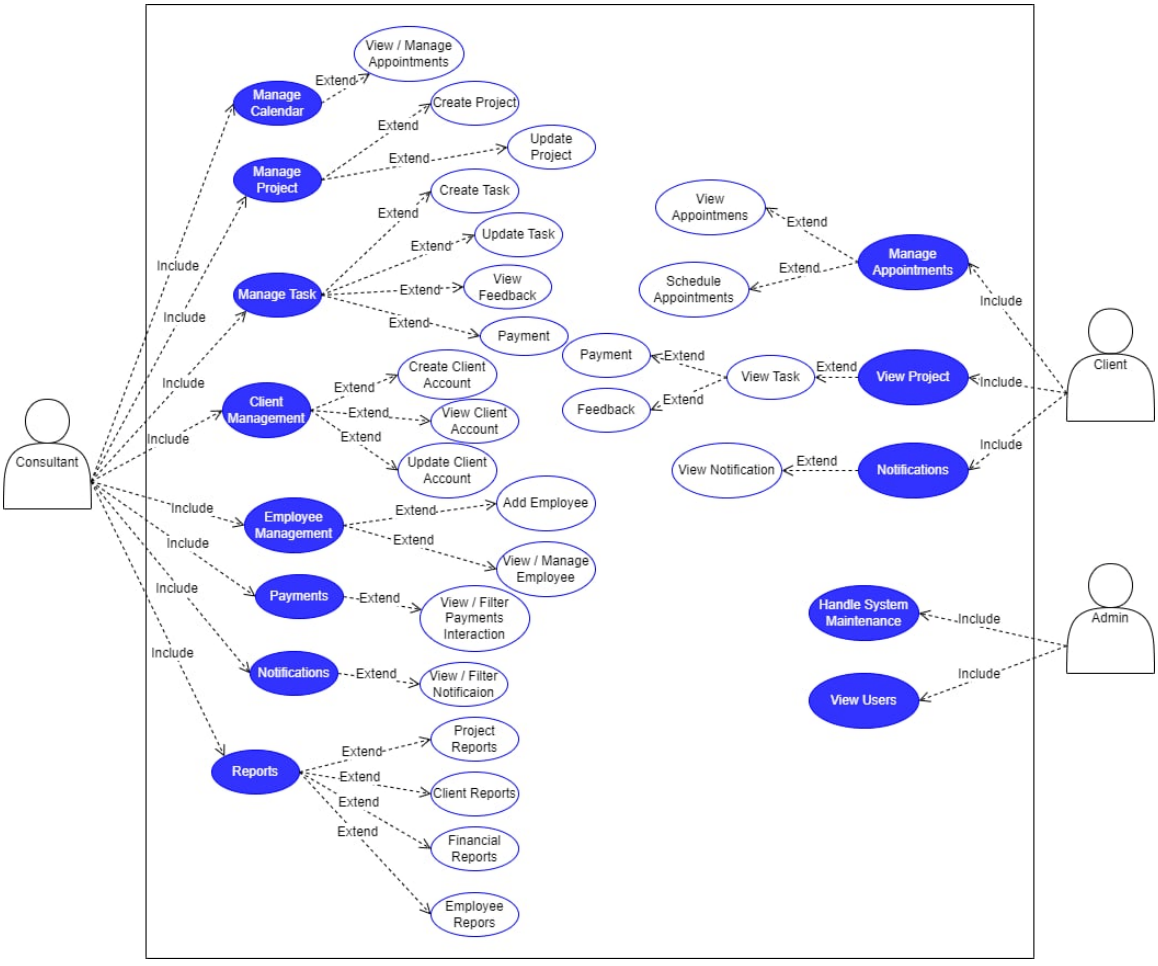


Figure 2 shows a detailed use case diagram of how iConsult works. iConsult, is an web-based application designed for business consultancy integrating features of project management and client management. This diagram provides a visual overview at who uses the system and what they can do with it.

Clients can easily look through available time slots and schedule their appointments. Furthermore, they’re able to view and follow the progress of the project. Within the "view task" section, clients can also access payment methods and share their feedback to evaluate the service quality. Moreover, the system keeps clients informed about important details, including upcoming appointments, deadlines, payment reminders, and any other important updates related to their project.

Furthermore, consultants handles all aspects of projects in iConsult. This includes managing appointments, creating project plans, assigning tasks, creating the client account, adding employees, and monitoring the notifications and project progress to ensure everything stays on track. Consultants can also utilize iConsult's reporting features to analyze project performance and gain valuable insights to guide decision-making.

Finally, the admin oversees system maintenance and manages all technical aspects to ensure smooth operation. They have access to view all users within the system.

Figure 3

Module Hierarchy Chart

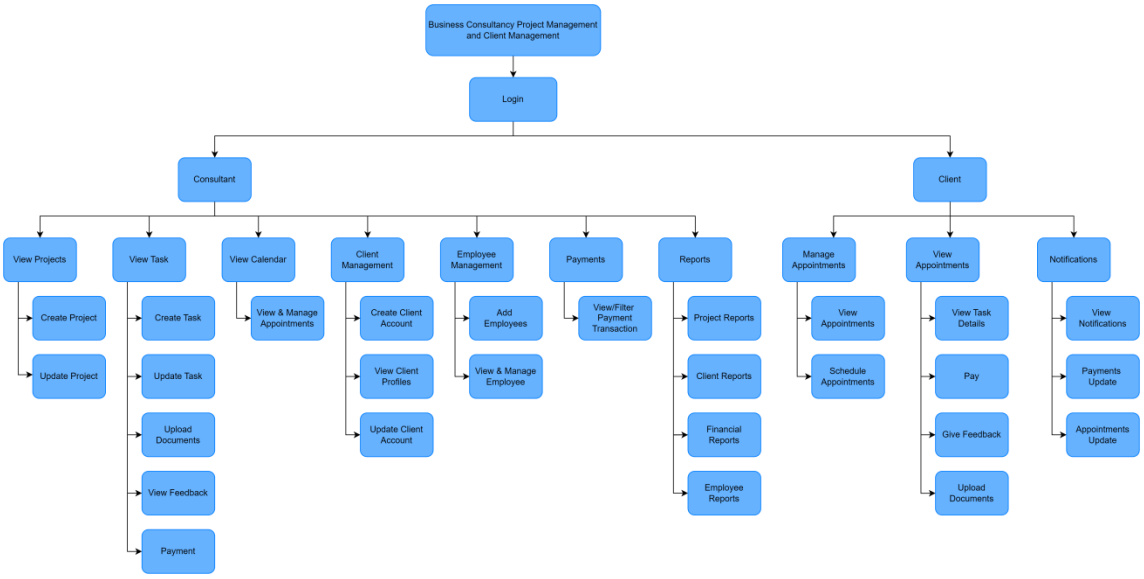


Figure 3 illustrates the hierarchical structure of the Business Consultancy Project Management and Client Management system. It depicts the interconnectedness of modules through the central Login functionality, which is accessed with consultant-provided credentials.

Consultants navigate key modules like View Projects, View Tasks, View Calendar, Client Management, Employee Management, Payments, and Reports, each catering to specific operational needs. Clients utilize (1) Manage Appointments for viewing their appointments and scheduling their appointments to manage their consultations, (2) View Appointments for tracking project details payments, giving feedback to the consultants, and uploading transaction documents. Notifications for receiving alerts regarding the status of the project, payment history, and appointment updates.

This diagram effectively conveys the organization and interrelationships between

modules by visualizing the module hierarchy. This clarifies the roles and responsibilities of consultants and clients within the iConsult system.

Data Flow Diagram

Figure 4

Context Level Data Flow Diagram

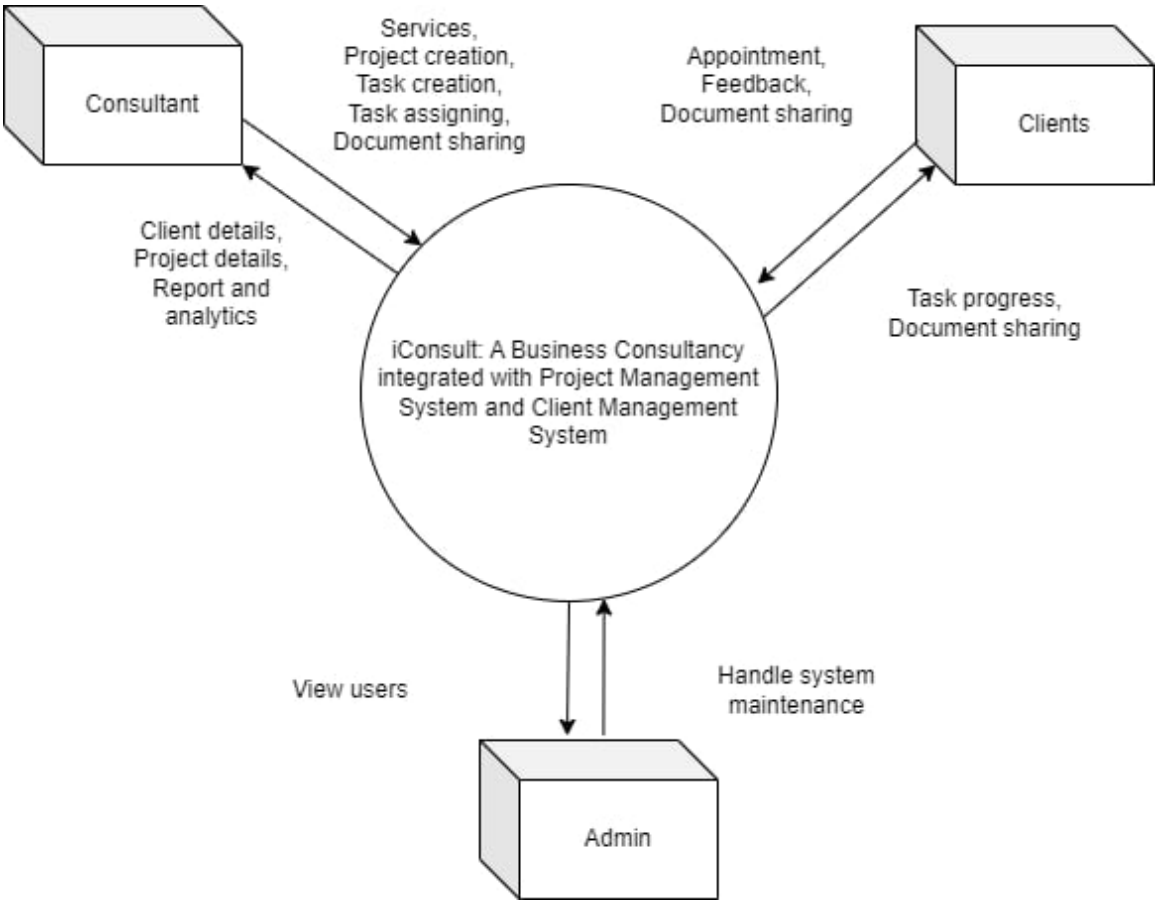


Figure 4 shows the Data Flow Diagram. It shows the flow of data between external entities and the whole system.

The system has three external entities: Clients, Consultants, and Admin. Interacting with it, clients schedule appointments, including their personal information, for the

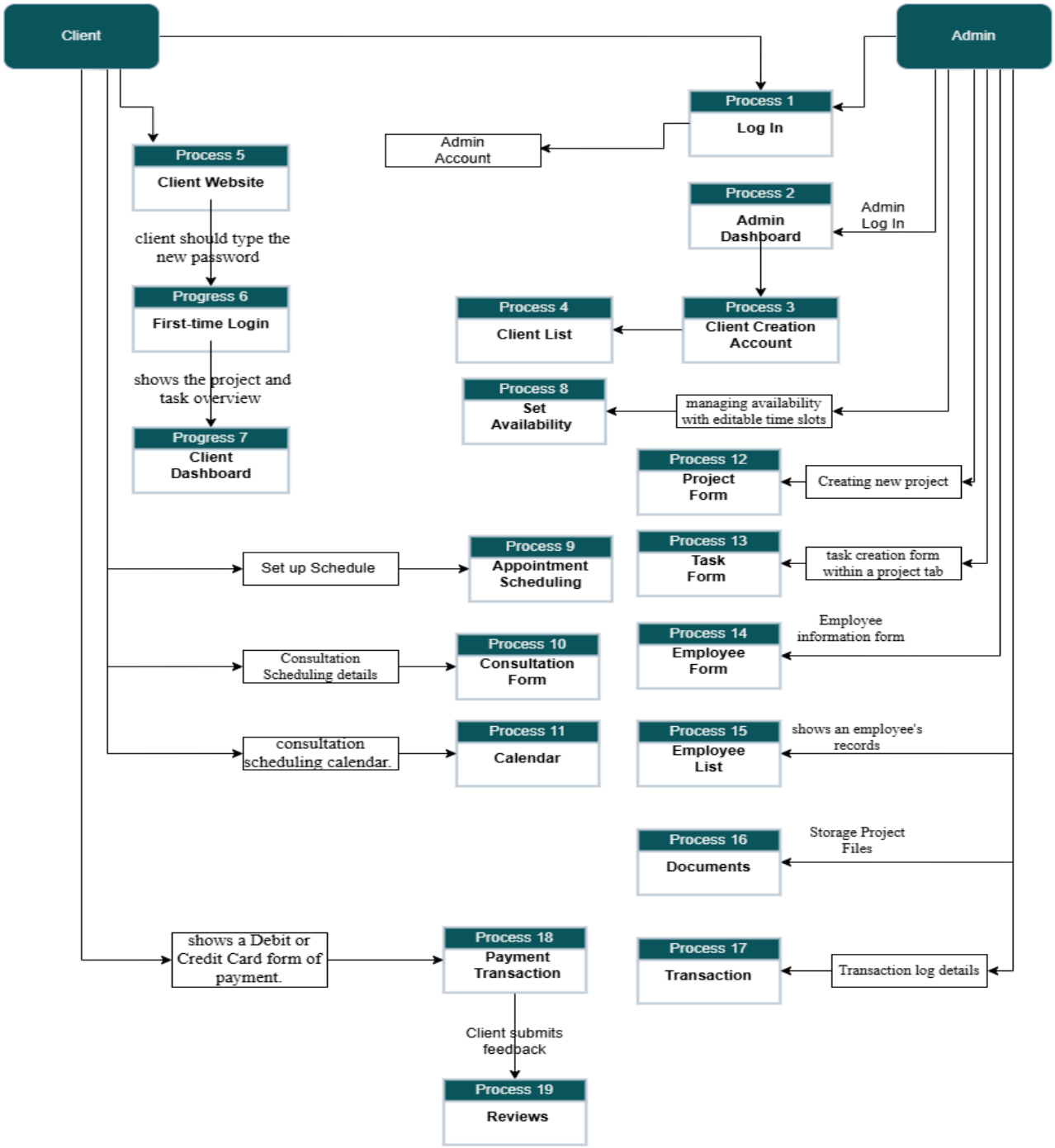
consultation. Next, clients need to upload relevant documents based on the discussion or requirements provided by the consultants. Lastly, clients can provide feedback on their experience and the quality of service they received, helping the system to improve and address any issues. They receive business services and are consulted through the system. Next, they can view task progress and also upload documents.

Consultants give customer service details, project creation, task creation, task assigning, and document sharing to the system. They can view client details, reports and analytics, and project details in the system.

The admin is responsible for maintaining the overall functionality and ensuring everything runs smoothly. The admin can also view all users, including consultants and clients, from the system.

Figure 5

Top Level Data Flow Diagram

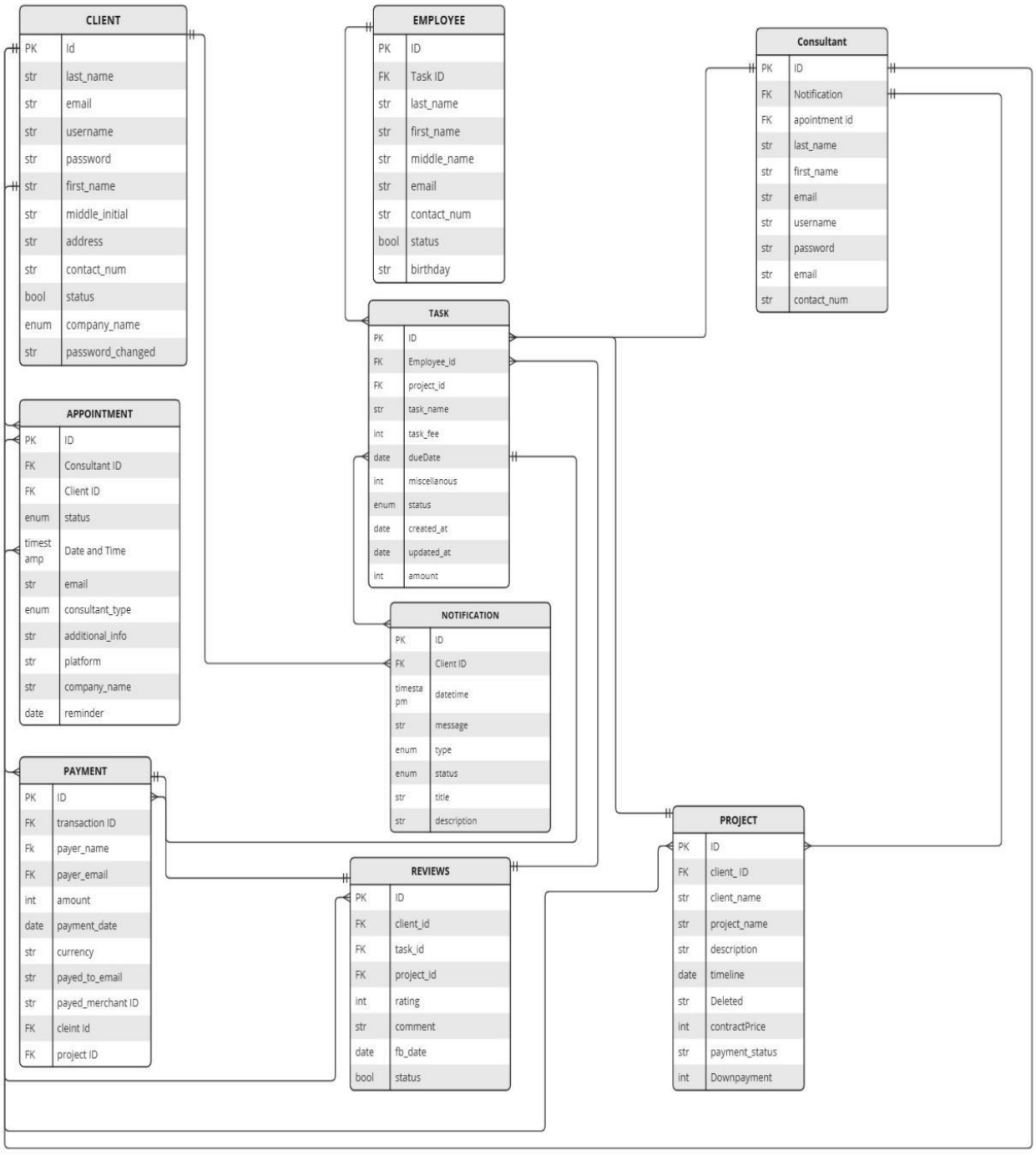


The system has two main entities: Client and Admin. Clients schedule appointments, manage profiles, and make payments, while Admins handle user

management, tasks, and projects. The System ensures seamless interactions. The Login Process authenticates users; new users must change their password upon the first login. Admins add clients, and clients manage availability. Appointment Scheduling & Consultation allows clients to book consultations, which are recorded in the system. Admins manage Projects & Tasks, assign employees, and track progress. Employee Management keeps staff records. Clients make payments, submit feedback, and leave reviews. Document Management allows file uploads and storage. Reports & Data Visualization provide insights into appointments. Databases store user credentials, project details, schedules, transactions, and documents, ensuring smooth operations and secure data flow.

Figure 6

Entity Relationship Diagram



This Entity-Relationship Diagram (ERD) as shown in Figure 6, represents the structure of a business consultancy management system with key entities such as Client, Appointment, Payment, Project, Task, Document, Feedback, Employee, User, and Notification. Each entity has its own attributes, with relationships established through foreign keys (FK).

The Client entity is connected to multiple entities such as Appointments, Payments, Projects, Feedback, Documents, and Notification. For instance, clients can schedule appointments, make Payments, and provide Feedback related to tasks. The Project entity links to Task and Document entities, allowing clients and consultants to manage and track tasks and upload relevant project documents. The Notification entity alerts clients about various updates (appointments, payments, etc.), and the Task entity assigns employees to specific projects, tracking the task's due dates and statuses. Feedback allows clients to rate and comment on services, and employees are linked to tasks through the Employee entity while the User manages login credentials for the system. The system is designed to efficiently handle various business processes, such as client interactions, project management, payment tracking, and feedback collection.

System Flow Chart

Figure 7

System Flowchart (Client)

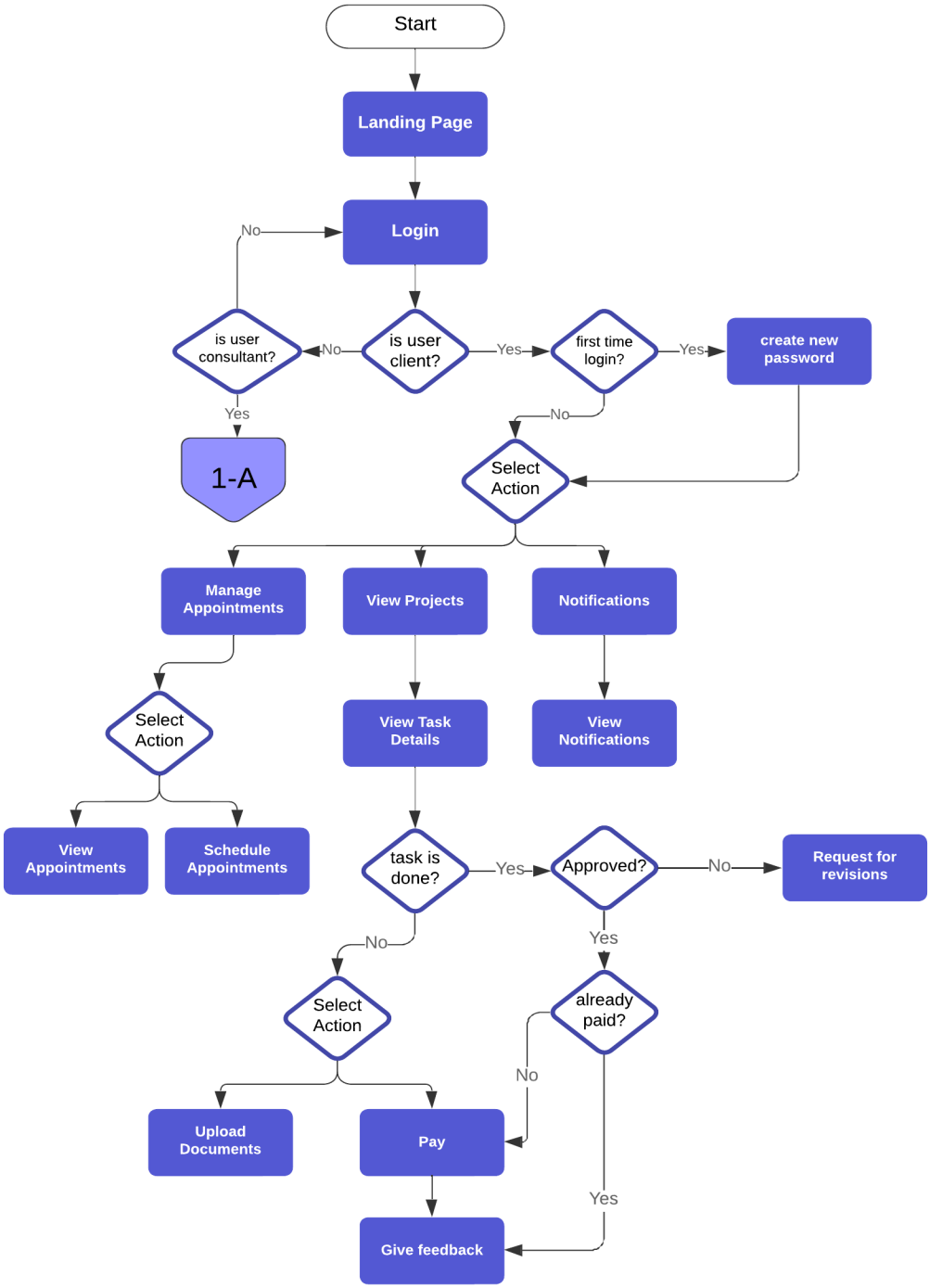


Figure 7 shows the flow of the "iConsult" from the landing page to the clients interacting with the system. The client begins by logging in, and if it is their first time, they will be prompted to create a new password. After logging in, the client can manage appointments (view or schedule), view project tasks, upload documents, and pay for services. Additionally, they can view notifications and provide feedback after completing tasks or making payments. This process ensures clients can efficiently manage their appointments, monitor project progress, and stay informed through notifications.

Figure 8

System Flowchart (Consultant)

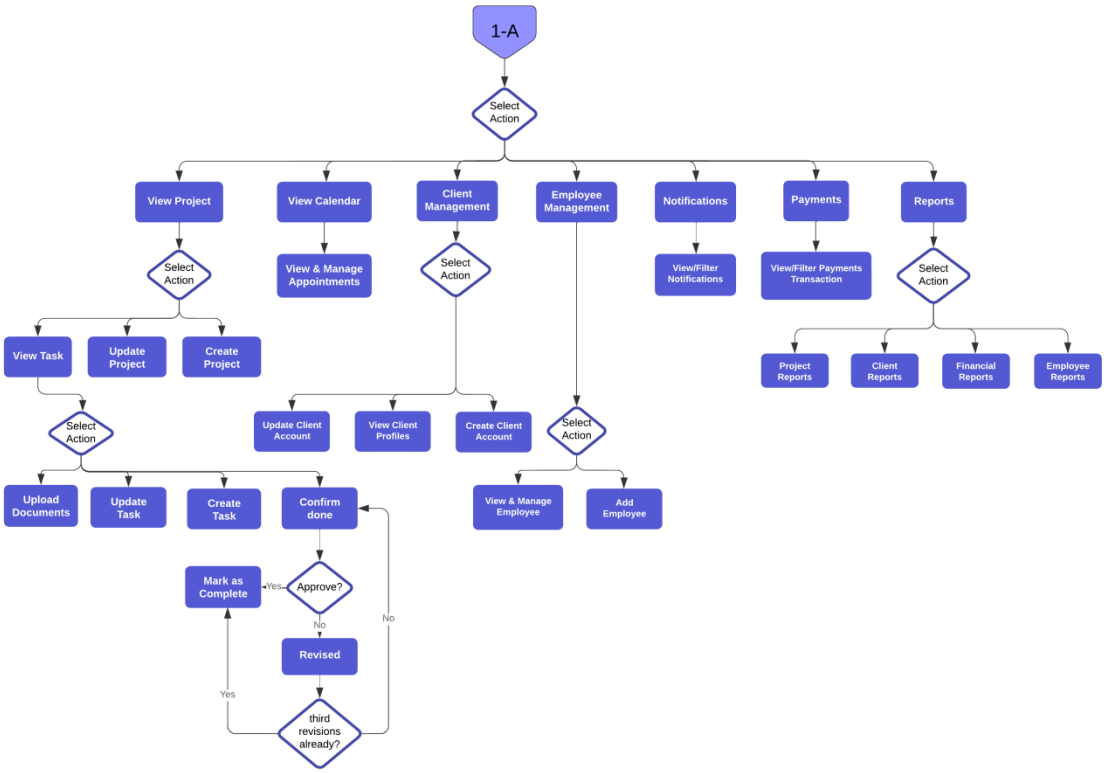


Figure 8 shows the flow of the "iConsult" from the landing page to the consultants interacting with the system. The consultants have several key responsibilities, including managing client appointments, managing projects by creating projects based on the services clients consulted for, breaking these projects into manageable tasks, and assigning tasks to specific employees. Additionally, the consultants can generate comprehensive reports, including project-specific reports, financial reports, employee reports, and client reports for comprehensive insights, as well as create accounts for clients. Once clients have been given their accounts, they can schedule appointments directly through the web interface, view updates on their projects or services, directly pay through the web application, view and upload documents related to the tasks, and provide feedback after a service is completed, contributing to the continuous improvement of service quality.

Wireframe

Figure 9

Login

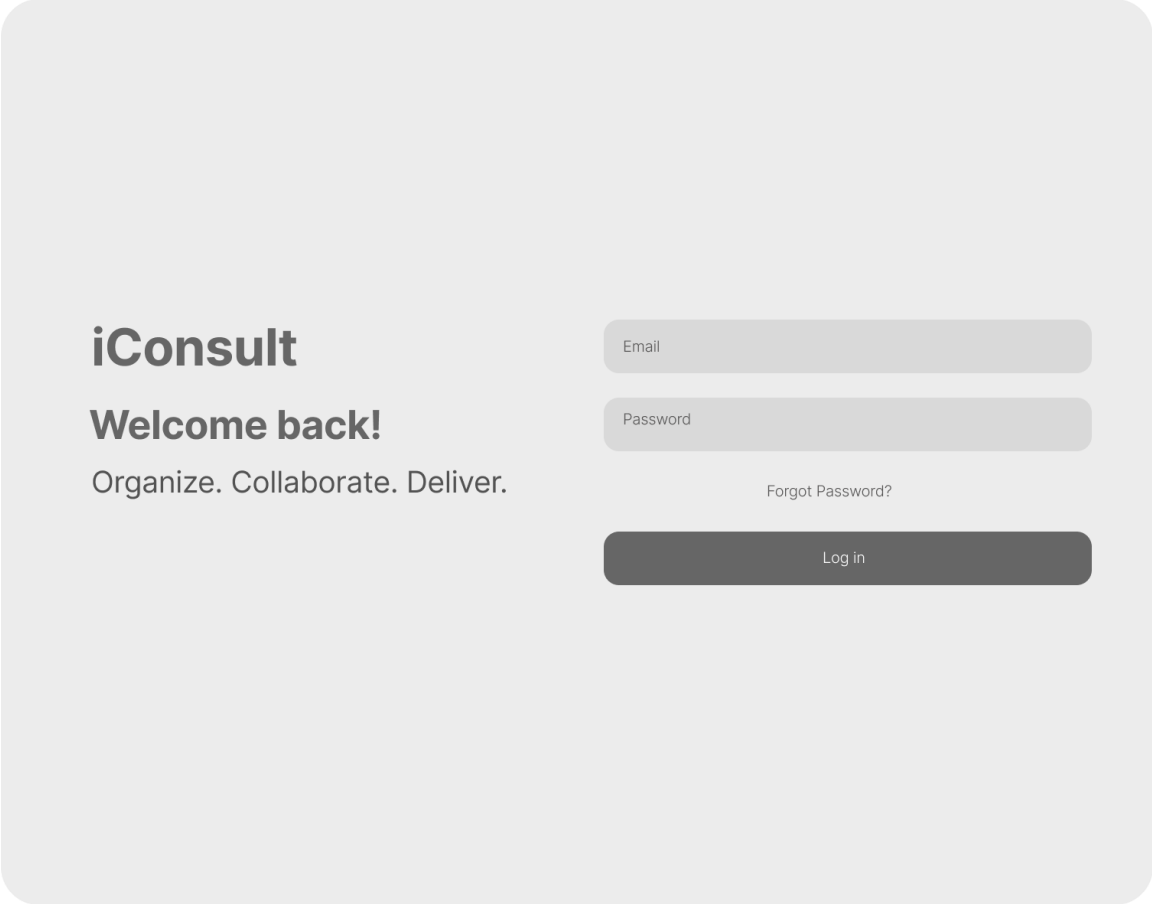


Figure 9 illustrates where consultants and clients can log in to access their accounts.

Figure 10

Consultant Dashboard

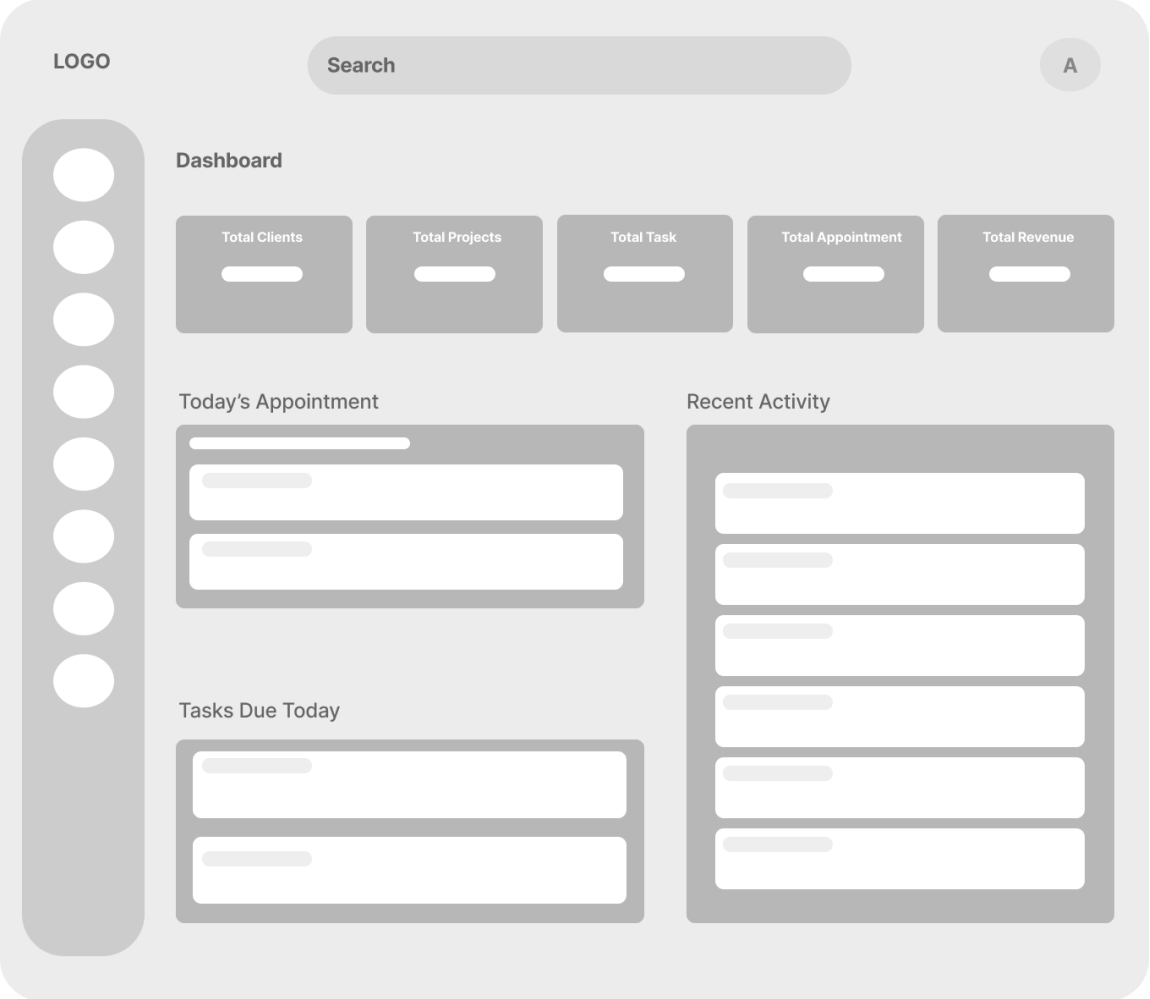


Figure 10 shows where the consultant can view real-time analytics and monitor the overall performance of the business.

Figure 11

Consultant Calendar

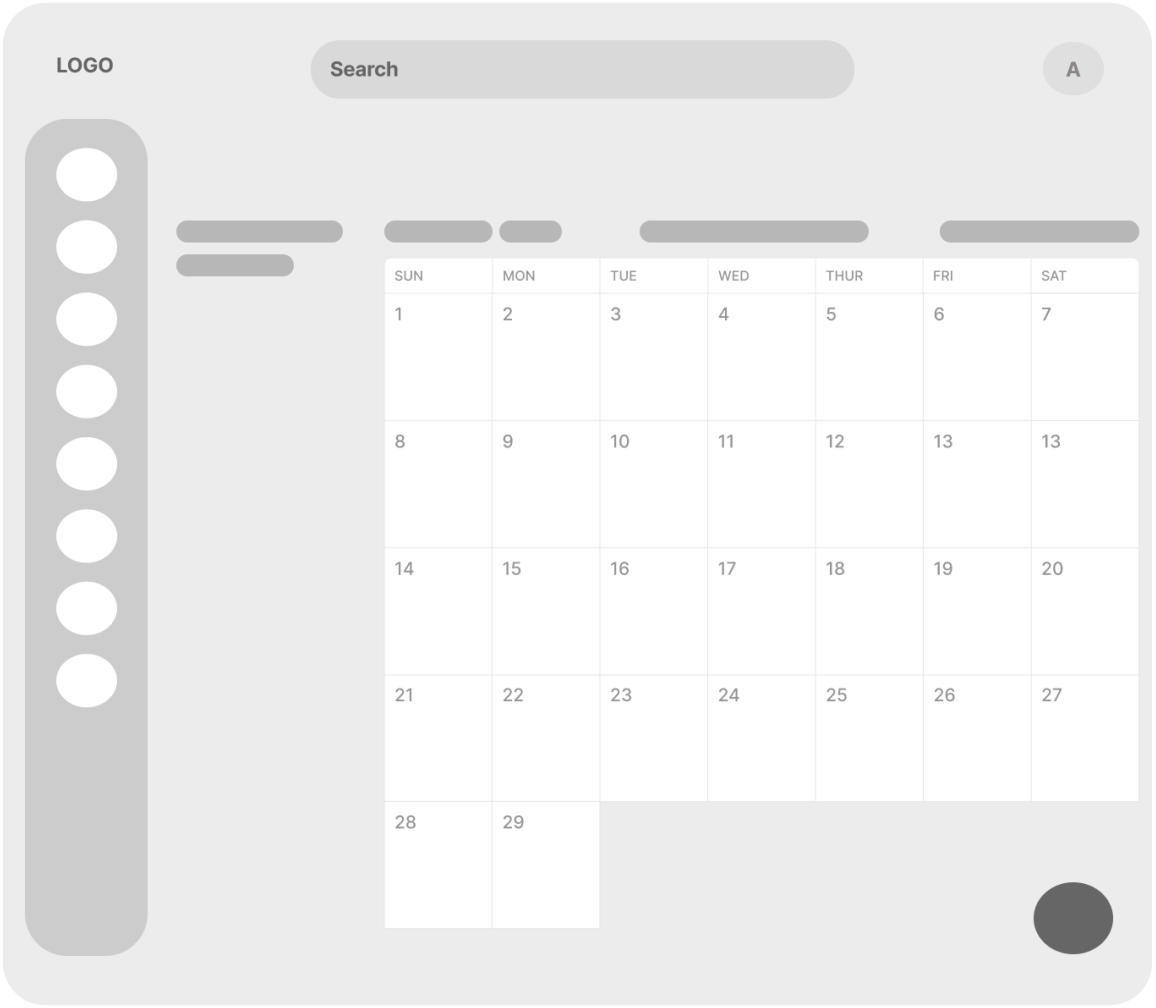


Figure 11 is where the consultants can manage their scheduled appointments.

Figure 12

Consultant Appointment Form - Availability

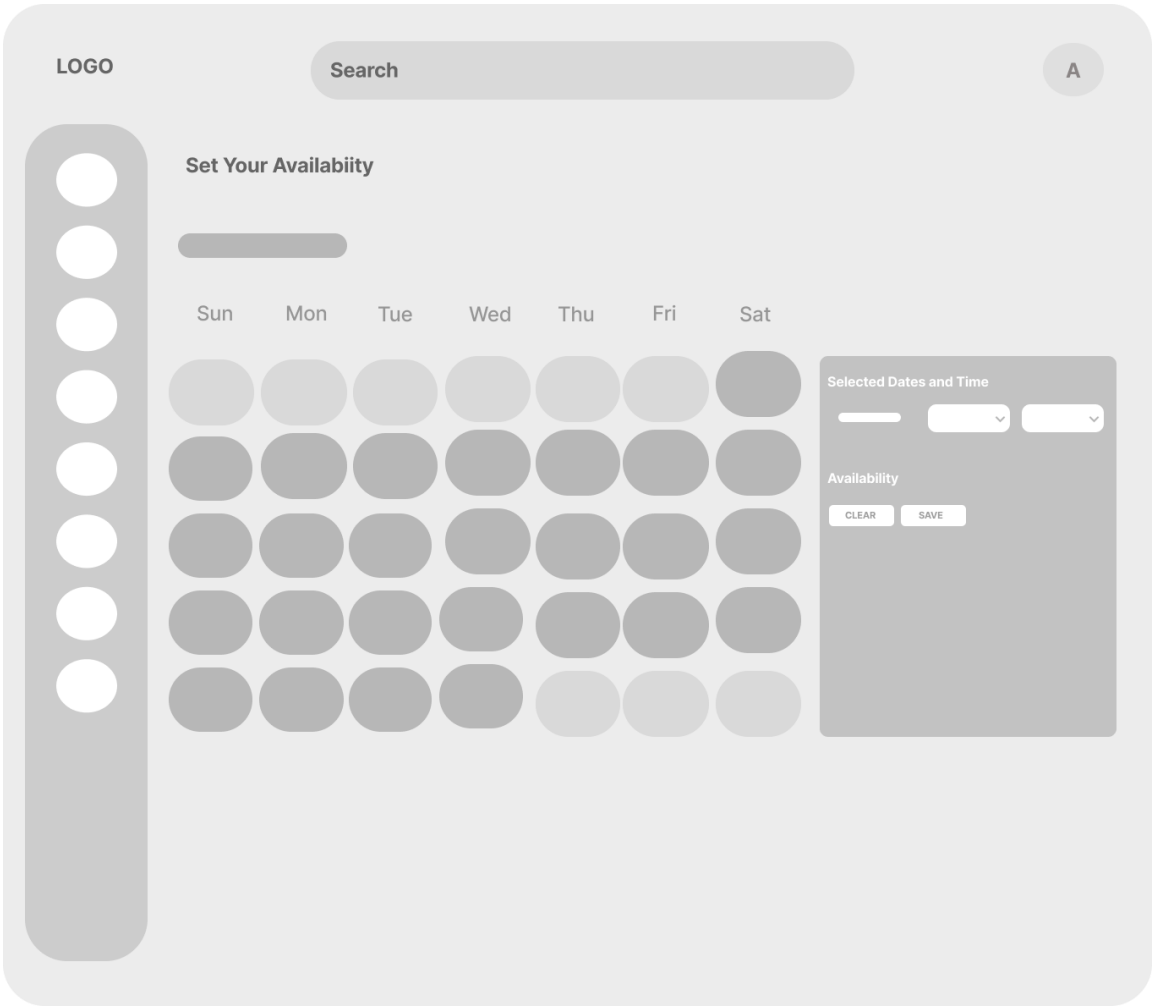


Figure 12 displays available time slots for scheduling appointments, ensuring efficient booking management.

Figure 13

Consultant Appointment Form - Date and Time

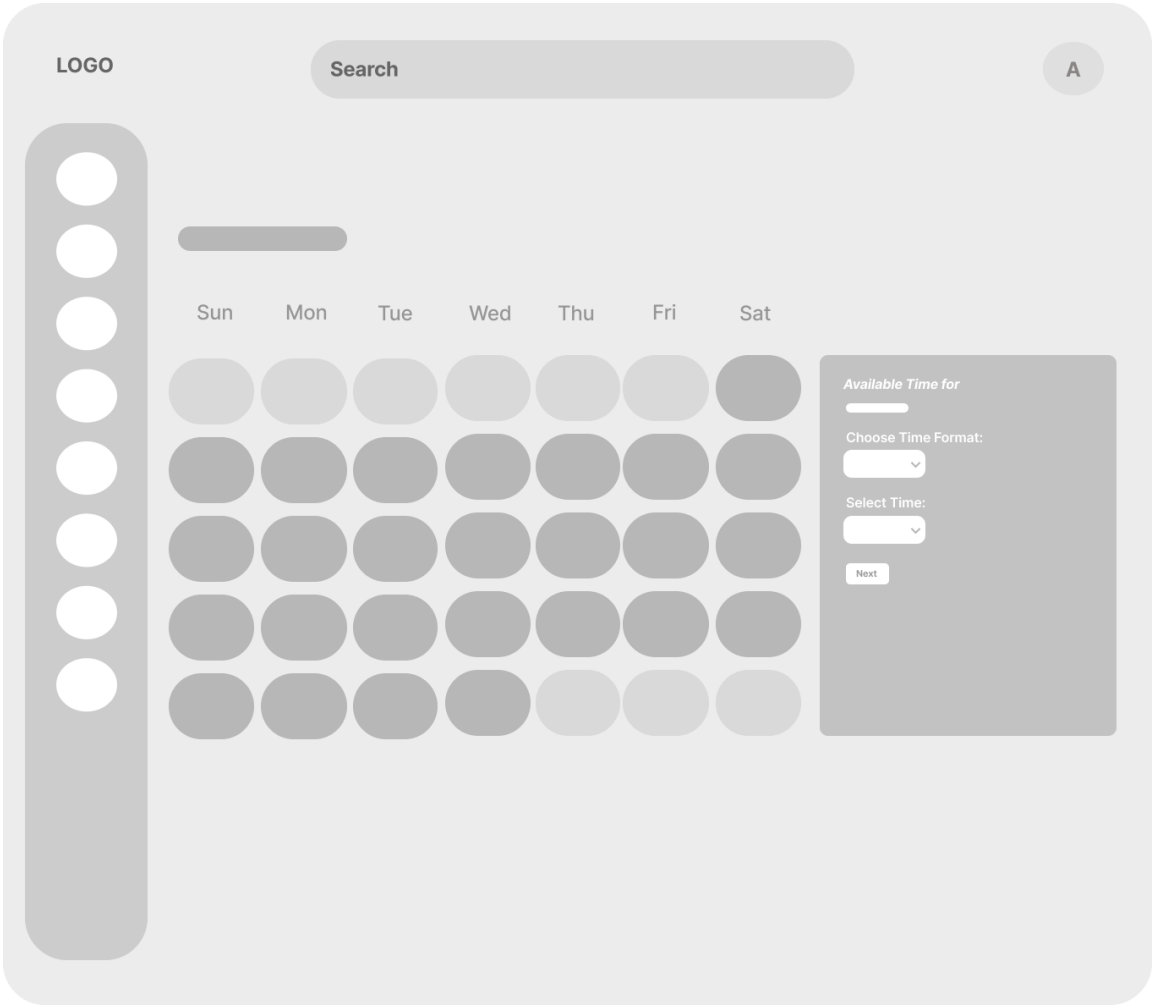


Figure 13 shows a meeting time and date, guaranteeing effective and well-organized appointment scheduling.

Figure 14

Consultant Appointment Form - Consultation Details

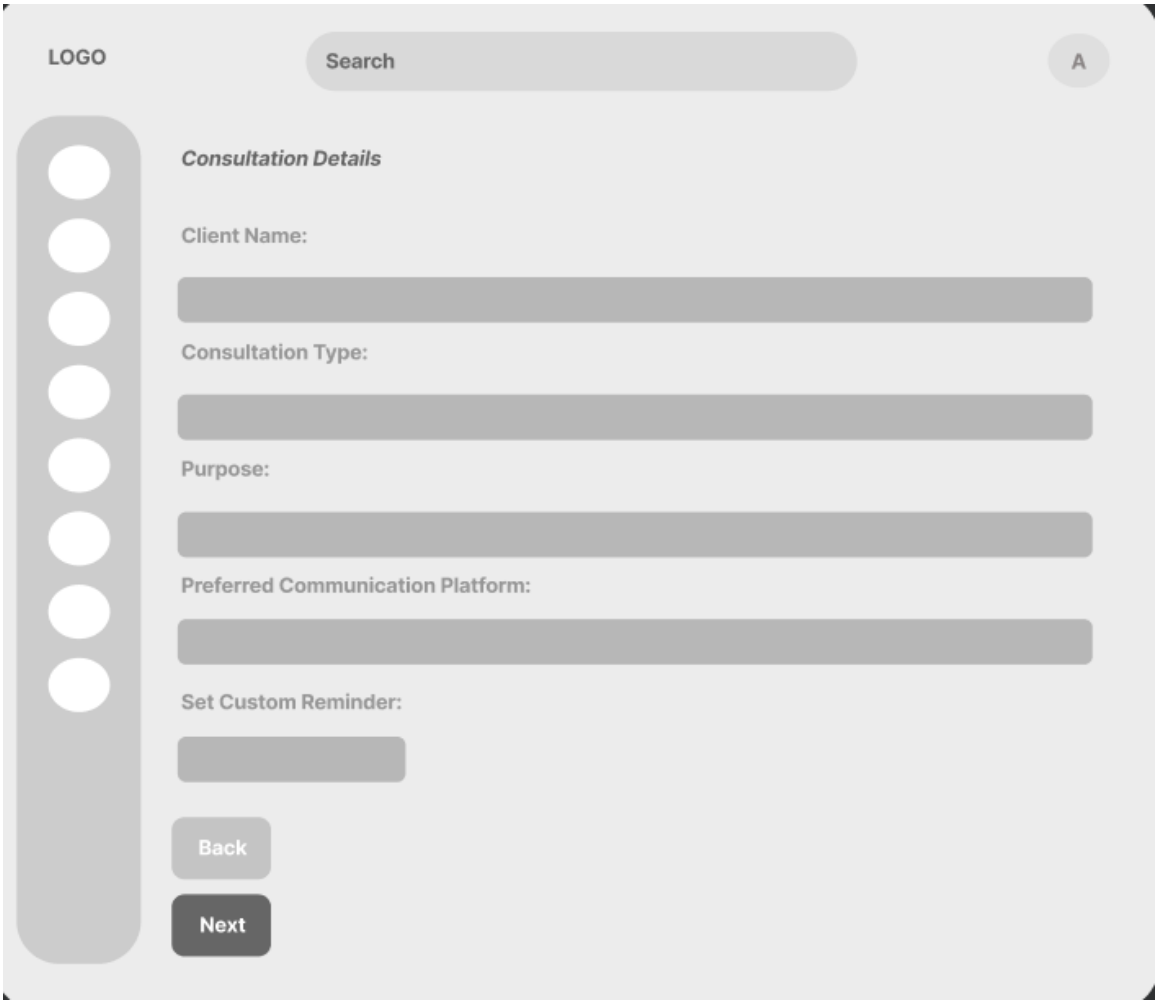


Figure 14 presents where clients can request appointments by specifying their desired date, time, and purpose.

Figure 15

Consultant Appointment Form - Confirmation

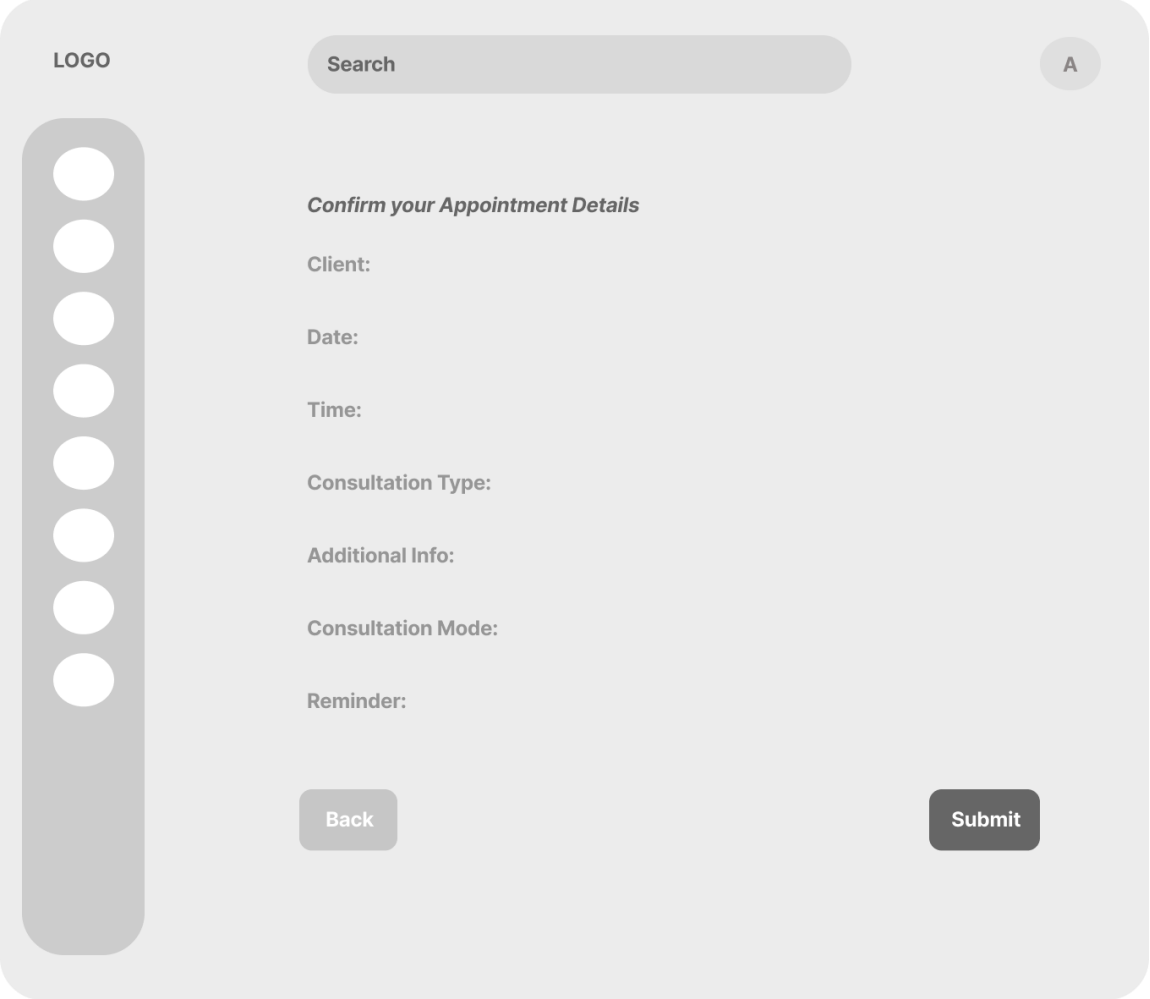


Figure 15 displays the Consultant Appointment Form of Conformation, where the consultant can confirm their appointment details.

Figure 16

Consultant - Notification

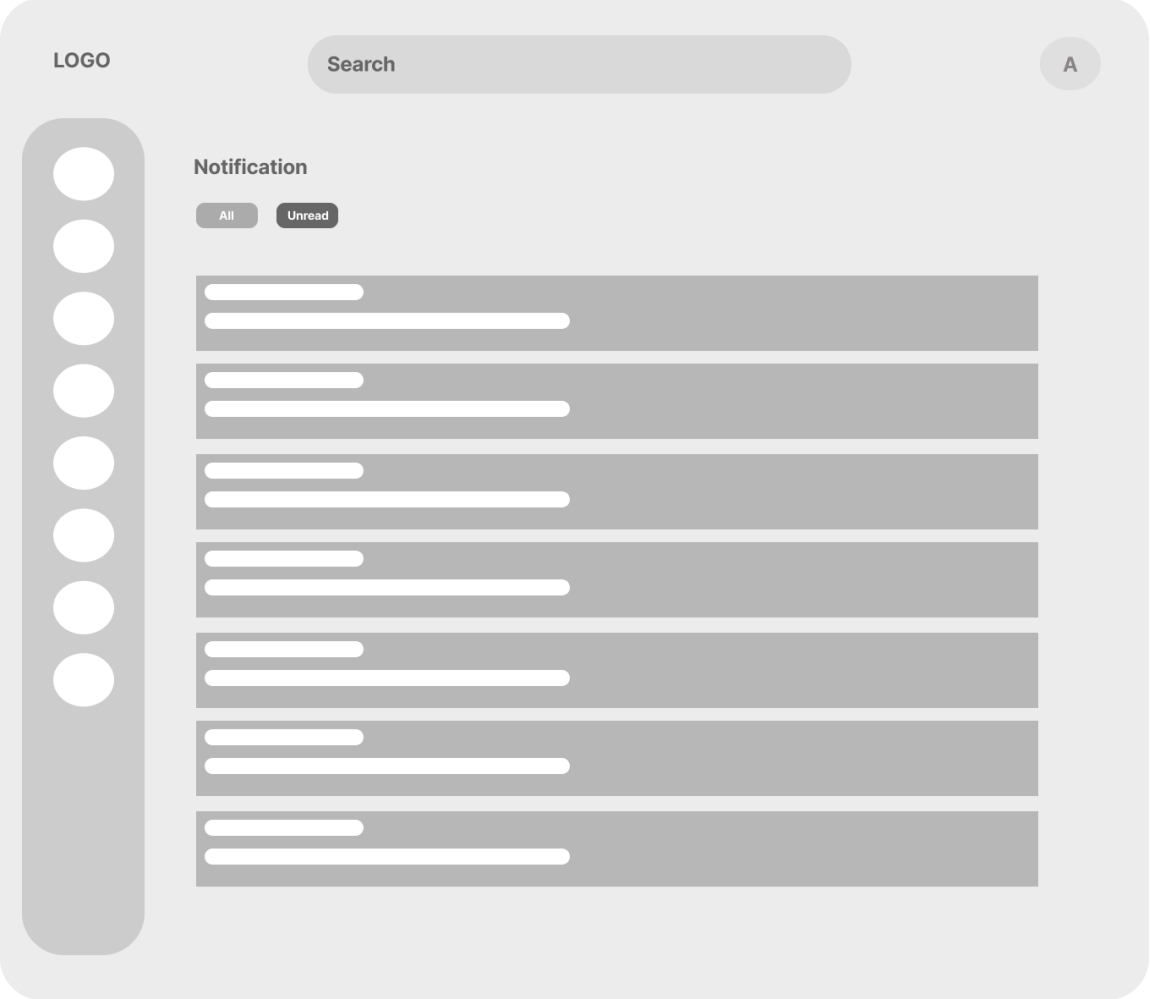


Figure 16 shows the Consultant - Notification, where consultants receive updates and alerts regarding their appointments or consultations.

Figure 17

Consultant Projects List

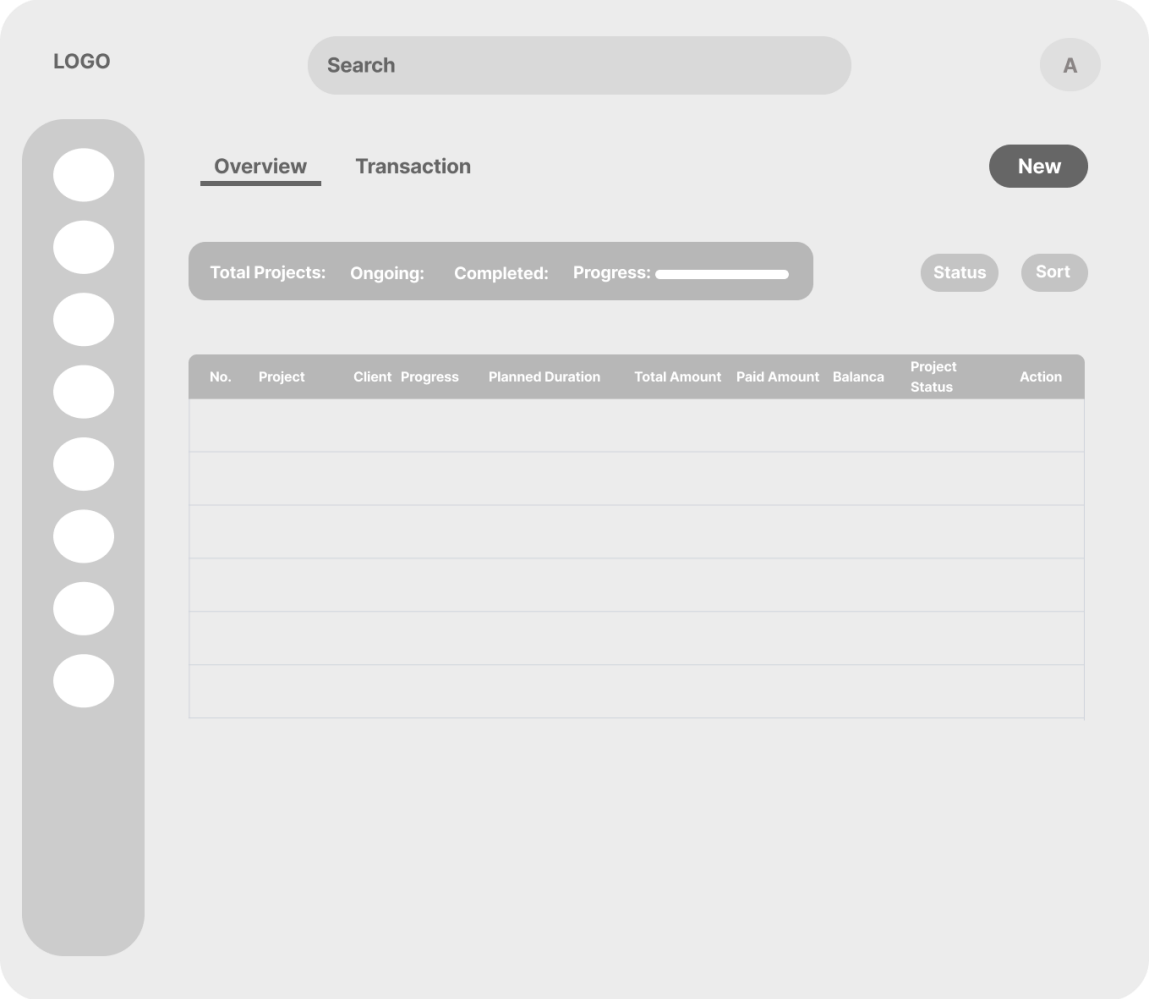


Figure 17 shows where the consultant can view all the projects, see a summary of their statuses, add new projects, and update project information.

Figure 18

Consultant - Project Form

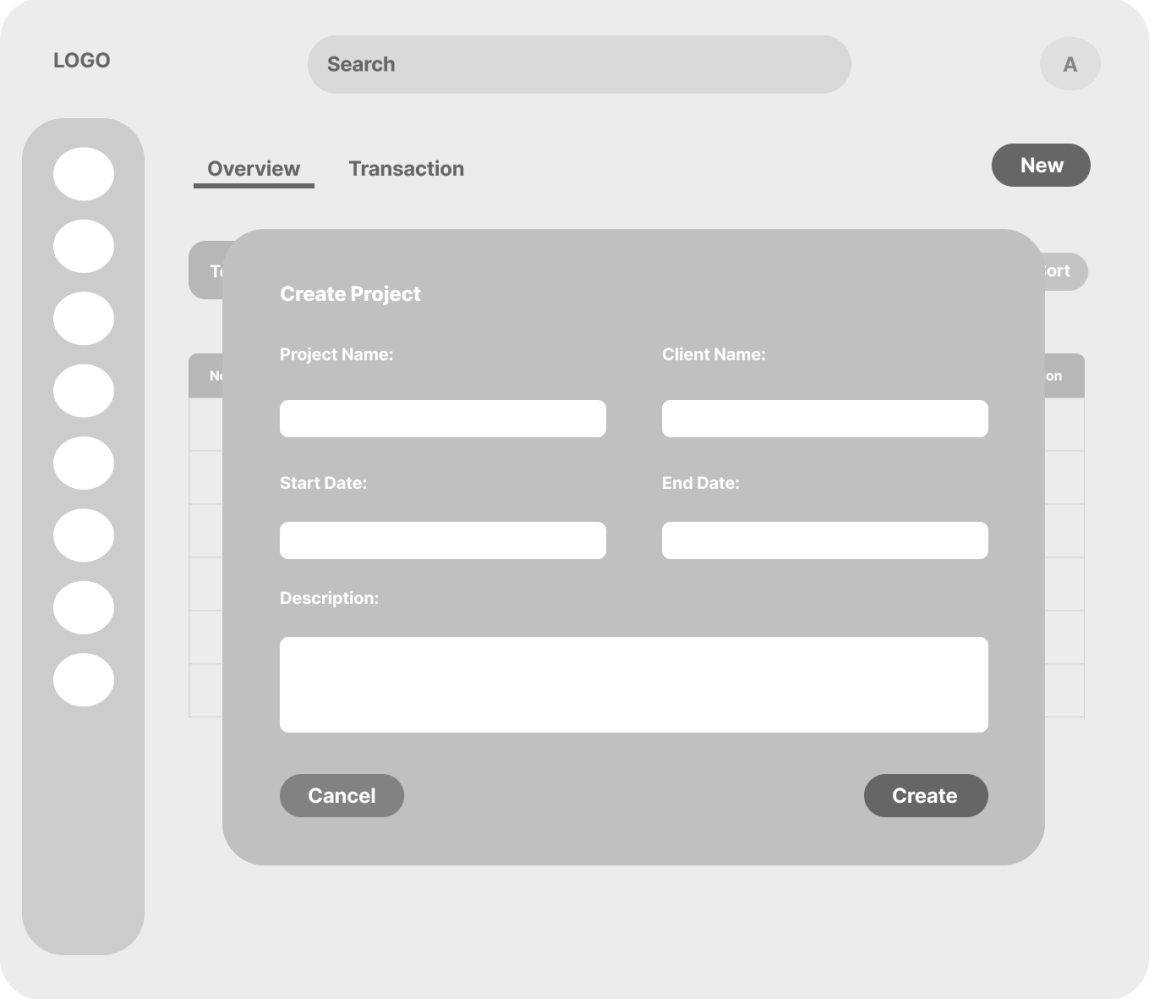
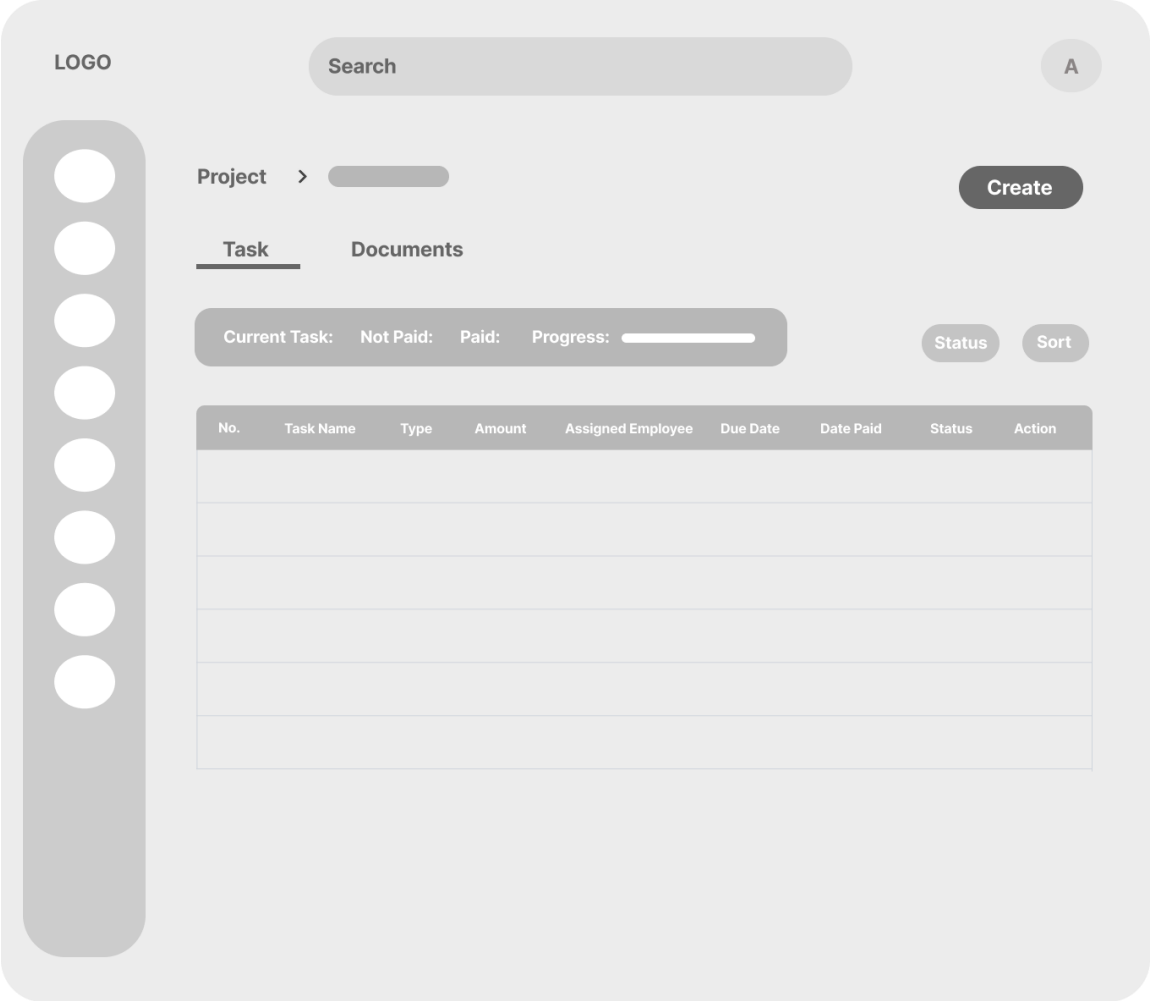


Figure 18 shows the form that collects project details (name, client, dates, description) for a consultant to create a new project.

Figure 19

Consultant – Project Task List



This Figure 19 presents the task list for a certain project.

Figure 20

Consultant – Project Task form

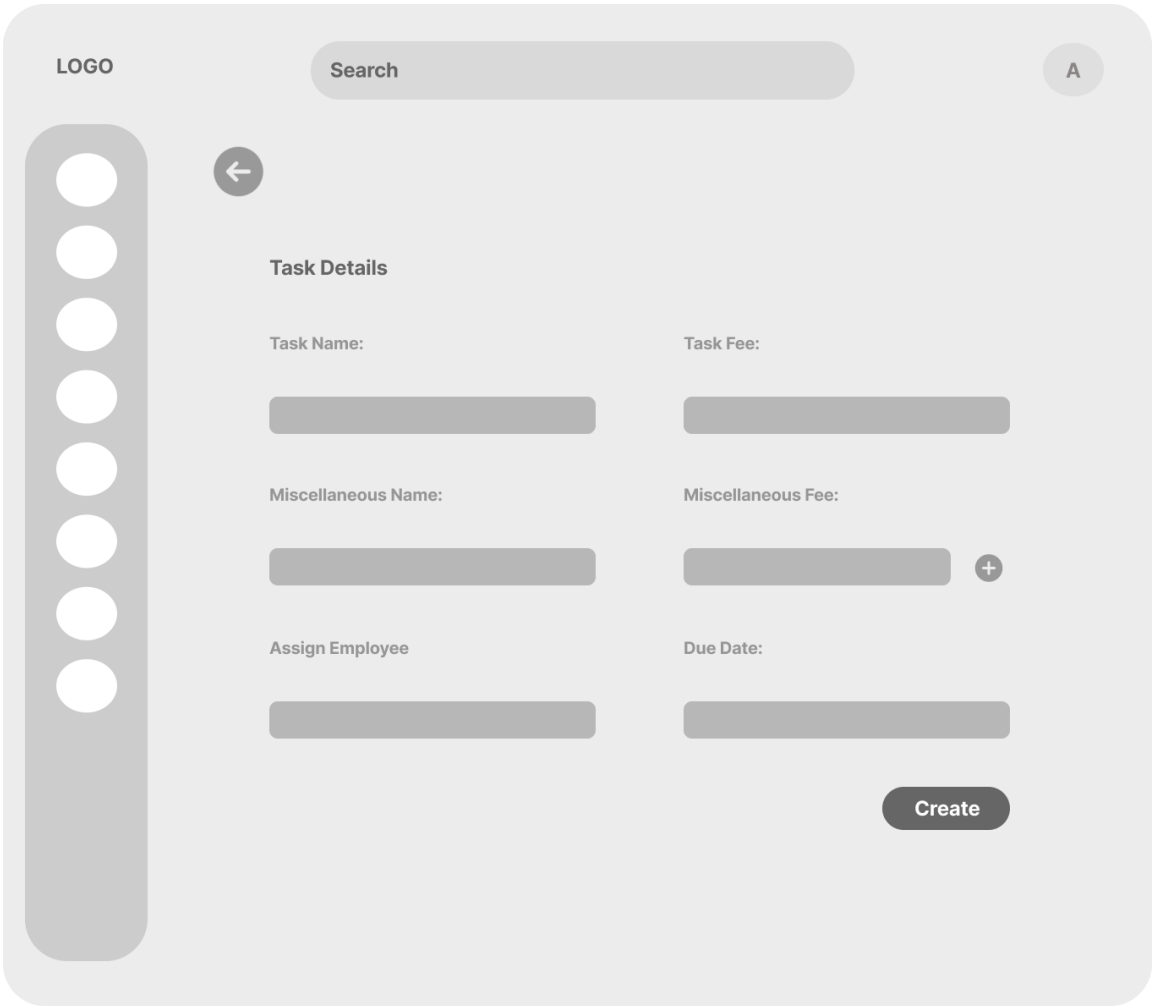


Figure 20 shows where the consultant can create a task for an employee to view, edit, and delete tasks, which can be accessed when a certain project is clicked.

Figure 21

Consultant - Transaction

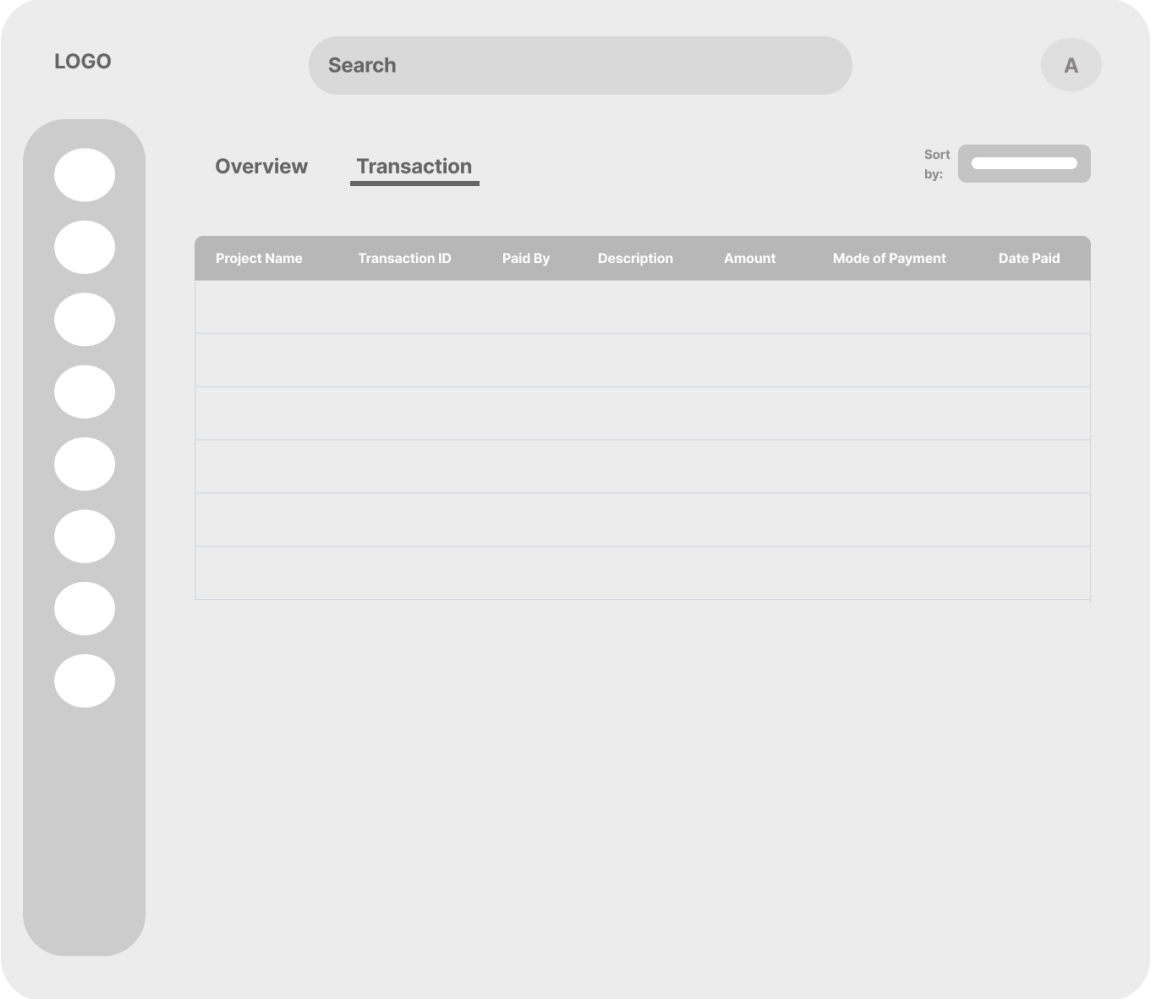


Figure 21 displays detailed payment information for each project. It allows the consultant to view the status of payments related to different projects by listing transaction details such as Project Name, Transaction ID, Paid By, Description, Amount, Mode of Payment, and Date Paid. This tab helps track and monitor the financial activity and payment status associated with each project.

Figure 22

Consultant – Documents

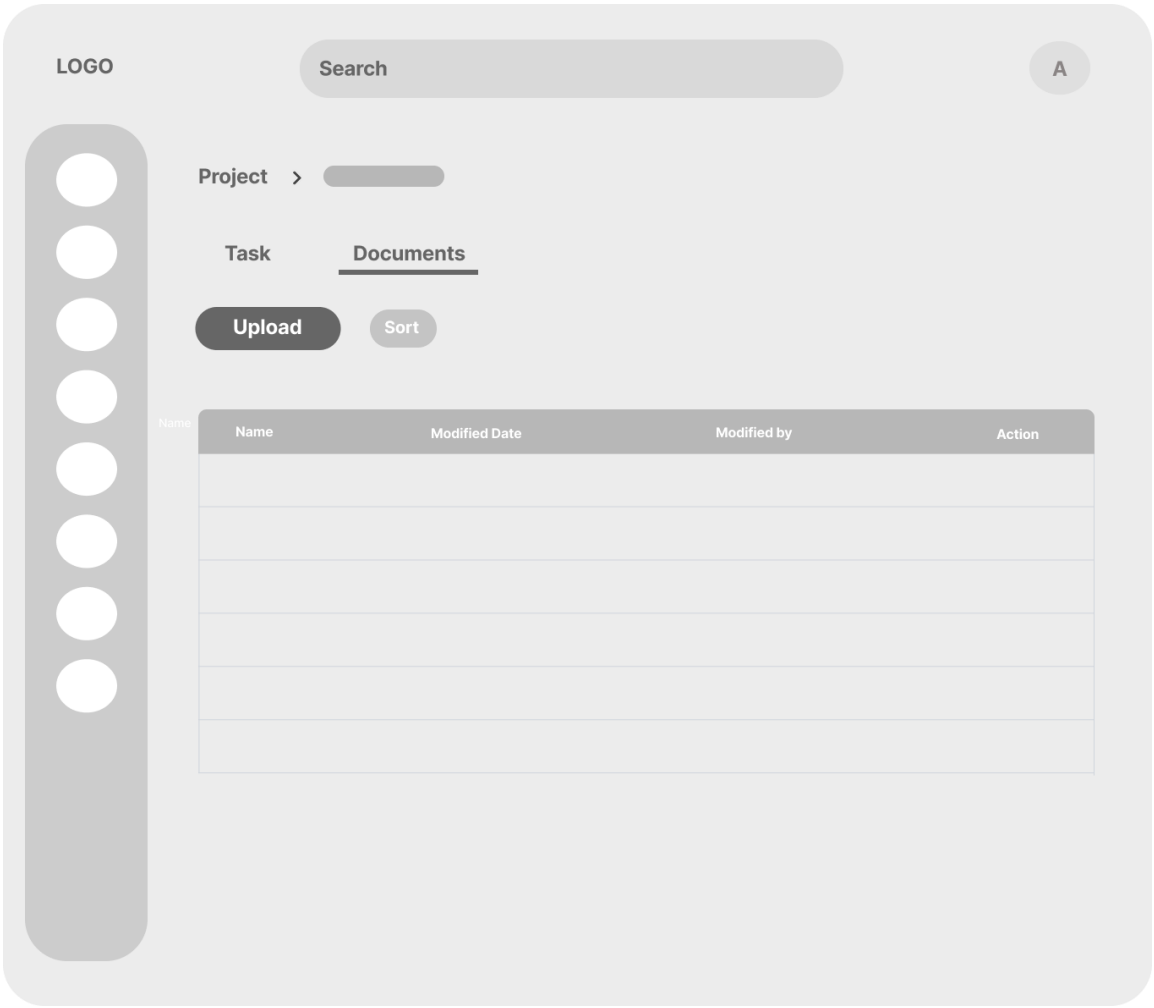


Figure 22 is where the consultant can view both the documents uploaded by them and the client.

Figure 23

Consultant – Client List

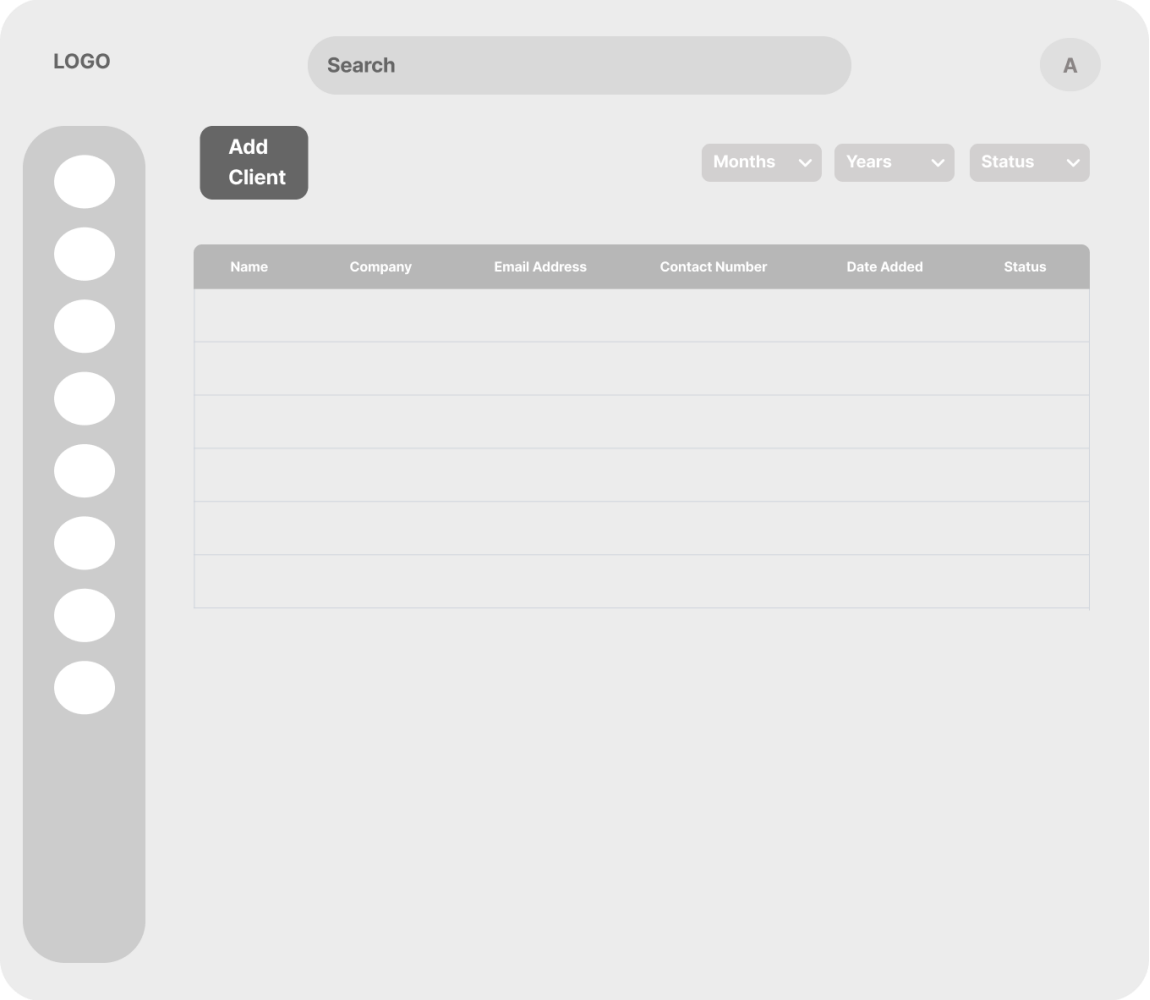


Figure 23 illustrates a system interface enabling consultants to manage client information, including creation, viewing, updating, and historical tracking.

Figure 24

Consultant – Client Account Creation

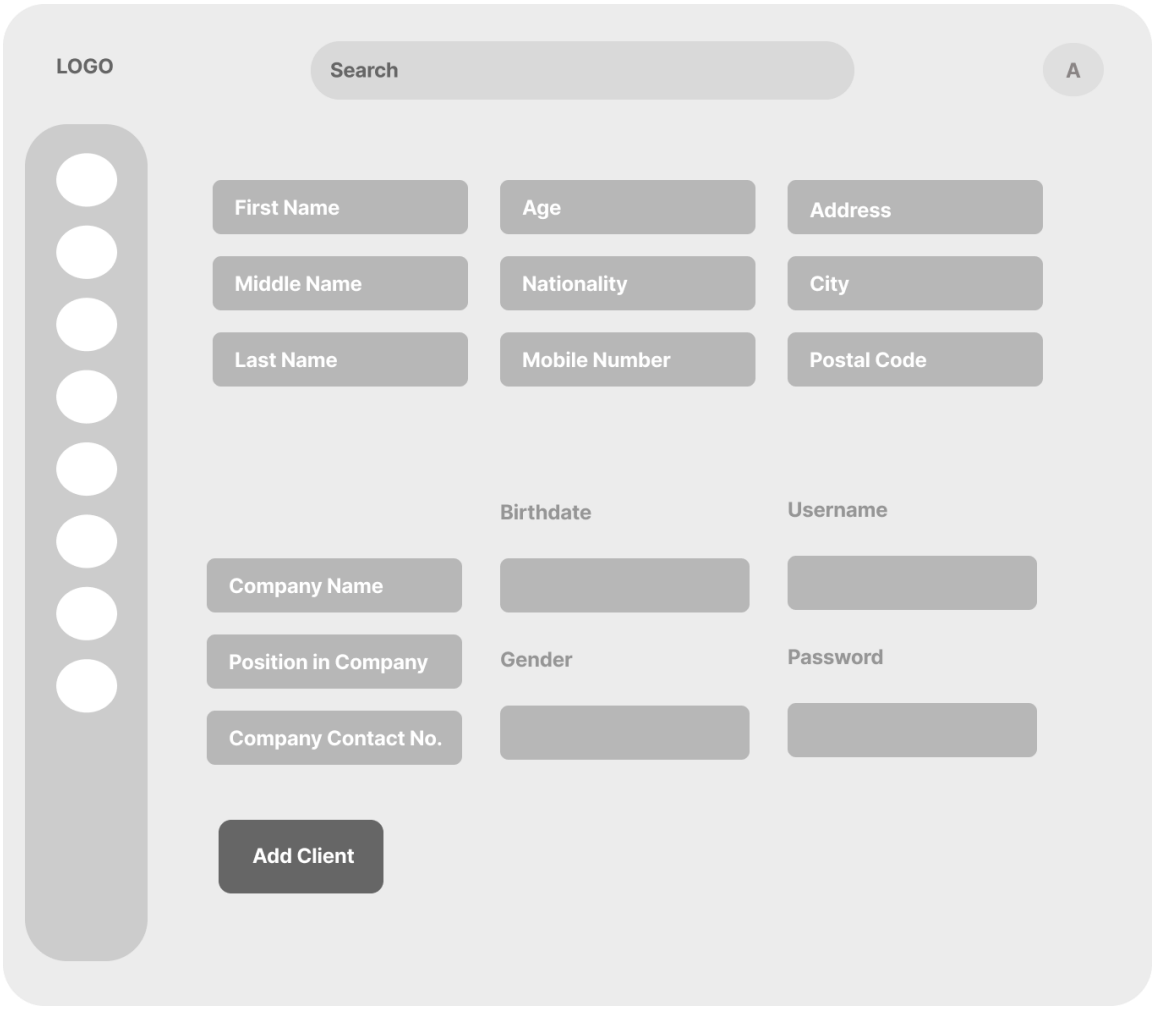


Figure 24 shows the form used to create a new client, including fields for personal information.

Figure 25

Consultant – Client Overview

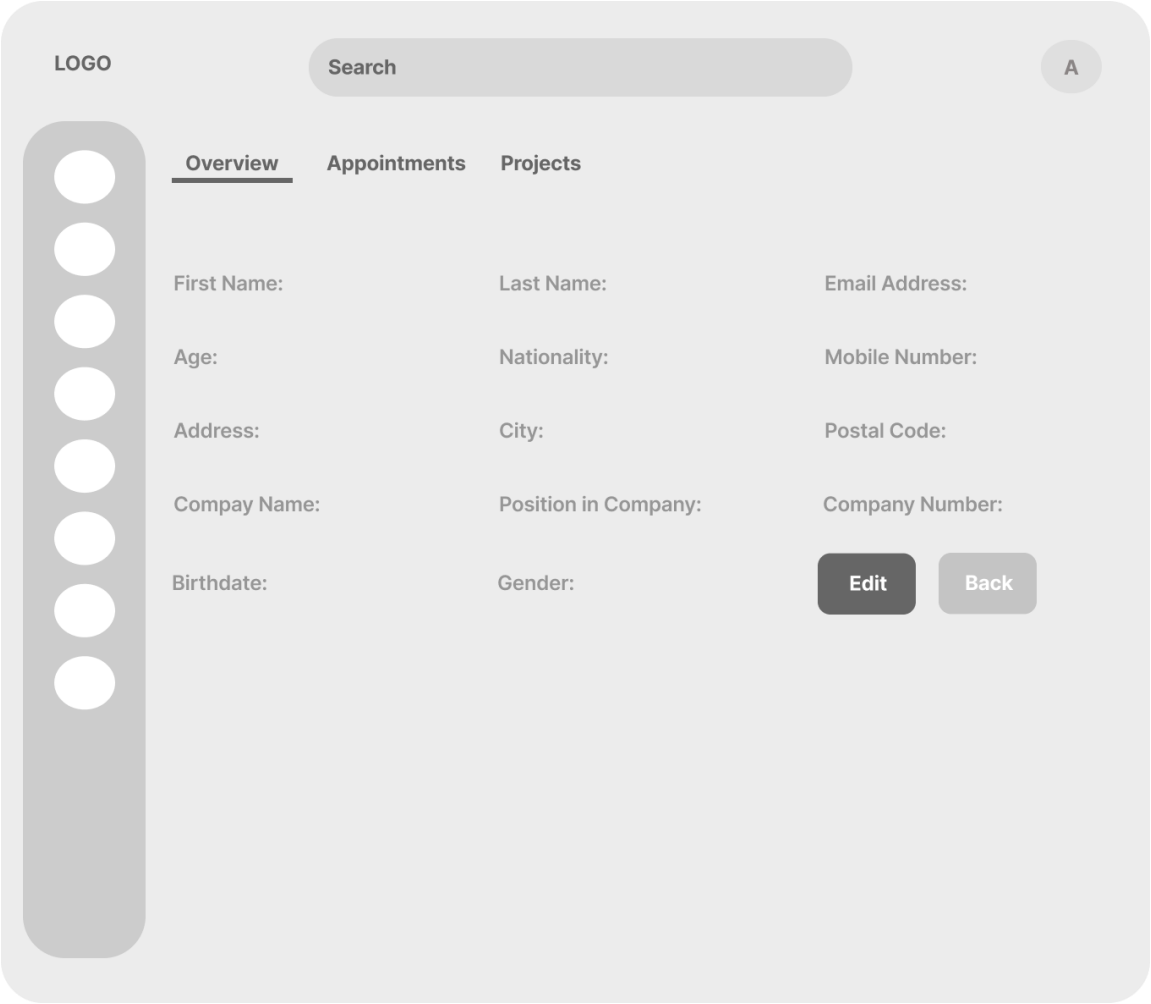


Figure 25 displays the Client Overview, enabling users to access and review the personal details of each client.

Figure 26

Consultant - Employee List

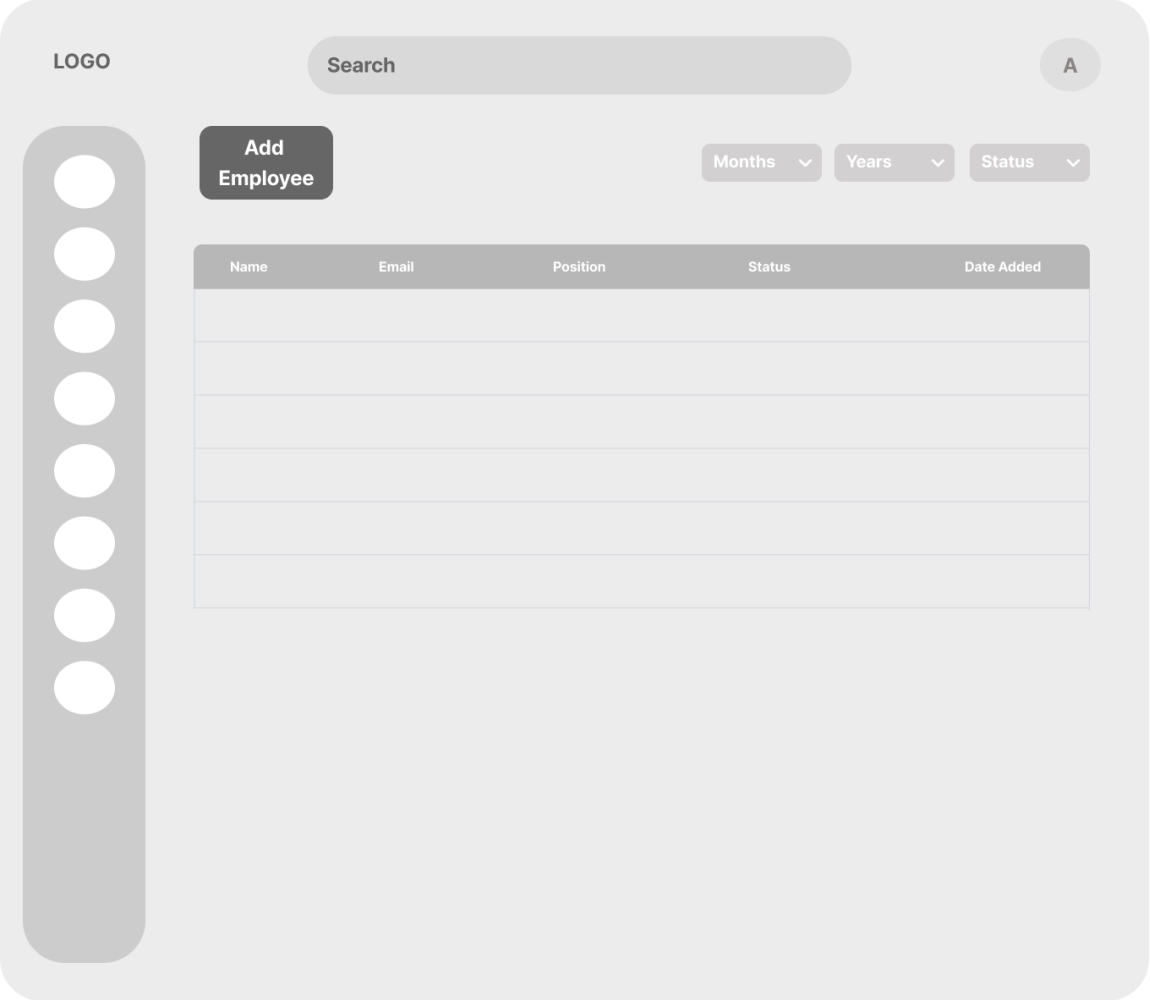


Figure 26 is where the consultant can create, view, and update employee's information.

Figure 27

Consultant – Account Employee Creation

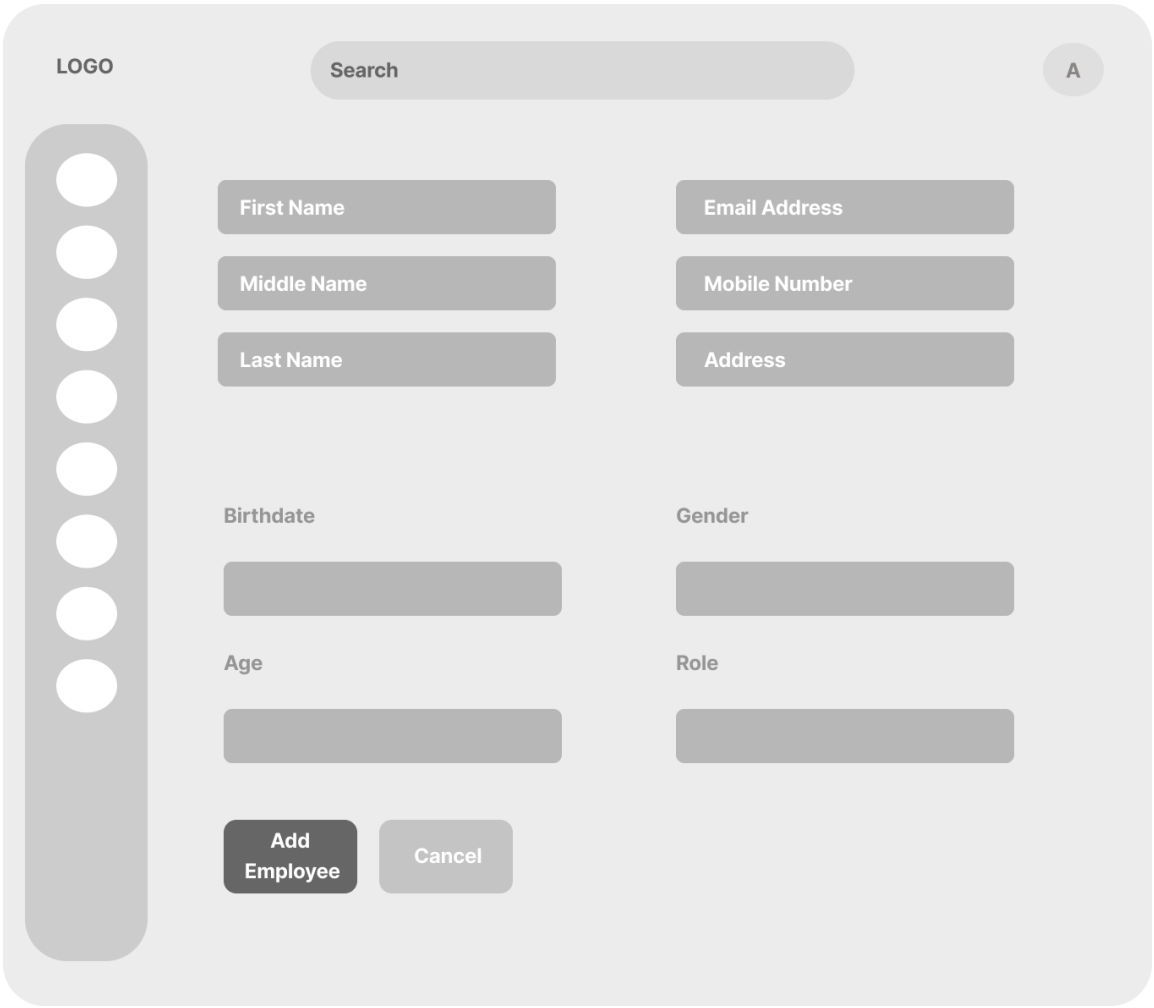


Figure 27 provides simple employee record administration by gathering vital information such as name, contact, address, and position.

Figure 28

Consultant - Employee Overview

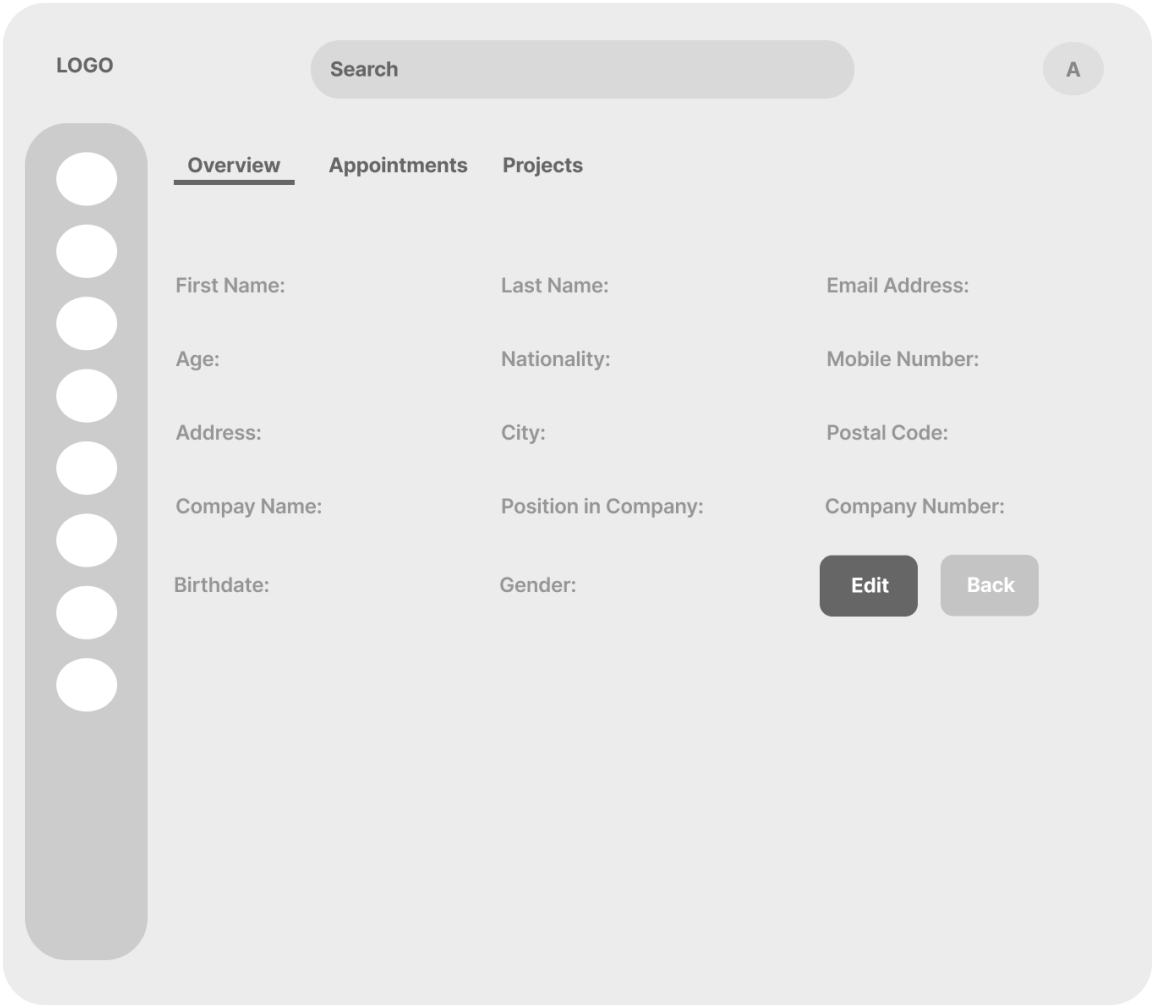


Figure 28 shows that it gathers contact information, personal data (name, age, address, email, nationality), and work-related data.

Figure 29

Client - Calendar

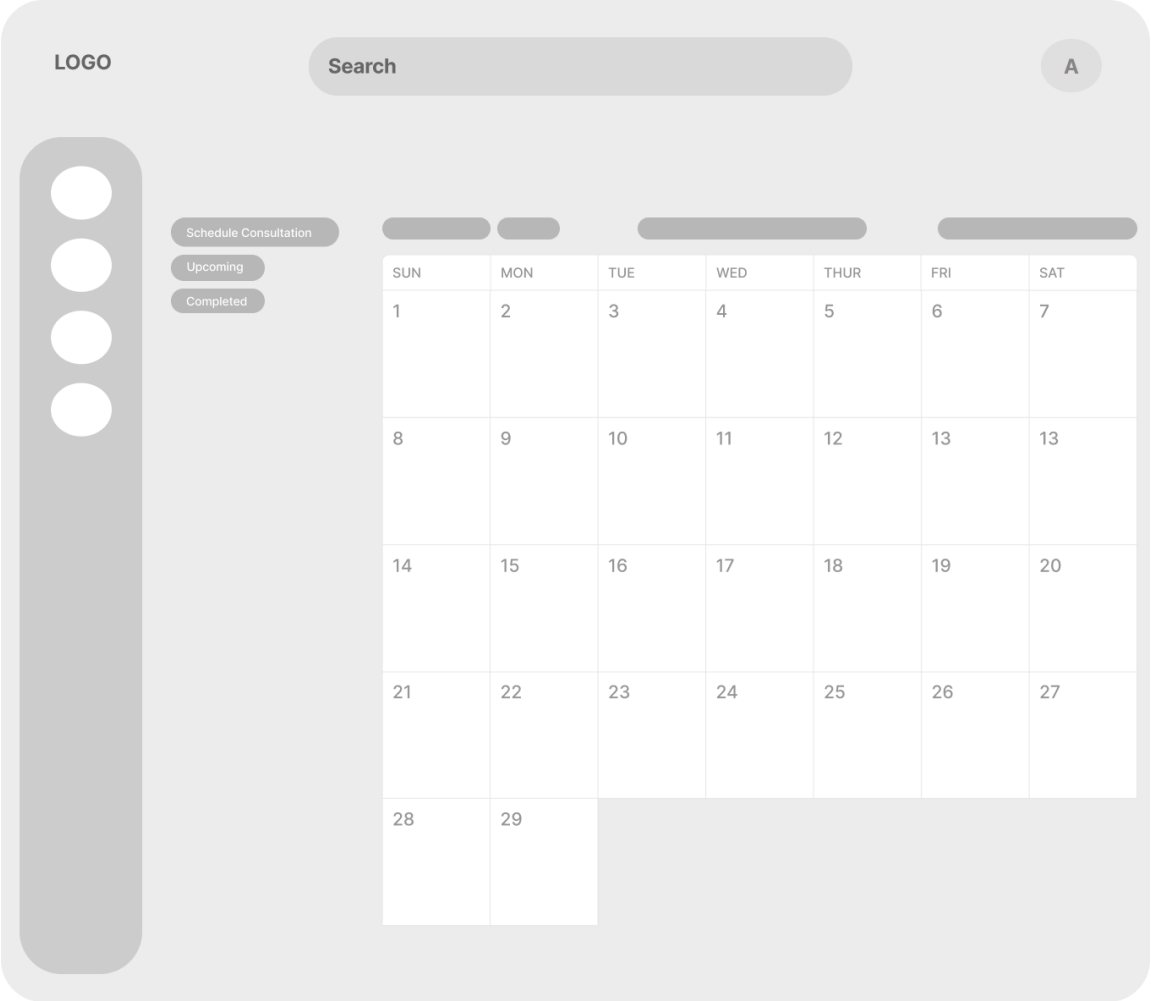


Figure 29 shows a monthly calendar grid with daily meeting entries. There's a section for upcoming and completed meetings and a button to schedule a consultation.

Figure 30

Client Appointment Form - Date and Time

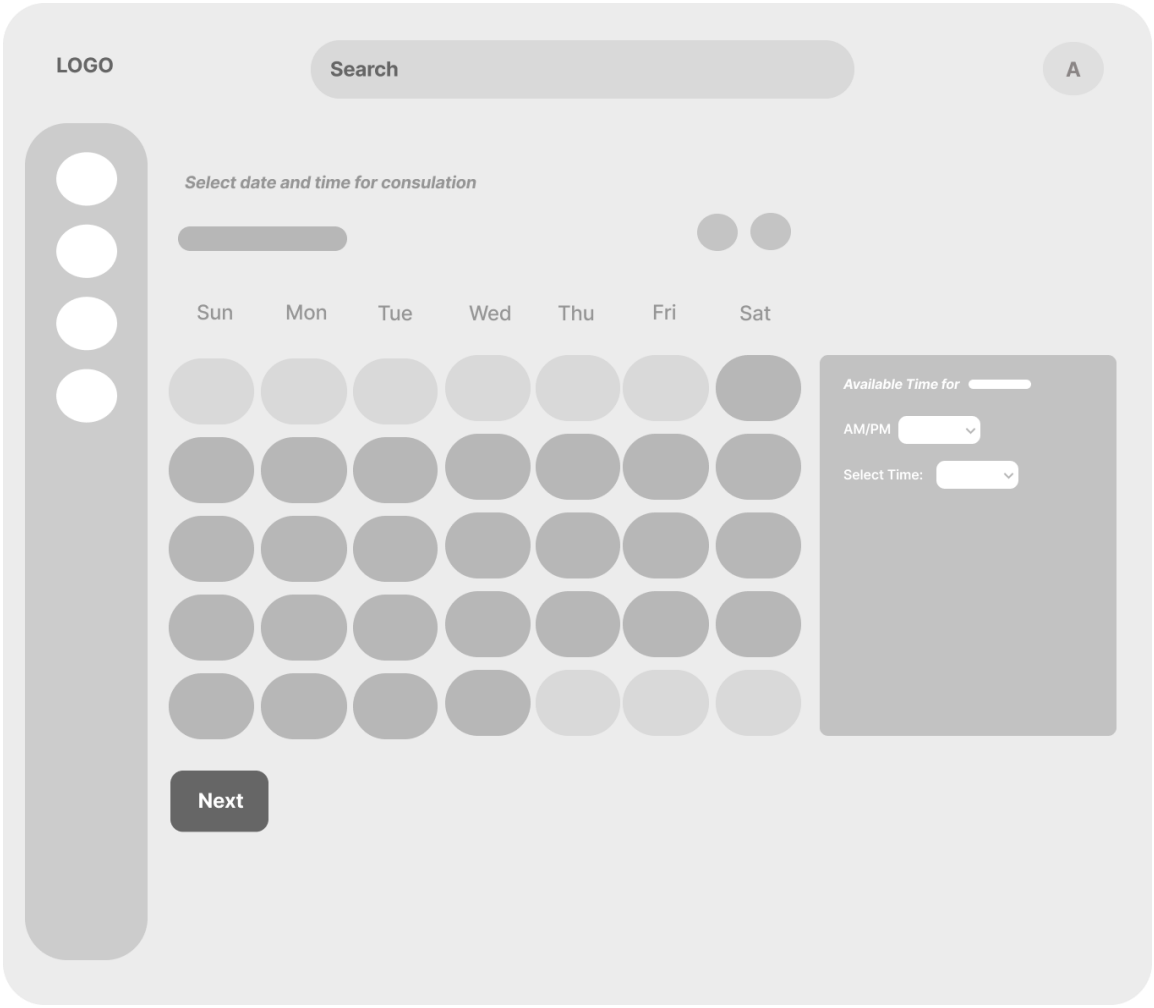


Figure 30 shows how users select a date and time for their consultation.

Figure 31

Client Appointment Form

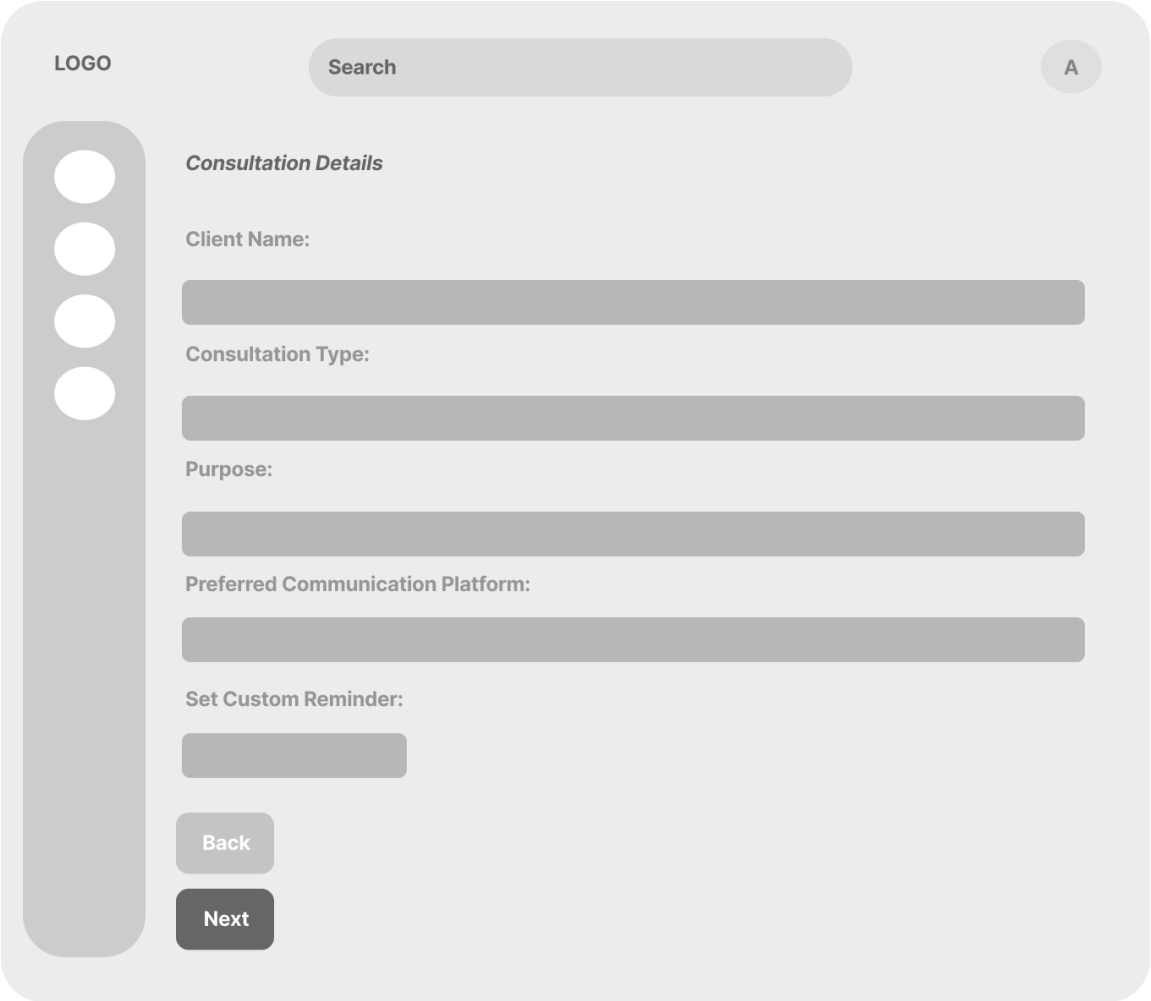


Figure 31 shows how users select a date and time for their consultation.

Figure 32

Client Appointment Form - Confirmation

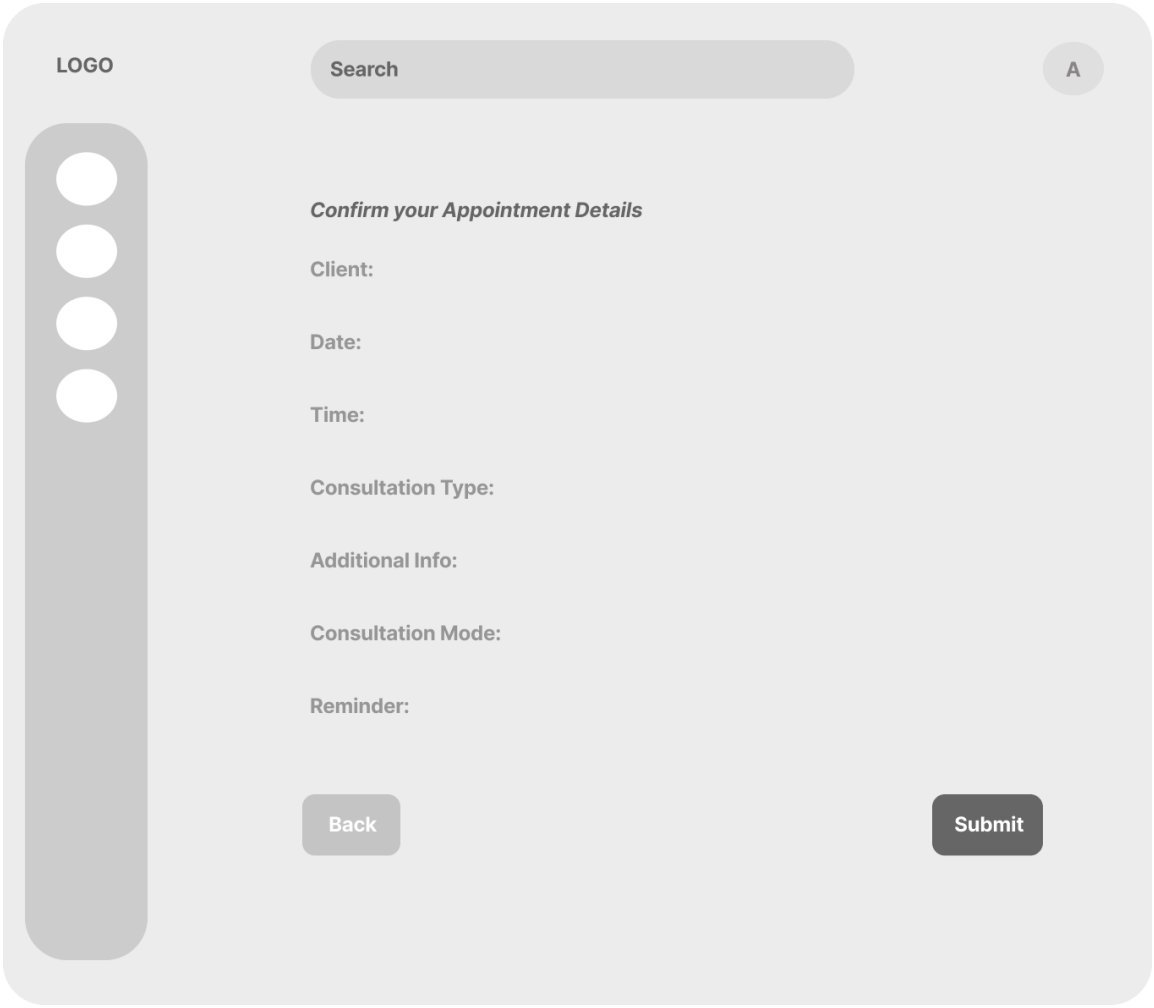


Figure 32 displays the confirmation of the client's selected appointment for the meeting.

Figure 33

Client Notification

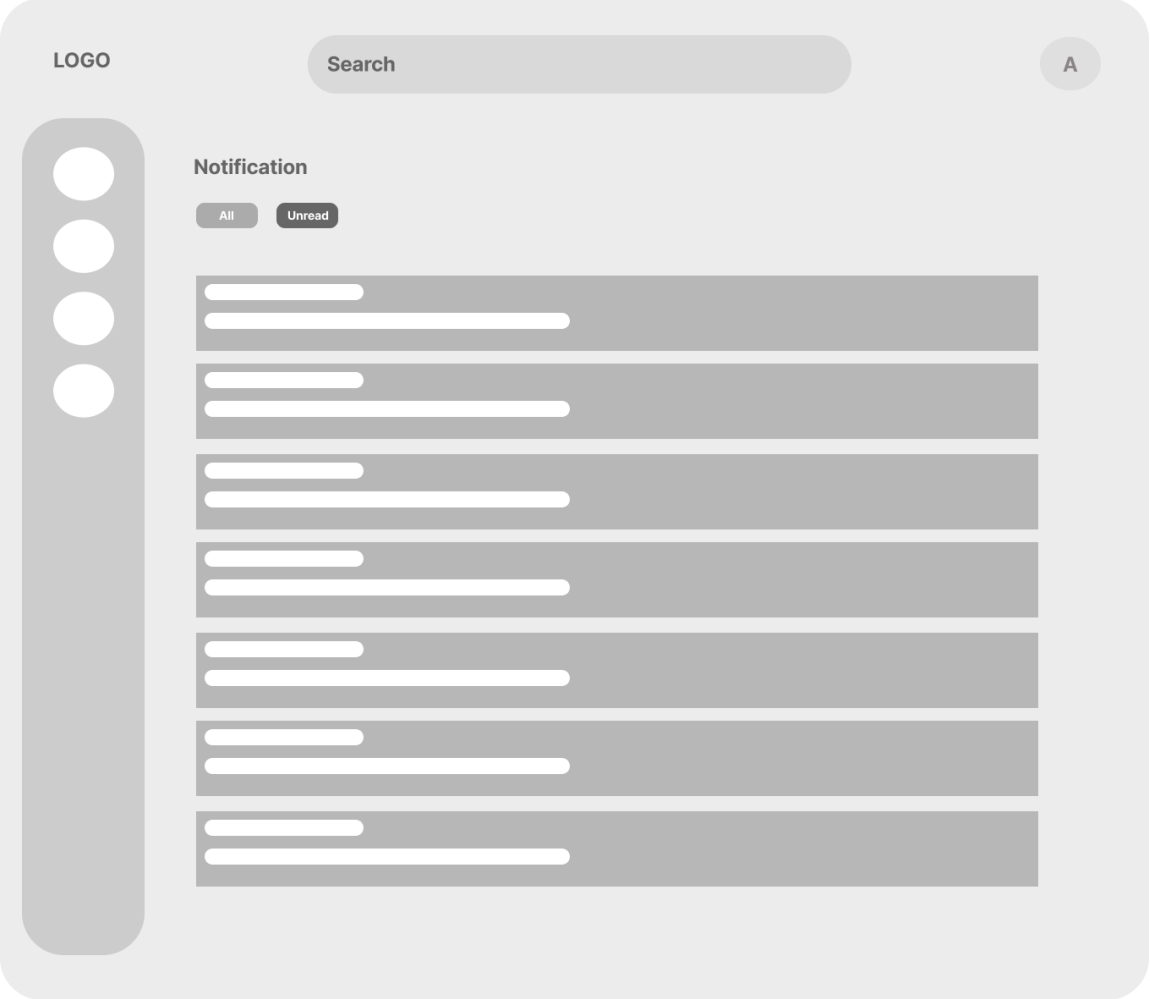


Figure 33 shows the notification on the client side, where clients receive updates such as appointment confirmations, document uploads, and other important alerts.

Figure 34

Client – Project Tab

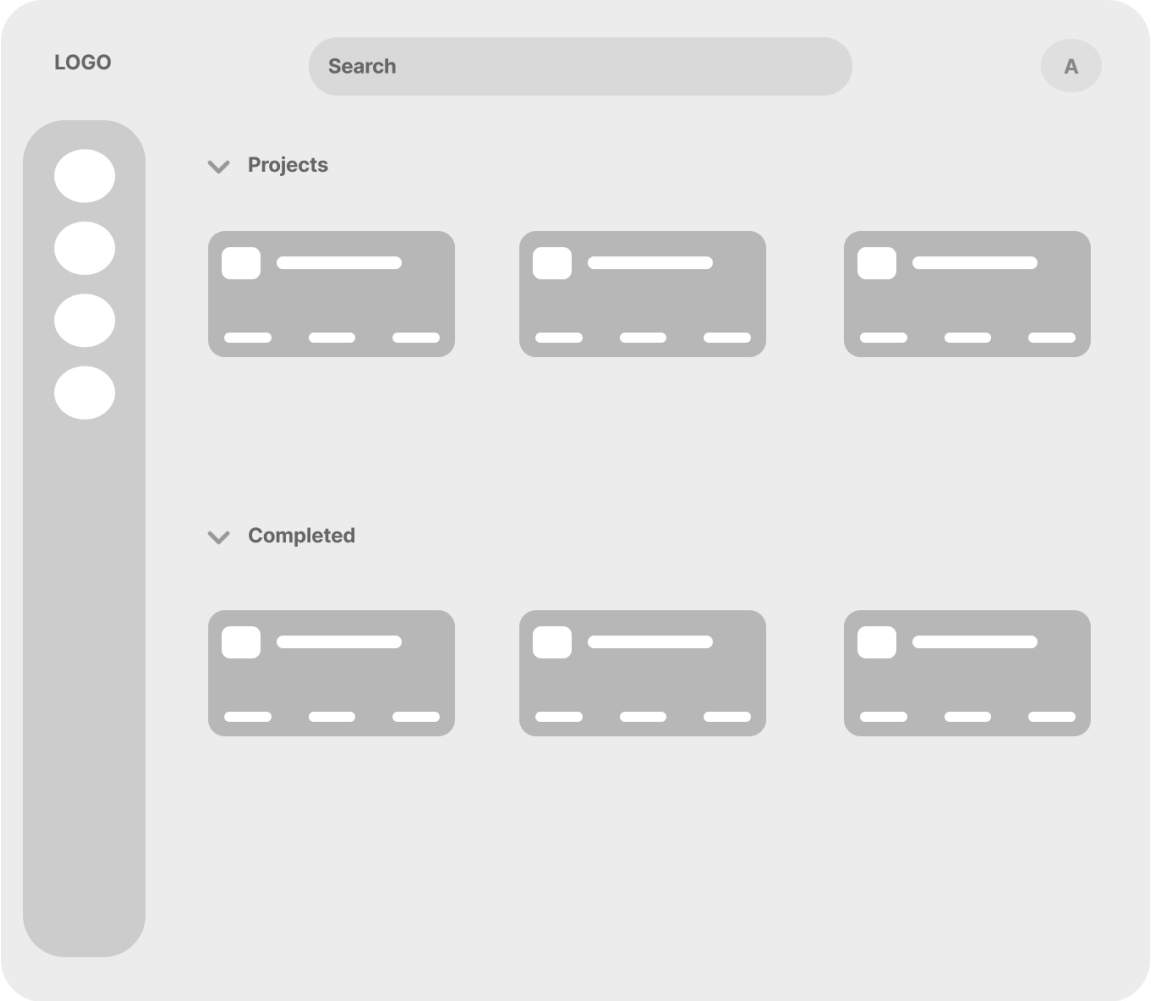


Figure 34 shows the "Projects" and "Completed" sections, with each project showing its name, status (Ongoing), and progress.

Figure 35

Client – Project Overview

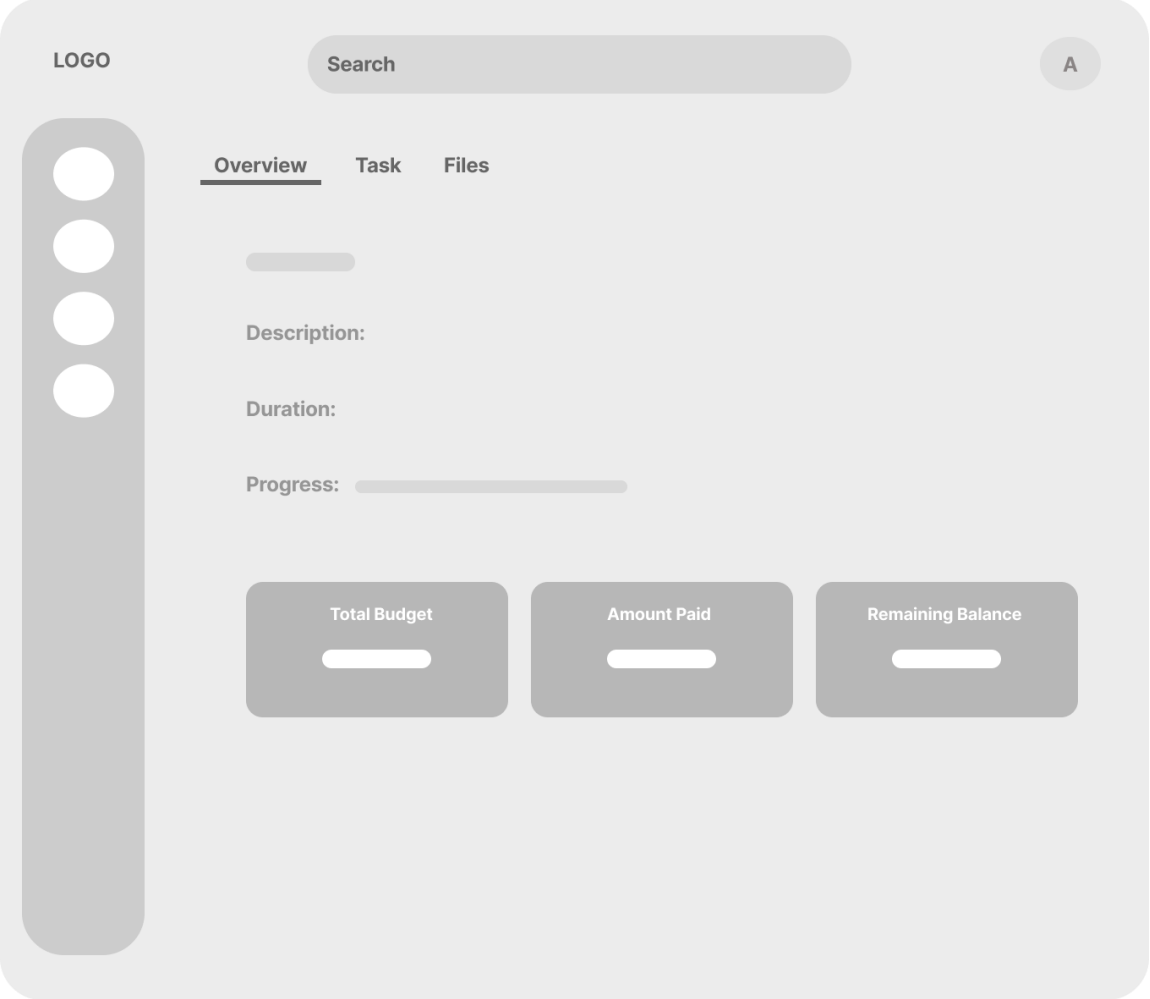


Figure 35 shows the project overview, including each project's description, duration, and progress.

Figure 36

Client – Project Task Overview

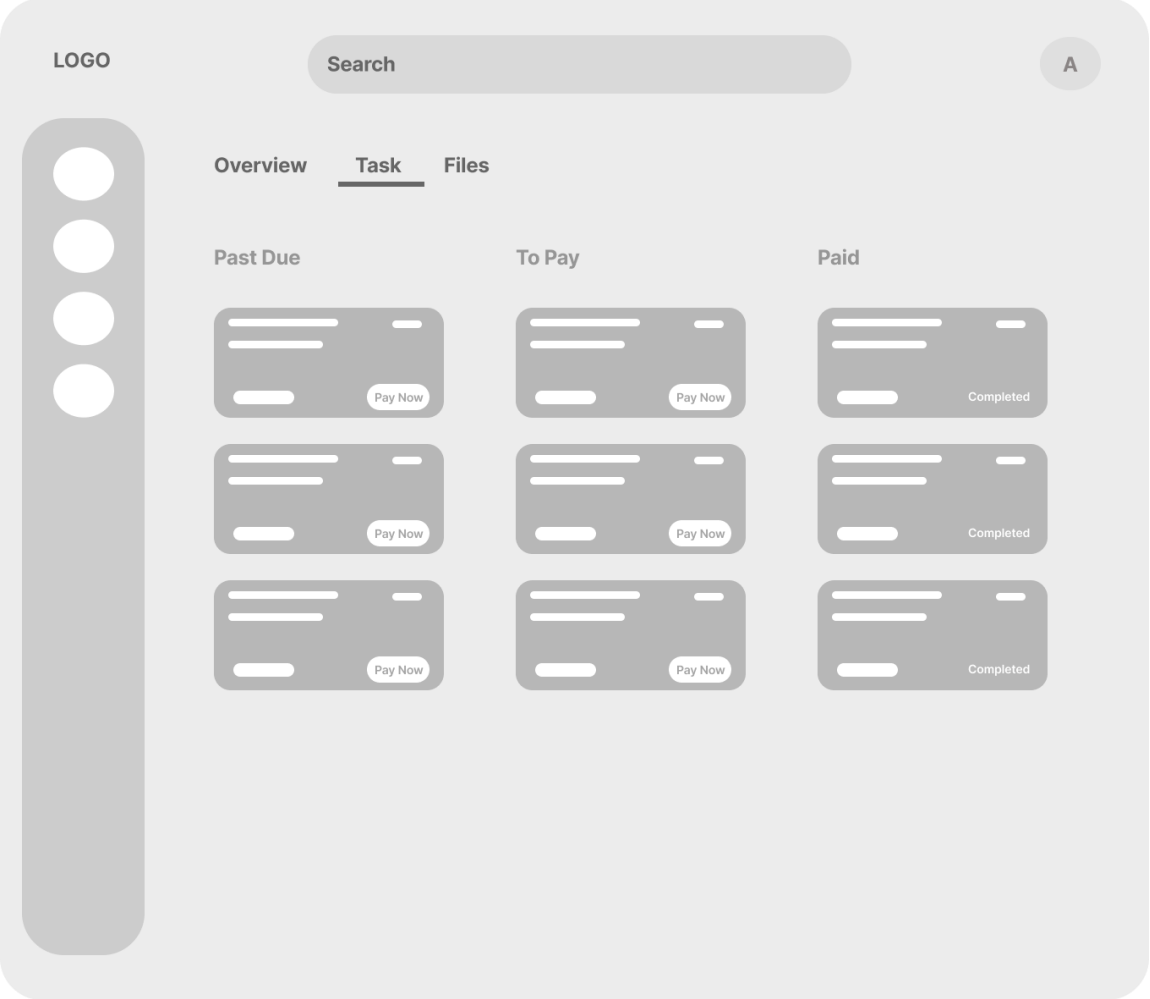


Figure 36 shows the project task overview, including a past due, to pay, and paid every task.

Figure 37

Client – Project files upload and view

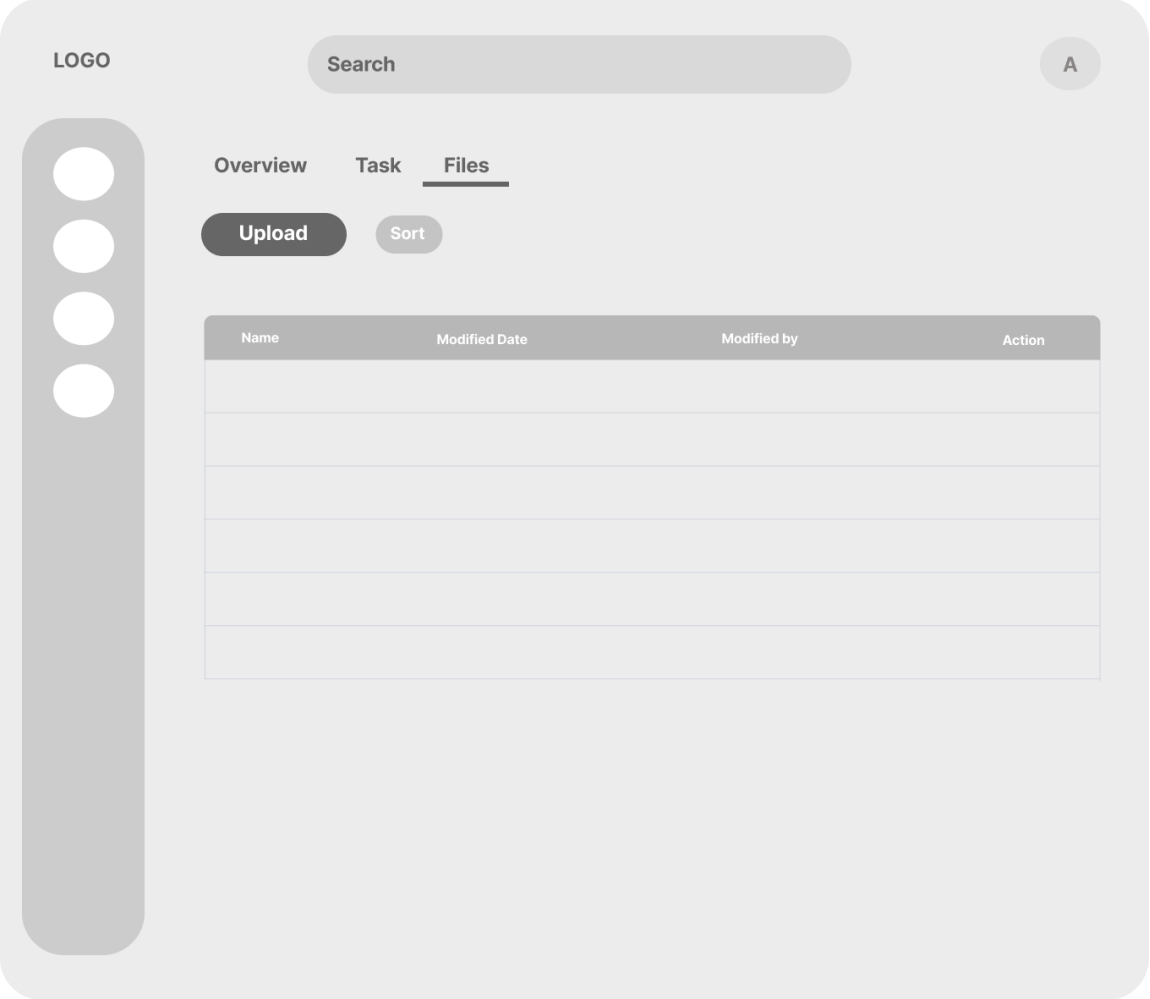


Figure 37 shows the Project Files, where clients can upload and view project files.

Figure 38

Client – Payments

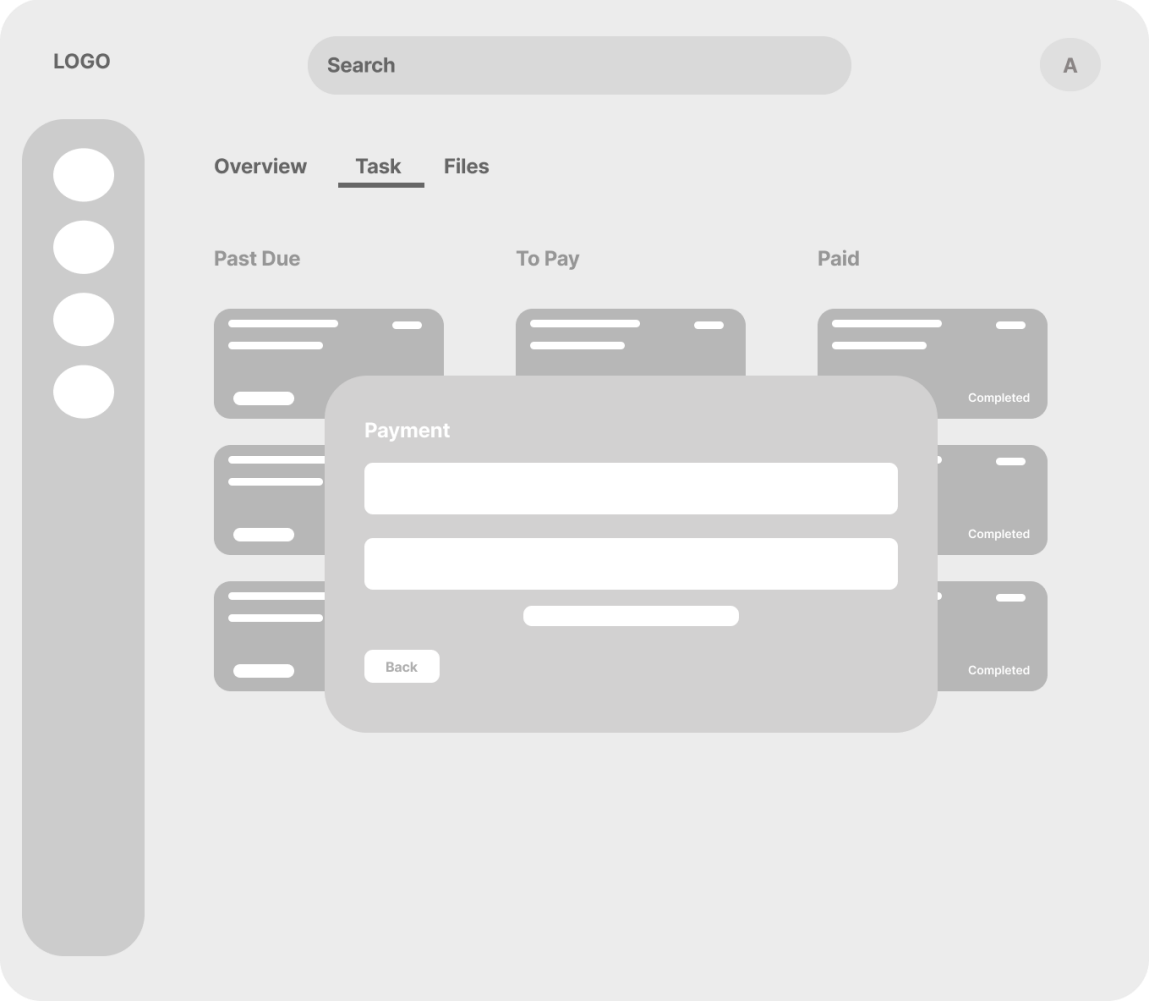


Figure 38 shows the Payment button, where clients can make their payments.

Figure 39

Client – Review

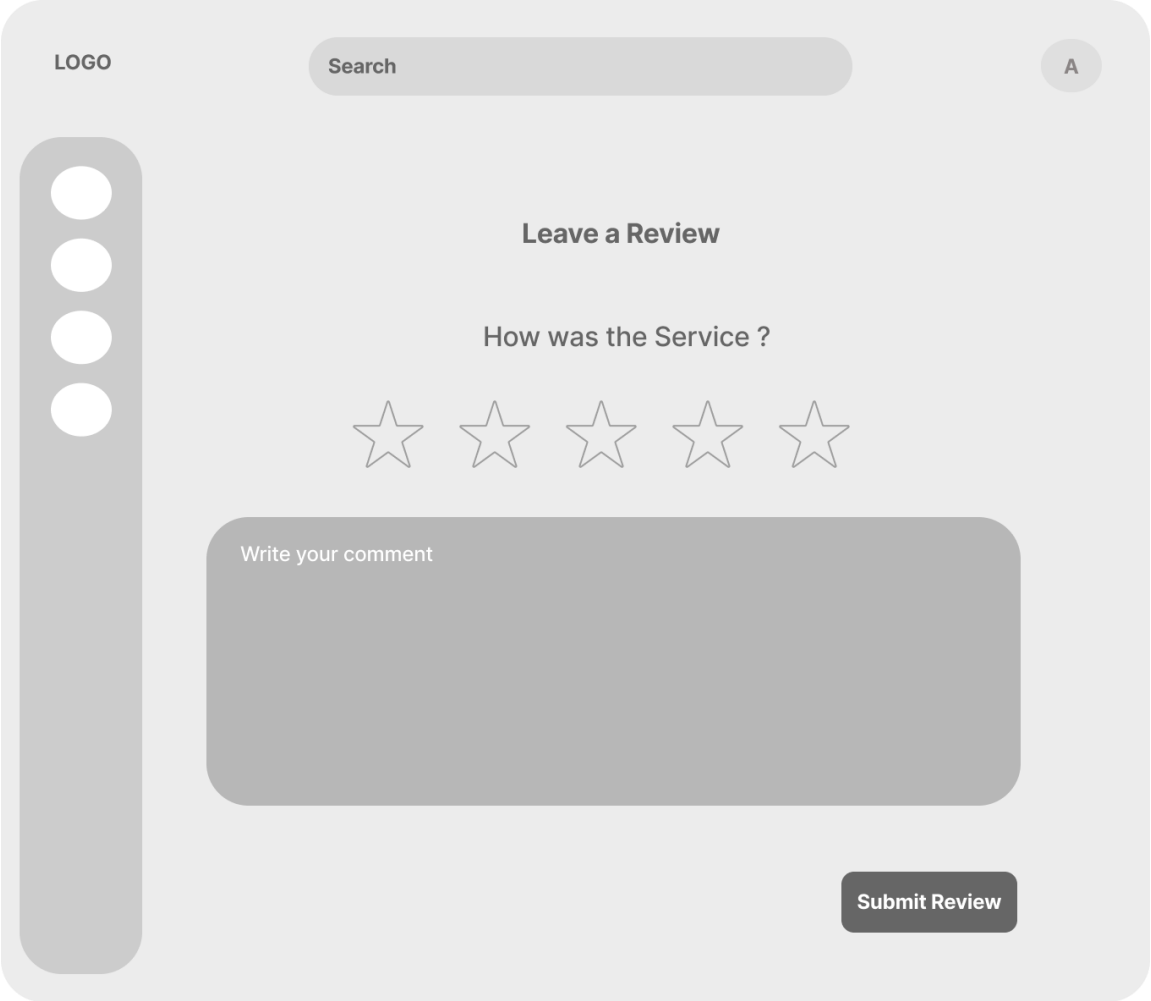


Figure 39 shows that Users rate their experience using a five-star rating system and provide a written comment. A "Submit Review" button finalizes the submission.

Project Development

The Web-based Business Consultancy with Project Management and Client Management System was developed using the Scrum Agile Development Methodology that helped the researchers to determine specific modules and features that are accomplished during a sprint of a given time frame, usually lasting one to one to four weeks. With the given methodology, the researchers easily identified the problems encountered by the web-based ordering system and provided solutions to these problems. This allowed the researchers to progress efficiently and effectively during the project development.

Scrum is a widely used framework and a popular way to organize and handle complicated projects, especially in software making (Bhaskar, 2024). In this study, the researchers employed the Scrum technique to guide the method of development. This agile framework technique focuses on iterative development, collaboration, and adaptation. Our process will include breaking down the project into smaller tasks known as 'user stories'. Sprints are also part of this system. It is the amount of time given to complete a set of activities. There are also daily Scrum meetings or stand-ups to track and review the project's progress, challenges, and requirements. At the end of the sprint, the researchers conducted a sprint review to demonstrate the completed work and gather feedback. These sprint reviews will provide feedback that can be used to know how well we did and assess the parts of the project that still need improvement. Using this methodology was very helpful for us researchers in developing an effective, efficient, and user-friendly website or platform for our client.

Figure 40

Scrum Agile Methodology Framework



Phases:

Sprint Planning Meeting - The development team held meetings to determine which product backlog module to focus on in the upcoming sprint. The length of the next sprint was also determined during this phase.

Daily Scrum Meeting - The scrum team held a 20-minute meeting every day to assess the project's progress and ensure that each team was doing its assigned tasks.

Sprint Review - The sprint review is done at the end of every sprint, where the finished work in a sprint is presented to the product owner. The product owner reviewed the finished work in the sprint and assessed whether changes would be made or the finished work would be accepted.

Sprint Retrospective - The goal of a sprint retrospective is to identify what worked well in the previous sprint and what can be improved for the one that follows. Sprints are continuously improved through sprint retrospectives, making each one more successful than the one before it.

Roles:

Product Owner - The client who utilized the software is represented by the product owner. The product owner decides what features and modules will be added to the product, what changes the product have, and whether to accept or reject the finished product.

Development Team - a group of people in charge of delivering the promised and desired product increments.

Scrum Master - The development team's efficiency and productivity were under the scrum master's control. They are in charge of ensuring the team comprehends scrum theoretically and practically and providing support and direction as the team advances through its sprints.

Tools:

Product Backlog - The development team's efficiency and productivity were under the scrum master's control. They are in charge of ensuring the team comprehends scrum theoretically and practically and providing support and direction as the team advances through its sprints.

Sprint Backlog - It is a set of tasks from the product backlog identified during the Sprint Planning Meeting. These tasks need to be accomplished during the upcoming sprint.

Operation and Testing Procedures

The operations of the system were ensured to be correct by testing the system's features and functionality. The tests will verify every function in the system to ensure it functions properly as intended.

Functional Suitability Testing was conducted to test if the functionality of the features of the web-based application can achieve the expected output. The following procedures will be the steps accomplished in this test. An example of how the Operation and Testing Procedure sections could be defined for the given test case will be provided. A sample test case will be shown in Table 1.

Table 1.

Sample Test Case

Test Case ID				**UC Reference**	
Objective		To ensure that the system will be able to manage projects.			
Assumptions/ Preconditions		The admin is logged in as a valid admin account.			
Actions		**Expected Result**		**Actual Result**	
1. The admin signs into the admin account. 2. The admin clicks on "Projects" from the menu.		1. The system enables the admin to generate reports.		1. The system successfully allows the admin to manage projects.	
Status	PASSED	**Severity**		**Priority**	

The table serves as a standardized format for the test case document utilized by the researchers. Each row in the table corresponds to a specific test case and includes key

details such as the test case ID, objective, expected result, action procedure, test data, actual result, and status. The status column indicates the outcome of the test case, whether it passed, failed, was not executed, or was blocked. This organized layout guarantees comprehensive testing and validation of the web application's functionality.

Table 2.

Classification of Error Severity

Severity	Description
Critical	The problem signifies that the process has been completely stopped and cannot continue until it is resolved.
Major	The problem causes the system to crash. Nevertheless, some system components are still functional.
Minor	The problem does not result in any significant system failure.

The table provides a clear and comprehensive overview of error severity classification. It outlines three distinct levels: Critical, Major, and Minor. Each level clarifies the potential impact of the corresponding error on the system's functionality. This table is a valuable quick-reference tool, enabling stakeholders to categorize errors based on severity. This categorization facilitates efficient prioritization and resource allocation for error identification and resolution efforts.

Table 3

Classification of Error Priority Levels

Priority	Description
High	The problem needs to be resolved as soon as possible since it significantly affects the application.
Medium	The problem should be fixed throughout the normal course of development.
Low	The problem must be resolved when a more crucial feature is taken care of.

The table offers a structured classification scheme for error priority levels. It categorizes errors into three distinct levels: High, Medium, and Low. Each level includes a description that clarifies the urgency of addressing the error. This table serves as a valuable tool for prioritizing and allocating resources effectively. By utilizing this classification, developing teams can optimize workflows by assigning an appropriate priority level to each error. This ensures that critical errors are handles immediately, while less urgent ones are addressed later in the development cycle.

Table 4

Overall Summary of Functionality Test Cases

Use Case	No. of Test Cases
Verified User	
Admin	
Total	

This table provides a detailed set of functional test cases categorized by user roles: "Verified User" and "Admin." Each listed user role corresponds to a specific use case

within the system. Furthermore, a "Total" row at the end gives the overall count of test cases for all user roles. This table is designed as a template for documenting and structuring comprehensive test case details, which helps ensure a systematic approach to functional testing.

Table 5

Reliability Test Cases Summary

Test Case ID	Objectives

The test case ID serves as a unique identifier assigned to each test case, facilitating easy reference and tracking. The objectives of each test case include a description of the goals and purposes, specifically outlining the aspects of reliability being assessed or verified.

Table 6

Testing Procedure for Functionality Suitability

Modules	Steps to be taken	Expected Output
1. Project Management	1. Navigate to the "Projects" section. 2. Create a new project by adding the project title, description, and timeline. 3. View the project details and track its progress. 4. Update the project status.	1. The project should be created successfully, with accurate details displayed. 2. Updates should be reflected in real time, and clients should see the project on their dashboard.
2.Task Management	1. Navigate to the "Tasks" section within a project. 2. Create a new task, specifying the task name, description, payment, and due, and assign an employee. 3. Close the task once completed.	1. Tasks should be created, assigned, and updated correctly. 2. The progress and status changes should be immediately reflected by both the client and consultants and completed tasks should be marked as closed.
3. Calendar	1. Access the "Calendar" section. 2. Check the consultant's availability and view existing events. 3. Set availability for consultations and sync with the project deadlines.	1. The calendar should accurately display all projects due, tasks due, and consultation availability. 2. New events should be added successfully, and the availability of the consultant should be correctly updated.
4.Client Management	1. Navigate to the "Client" section. 2. Create account for new client with their details. 3. View client profiles, project history, and notes related to each client. 4. Update client details and log interactions.	1. Clients accounts should be added to the system, and their details should be accessible and up-to-date. 2. Any changes or new interactions should be saved and reflected correctly.
5. Notifications	1. Trigger a system event (e.g., task assignment, deadline approaching) to generate a notification. 2. View the notification in the	1. Notifications should be generated and displayed in real-time.

	notification center.	
6. Payments	1. Navigate to the "Payments" section. 2. View a list of pending and completed payments.	1. Payments should be listed correctly and have accurate statuses (pending, completed).
7. Reports	1. Navigate to the "Reports" section. 2. Select the type of report to generate (e.g., project progress, task completion rate). 3. Apply filters such as date range or employee to narrow down the data. 4. Generate and download the report in PDF or Excel format.	1. The report should be generated accurately based on the selected filters. 2. It should be downloadable in the specified format with the correct data reflected.
8. Project (client)	1. Access the "Projects" section. 2. View a list of all ongoing and completed projects associated with the client. 3. Select a specific project to see details such as progress, milestones, tasks, payment, and deadlines. 4. Download any related documents or reports shared within the project. 5. Provide feedback or comments on the project's progress.	1. The list of projects should be displayed correctly, with accurate details about progress and deadlines. 2. Clients should be able to view and download shared documents and provide feedback within the system.
9. Notifications (client)	1. Trigger notifications by system events (e.g., task completion, new consultation scheduled, document shared). 2. View notifications in the notification center upon logging in. 3. Click on notifications to be directed to the relevant project or consultation.	1. Notifications should appear in real time based on relevant system events. 2. Clients should be able to view, access relevant details by clicking.

10. Payments (client)	1. Navigate to the "Tasks" section inside a project. 2. Select a task associated with payments. 3. View the task details, including the amount due, payment methods, and deadlines. 4. Complete the payment using the available methods (e.g., credit card, PayPal). 5. Confirm the payment and view transaction details. 6. Download receipts for completed payments.	1. Tasks with associated payments should display payment details clearly. 2. Clients should be able to complete payments within the task, and the status should be updated in real time. 3. Receipts should be downloadable for completed transactions.

Reliability Testing was conducted to assess how the system performs under specified environmental conditions. This testing assesses how the system functions under various environmental conditions and performs over a defined period. The following table outlines a collection of test cases designed to examine the functionality of various system modules, including project management, task management, calendar, client and employee management, notifications, reports, and payments. Each row represents a distinct test case, detailing the procedures for execution and the anticipated results. Documenting and organizing test cases in a table format facilitates streamlined test process management and ensure comprehensive testing of each system capability against established procedures and expected outcomes.

Table 7

Testing Execution Summary

Text Execution	Expected Result	Active Result Cycle 1	Cycle 2
No. of Test Cases Results of Test Cases	100%		
Passed	100%		
Failed	0%		
No. of Test Cases Not Executed	0%		

The table provides a comprehensive summary of the test case execution process. It signifies the successful completion of all test cases, with no instances of unexecuted tests. While the "expected results" column currently remains unfilled, it is intended to be populated with the anticipated outcomes for each test case. The conducted tests have yielded a positive outcome, with all test cases passing without any failures. Overall, this chart serves as a concise overview of the testing status, confirming the successful and complete execution of all test cases.

Evaluation Procedure

Using ISO 25010, the system was assessed. A diverse group of evaluators, including students, faculty members, and professionals, was presented with the prototype. Random users from the College of Science, consisting of both web-based programming experts and students, will participate in the evaluation. A minimum of 30 students was included among the respondents. The system was demonstrated to the participants, who then received and

completed the surveys. The following outlines the planned evaluation process:

General Evaluation Process

1. Each respondent received a software assessment form to use in evaluating the effectiveness of the system.
2. The web-based application's functions were demonstrated, and its objectives were discussed to demonstrate how to use it.
3. The application was made available to all respondents.
4. According to Table eight (8), respondents were asked to evaluate the system in light of the ISO 25010 standard evaluation criteria, using a 4-point Likert scale where (4) is the highest rating and (1) is the lowest rating.
5. This project was evaluated using the following characteristics: functional suitability and reliability.
6. Based on the evaluation data collected, the overall weighted mean rating for each criterion and the grand weighted mean was determined.
7. The evaluation outcomes utilized the weighted mean value range shown in Table nine (9) and the qualitative analysis relating to that range.

Table 8

Four-point Likert Scale

Scale	Descriptive Rating
4	Highly Acceptable
3	Very Acceptable
2	Fairly Acceptable
1	Not Acceptable

The table presents a rating scale with numerical values (1-4) and corresponding descriptive labels. "Highly Acceptable" (4), "Very Acceptable" (3), "Fairly Acceptable" (2), and "Not Acceptable" (1) define the scale. This table facilitates evaluating something's acceptability. Higher numerical values indicate a more favorable rating. The descriptive labels help understand the acceptability level based on the numerical values.

Table 9

The Range of Mean Ratings and the Equivalent Descriptive Rating

Scale	Descriptive Rating
3.26 - 4.0	Highly Acceptable
2.51 - 3.25	Very Acceptable
1.76 - 2.50	Fairly Acceptable
1.00 - 1.75	Not Acceptable

The table shows a range of numerical values in each row and the corresponding descriptive ratings in the next column. This scale allows for assessment or grading based on acceptability levels. Each rating corresponds to a specific numerical range. For example,

if a value falls between 3.26 and 4.0, the table indicates it’s "Highly Acceptable". Similarly, other ranges correspond to "Very Acceptable," "Fairly Acceptable," or "Not Acceptable."

Chapter 4

RESULT AND DISCUSSION

This chapter presents the findings from the system evaluation and provides an analysis of the project. It encompasses the system's description, structural design, key capabilities, identified limitations, and a detailed discussion of the project evaluation results.

Project Description

The iConsult system is a web-based application designed to address the challenges of business consultancies by integrating project management and client management functionalities. This platform streamlines operations, enhances communication and transparency, and improves client satisfaction. iConsult provides business consultants with a centralized system to efficiently manage projects, clients, and appointments while offering clients a user-friendly interface to interact with consultants.

Therefore, to ensure a responsive and interactive user experience, the system was designed using Visual Studio Code as the main IDE, together with React.js, HTML, CSS, JavaScript, and Vite for front-end development. On the back end, MySQL acts as the database management system, and Node.js controls server-side functions. Server environments were simulated throughout development using XAMPP. External services like SendGrid for email notifications, PayPal Sandbox for safe payment processing, and the Google Calendar API for scheduling consultations are all integrated into the program.

For deployment, AWS and Heroku ensure the system is scalable and reliable. The UI/UX was designed using Figma, and version control was managed through GitHub, enabling efficient collaboration among developers.

Project Structure

This section provides a detailed overview of the project using screenshots of the web application. These images illustrate the user interface, highlighting essential elements, features, and interactions within the system. They offer readers a clear insight into the project's design decisions, functionality, and workflow.

Figure 41

Landing Page

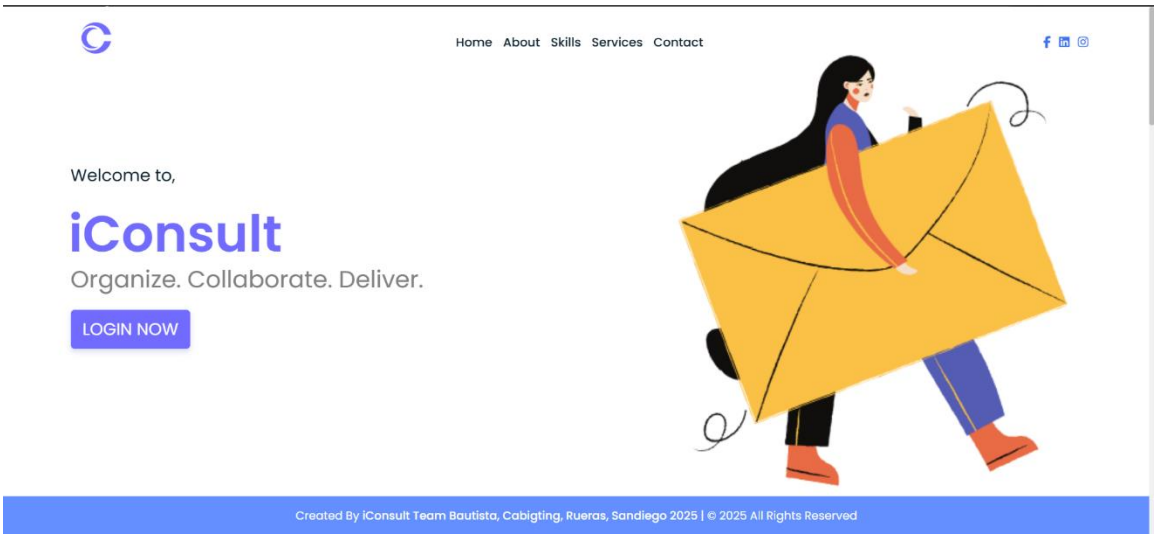


Figure 41 shows the Landing Page, which shows the user's login for the consultant and client.

Figure 42

Landing Page - About us

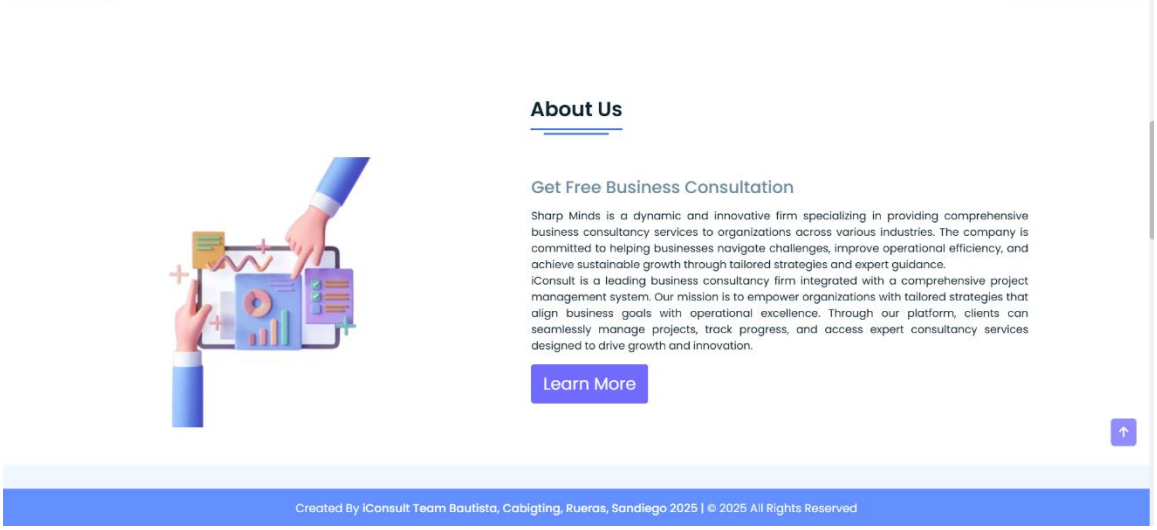


Figure 42 shows the Landing Page About Us section, presenting iConsult as a business consultancy firm with an integrated project management system outlining its mission and services.

Figure 43

Landing Page - OurSkills



Figure 43 shows the Landing Page Our Skills section, highlighting iConsult's expertise in accounting, financial management, and business consultancy.

Figure 44

Landing Page – Our Services

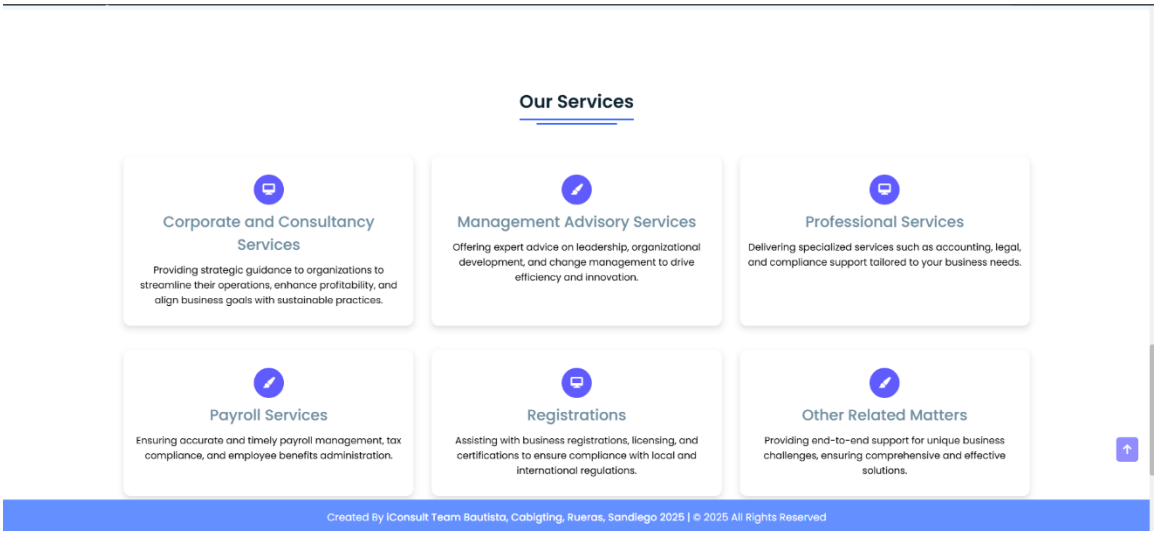


Figure 44 shows the Landing Page Our Services section, outlining iConsult's key offerings, including corporate consultancy, management advisory, professional services, payroll management, business registrations, and other specialized solutions tailored to support business growth and compliance.

Figure 45

Consultant/Client - Log In

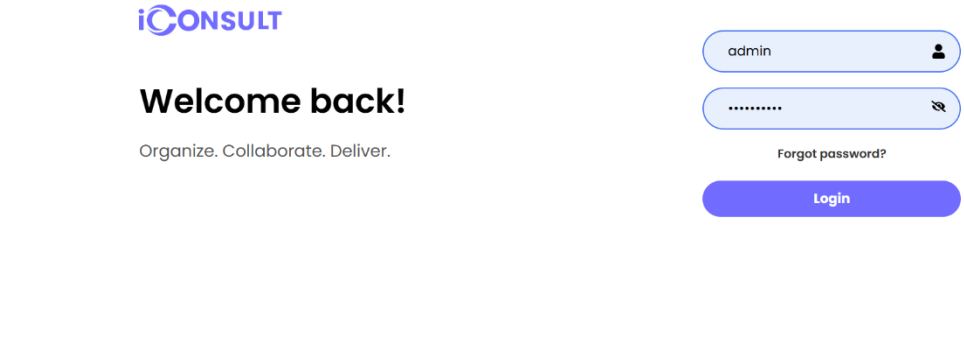


Figure 45 shows the login button where the consultant can type the username and password. There is also a button that says “Forgot Password” in case the consultant cannot remember the password and cannot log in to the website.

Figure 46

Consultant - Dashboard

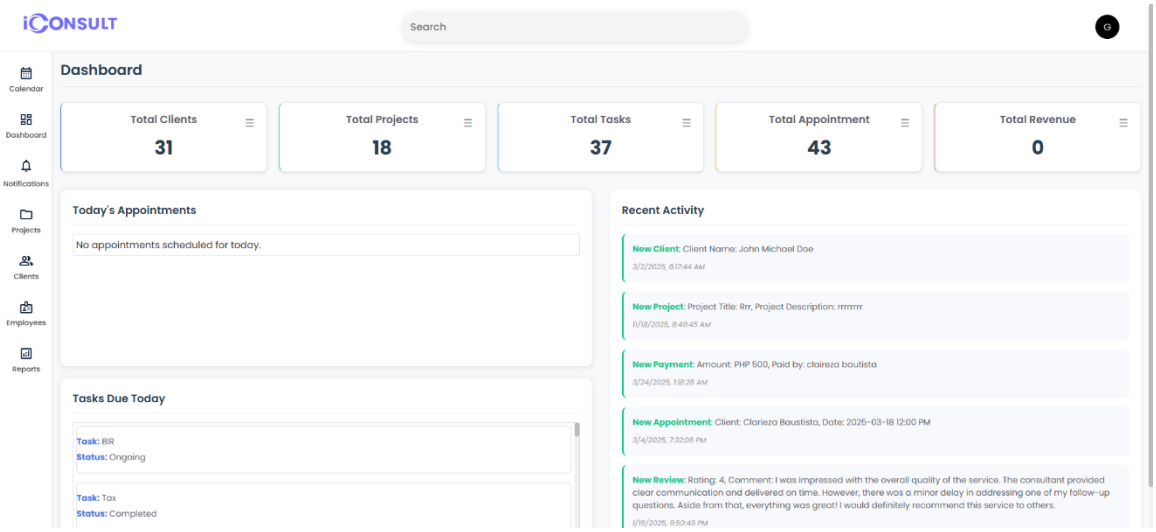


Figure 46 shows the Consultant Dashboard, providing an overview of key business metrics, including total clients, projects, tasks, appointments, and revenue. It also displays today's appointments, tasks due, and recent activities such as new clients, projects, payments, and reviews.

Figure 47

Consultant Calendar

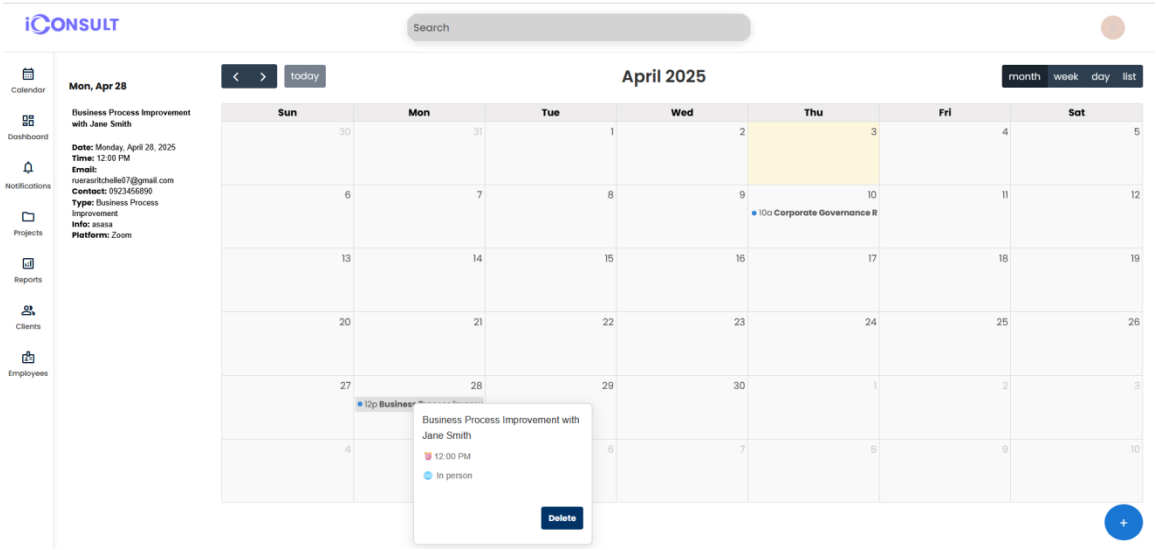


Figure 47 shows the Google Calendar designed for administrative or scheduling purposes. The main section shows the month view, with each date represented in its corresponding cell. Events scheduled for specific days are listed within these cells, including the activity's time, title, and brief description. The top navigation includes buttons to move between months, return to the current date, and switch between different calendar views, such as "month," "week," or "day." This tool helps manage appointments, consultations, and other events in a structured manner.

Figure 48

Consultant Availability

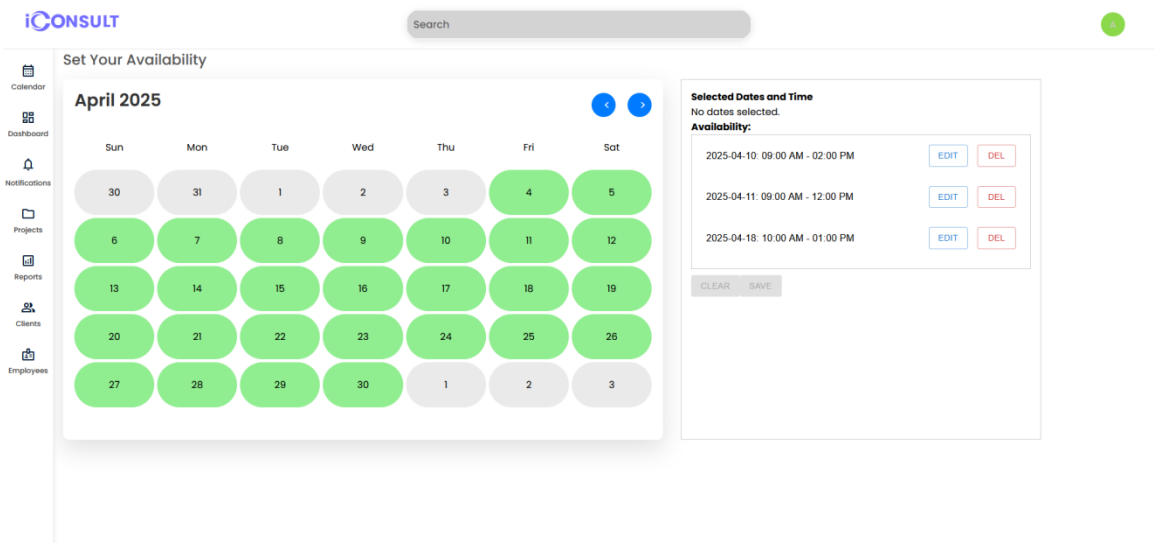


Figure 48 shows the image display for scheduling, featuring a calendar for selecting dates and a section for managing availability with editable time slots. The consultant can set specific time ranges, edit existing entries, and save their availability preferences.

Figure 49

Consultant Appointment Form - Date & Time

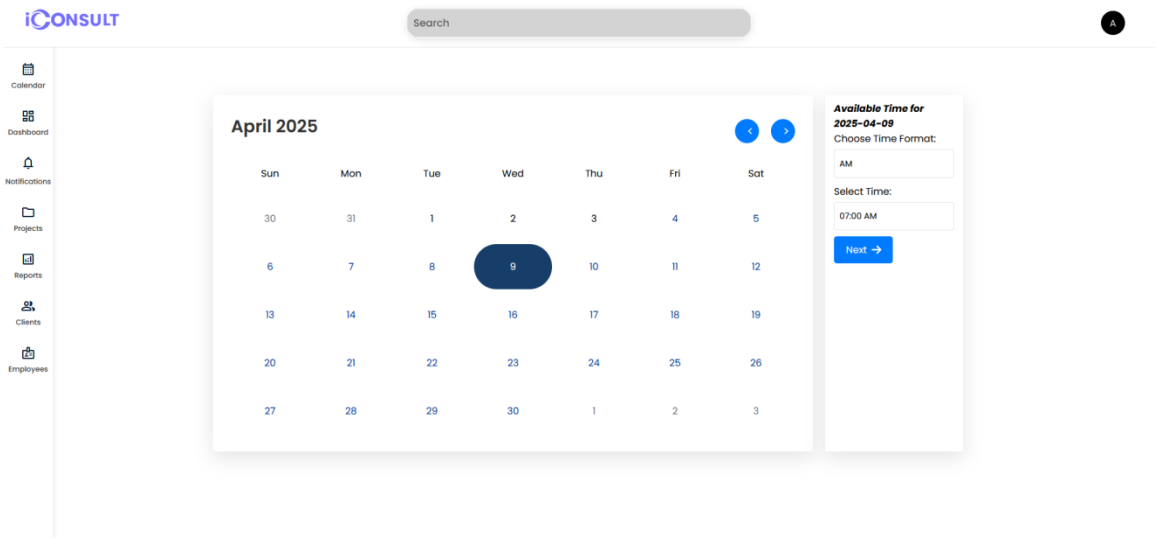


Figure 49 shows where the consultant can choose the Date and time for the consultation with the client.

Figure 50

Consultant Appointment Form - Consultation Details

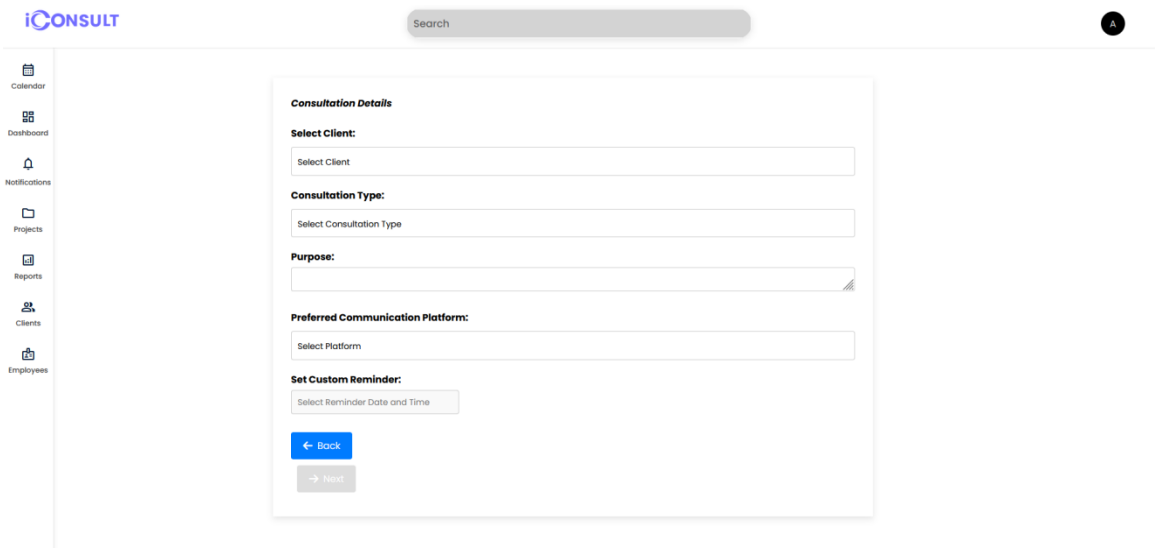


Figure 50 shows where the consultant can choose a specific client, what type of consultation they decided, and the purpose of the consultation. The consultant can also choose the preferred communication platform by the client. Another feature of this page is the reminder, which allows the employee to be notified one day before the consultation through email.

Figure 51

Consultant Appointment Form - Confirmation

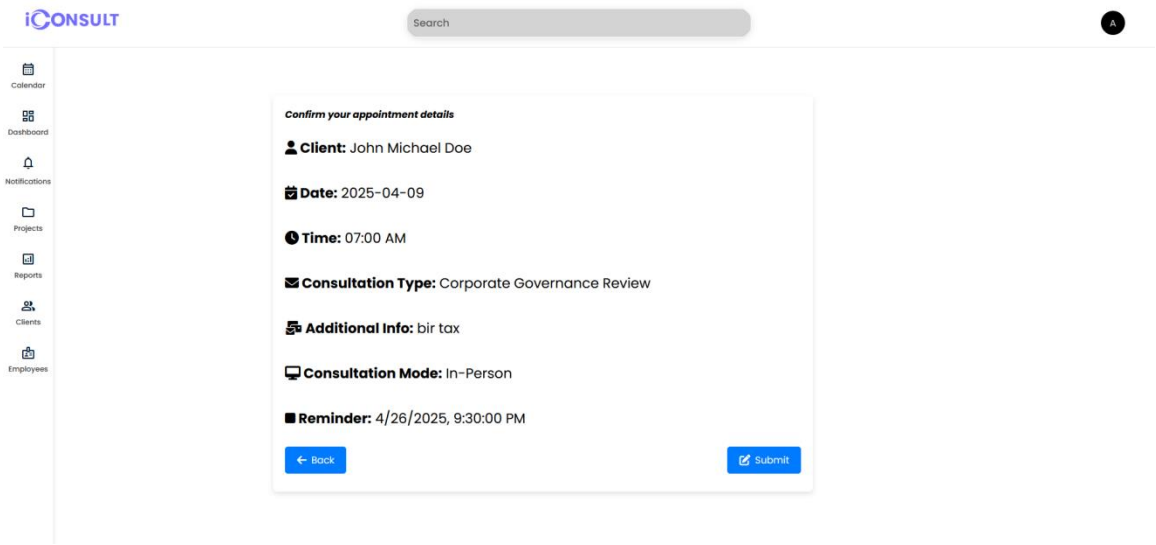


Figure 51 shows the Confirmation of the appointment details.

Figure 52

Consultant – Notification

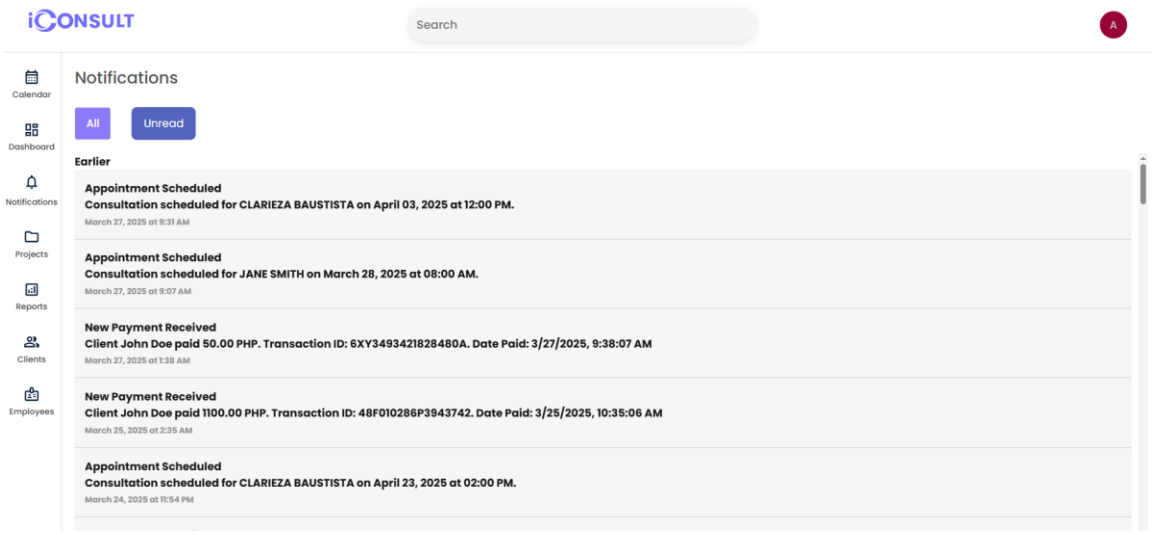


Figure 52 shows a consultant notification log. This displays a list of recent notifications, including appointment schedules and payment updates. The notification provides filtering

options for viewing all notifications or only unread ones. Each notification entry includes a title, description, and timestamp, allowing users to track important updates efficiently.

Figure 53

Consultant Project List

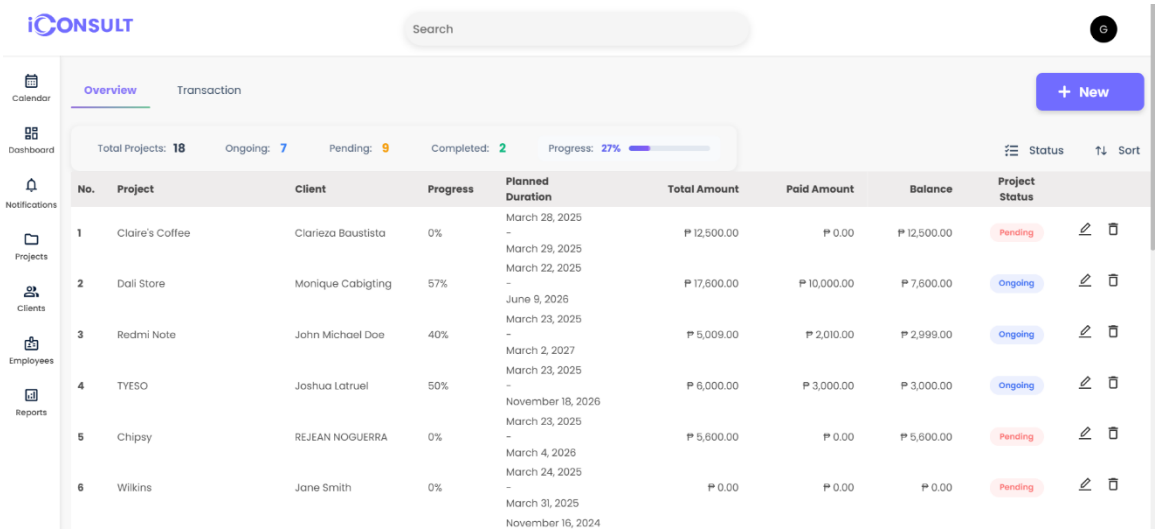


Figure 53 shows a tabular report detailing multiple projects. Each row represents a project and contains columns for the project name, the client's name, the project progress, the project timeline, the project status, the down payment amount, the total project cost, and the payment status. This table provides a concise summary of multiple projects, allowing for a quick assessment of their current state, timelines, and payment statuses.

Figure 54

Consultant- Project Form

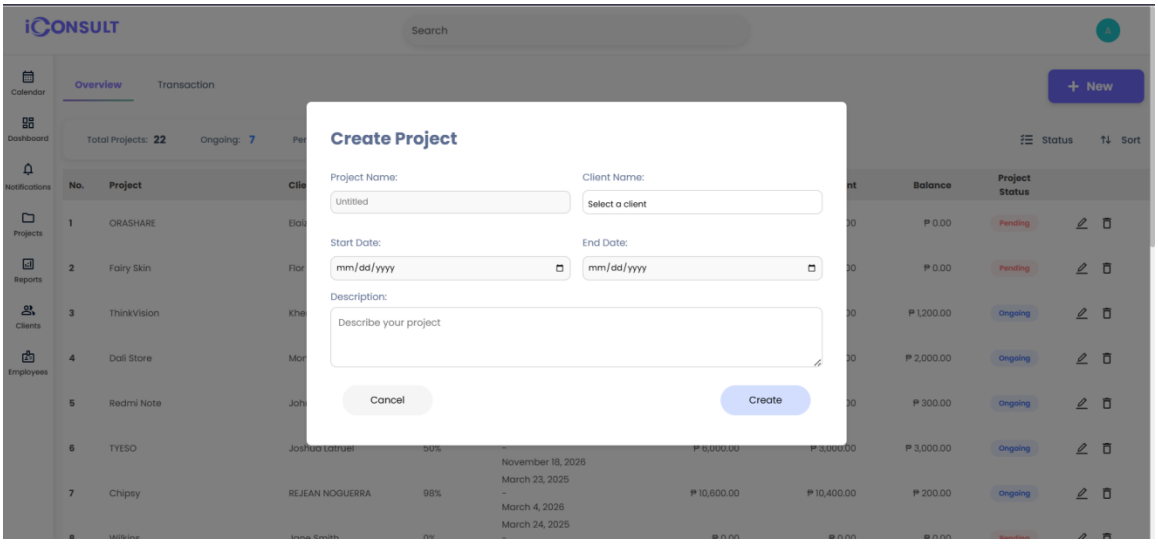


Figure 54 shows a form for creating new projects. To create a project, a user must fill in the fields for the project's name, select a client from a list, specify start and end dates, provide a description, enter the contract price, and optionally enter a down payment amount.

Figure 55

Consultant Task List

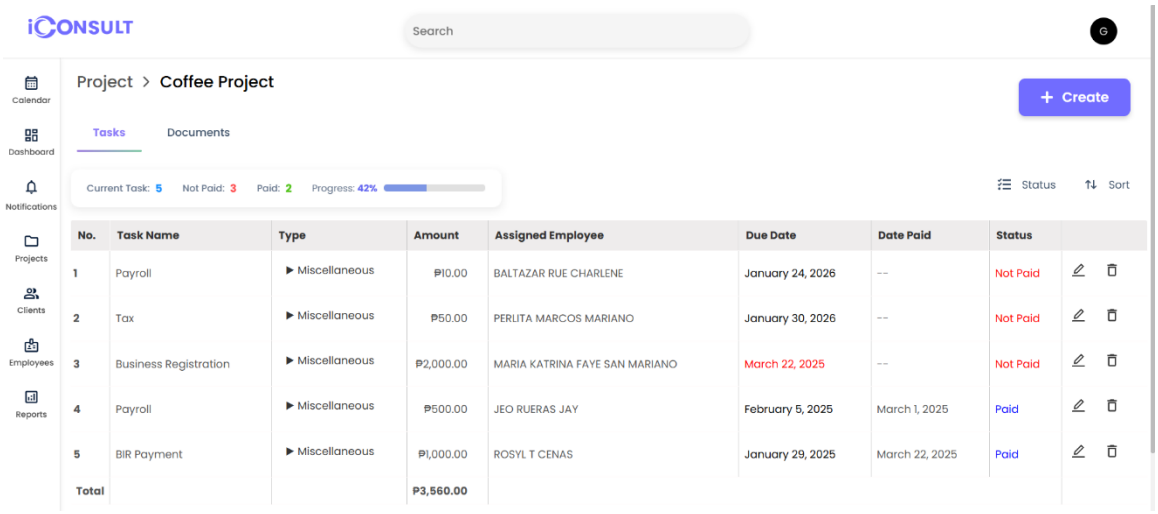


Figure 55 shows the task for the project the client availed, the employee who will do the task, the project status, the due date of the task, and the amount to be paid by the client for the said project.

Figure 56

Consultant - Task Form

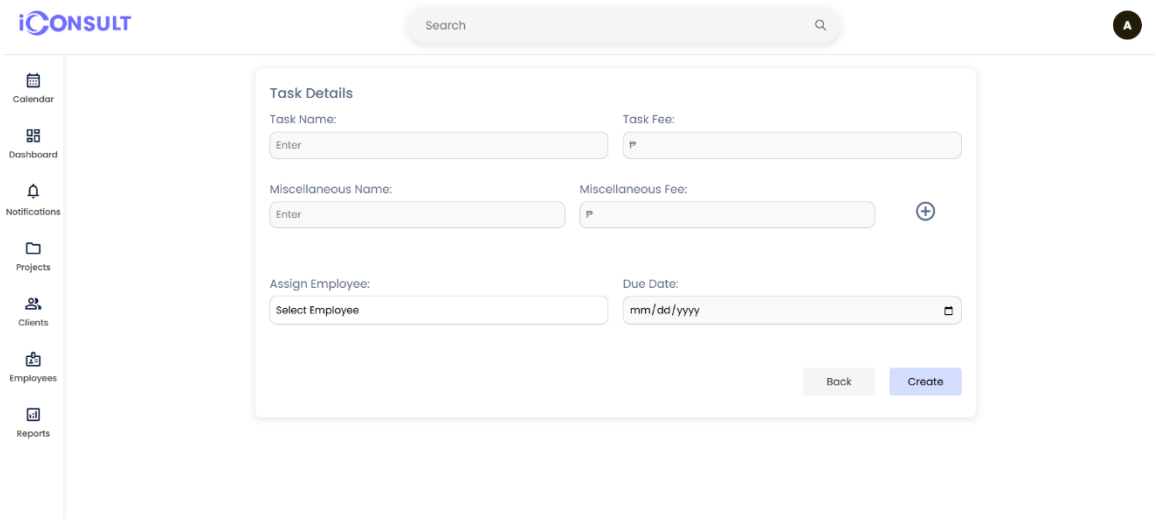


Figure 56 shows a task creation form within a project management system. The form requires input for task name, task fee, due date, and task assignment to an employee from a dropdown list. Additional miscellaneous fees can be added. Create Task and Cancel buttons are provided for submission or cancellation.

Figure 57

Consultant - Transaction

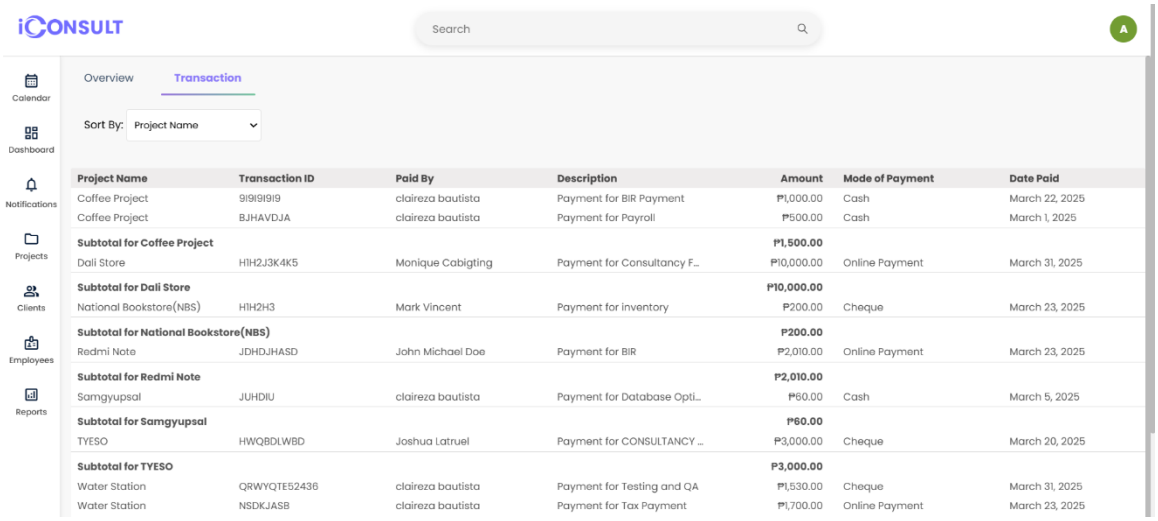


Figure 57 shows a client transaction. This displays a table of recent financial transactions categorized by project name. The table includes columns for Transaction ID, Paid By, Description, Amount, Mode of Payment, and Date Paid. Transactions are grouped under project names, with subtotals calculated for each category.

Figure 58

Consultant Documents

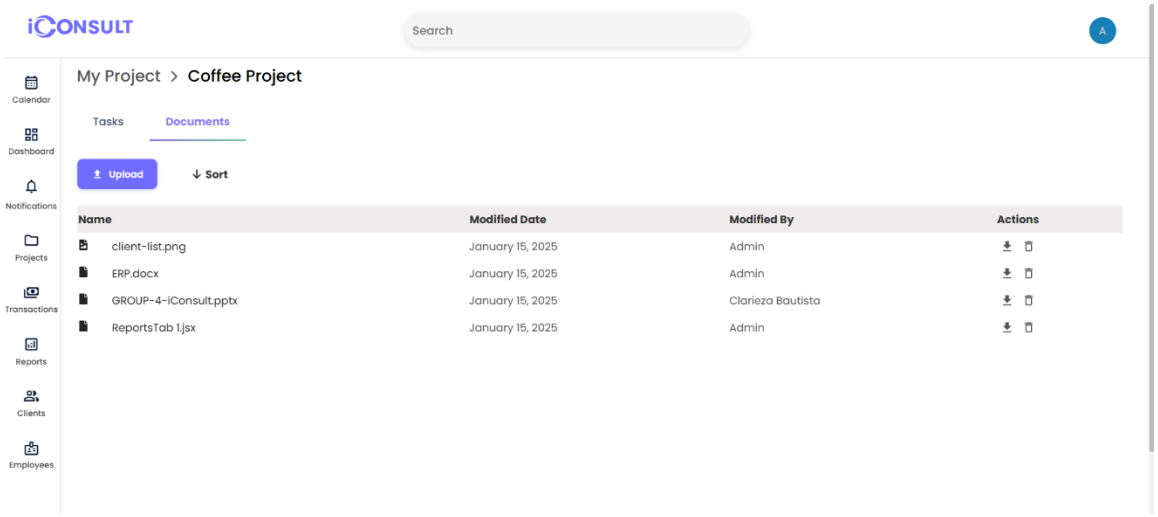


Figure 58 shows the project files. These files can be viewed as the services that the client has with the consultant. The consultant can upload the soft copy of the project directly from their devices. The uploaded project file can be deleted through the “Remove” button.

Figure 59

Consultant - Client List

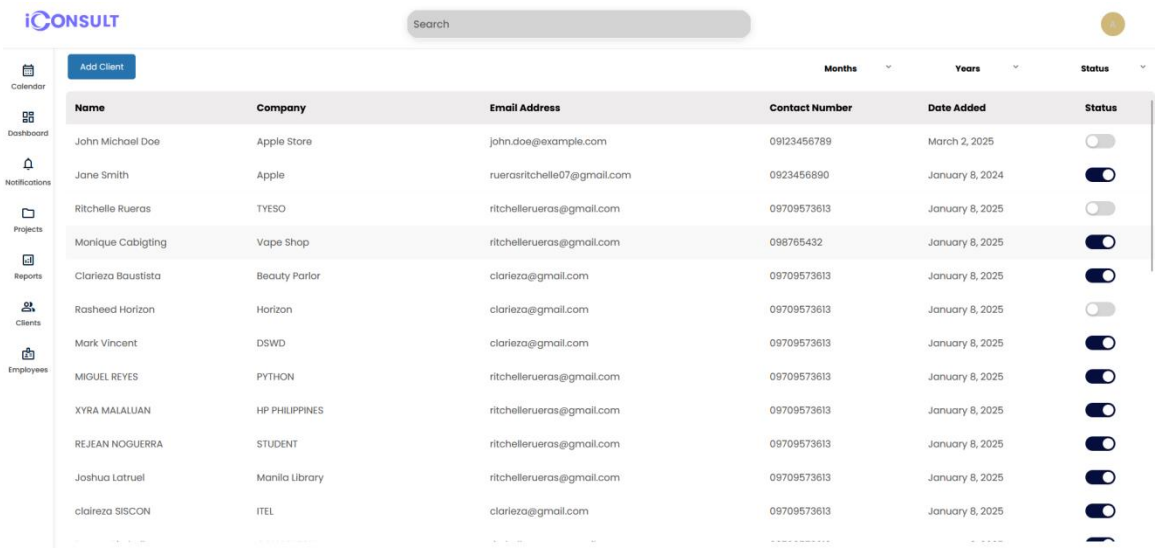


Figure 59 shows a client list, displaying each client's name and a status indicator. The status toggle switches indicate whether a client is active or inactive.

Figure 60

Consultant - Client Account Creation

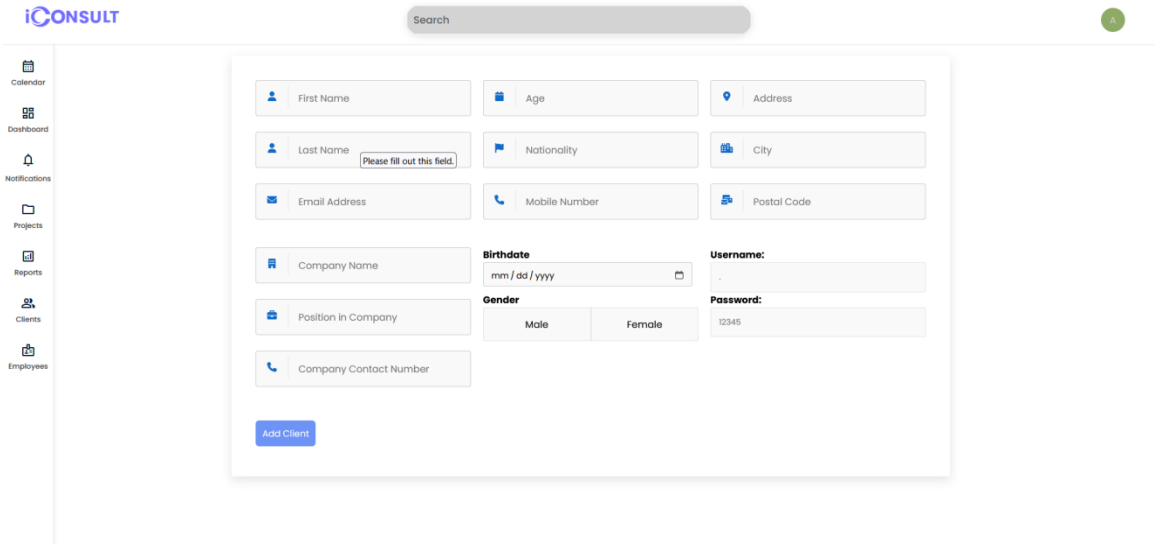


Figure 60 shows the image of a client account creation form within a web application. The form requests information such as last name, first name, middle initial, date of birth, username, password, address, mobile number, email address, company name, and status, with an "Add Client" button to submit the new account details. A client account creation form is accessible only to authorized consultants with the necessary permissions to add new client records to the system.

Figure 61

Consultant - Client Overview

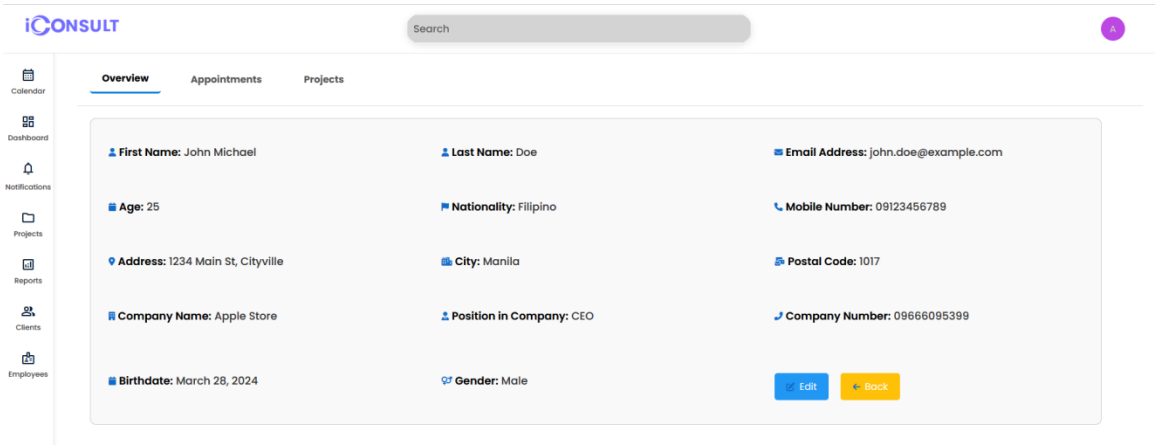


Figure 61 shows a client details page. It includes a section for the client's core information, followed by historical records. These records are often presented in tabular format, with separate sections for appointment history, project history, and payment history (tracking financial transactions and their status).

Figure 62

Consultant- Employee List

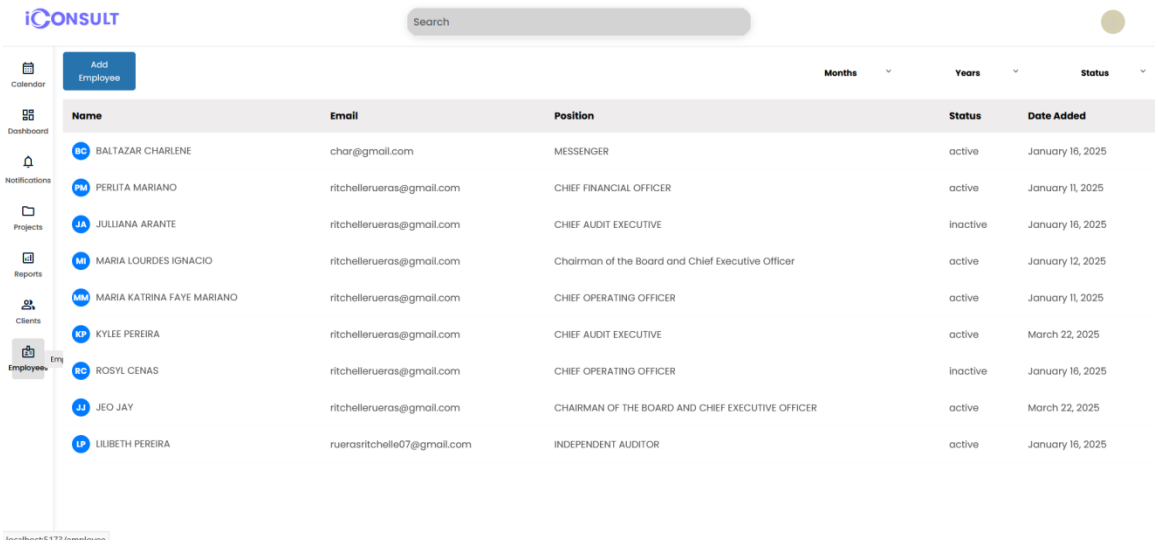


Figure 62 shows employees' records, detailing their personal information, contact details, and employment status.

Figure 63

Consultant - Employee Form

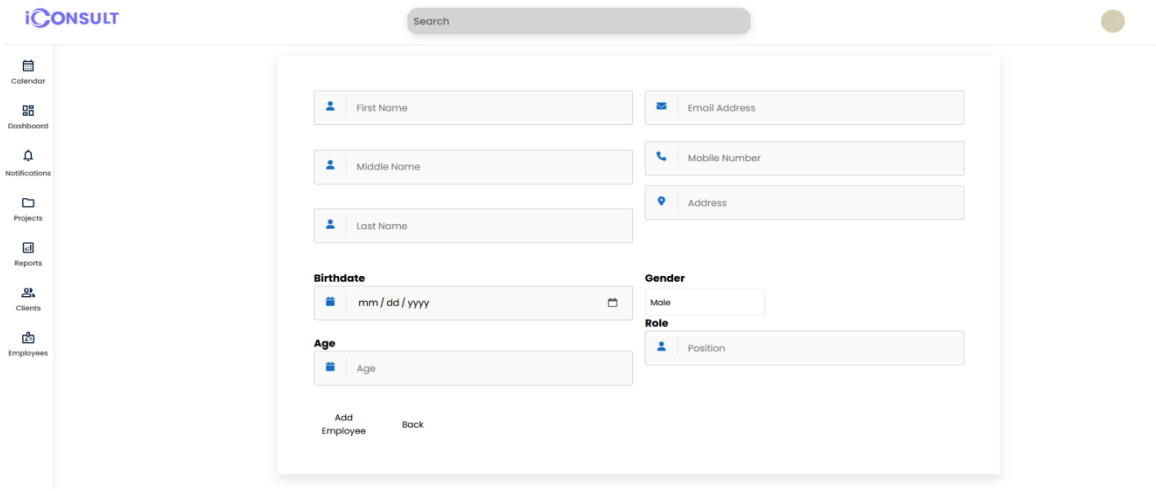


Figure 63 shows an employee information form. It allows for inputting an employee's last name, middle initial, first name, birthdate, address, mobile number, email address, and employment status (active or inactive). An "Add Employee" button submits the data.

Figure 64

Consultant - Employee Overview

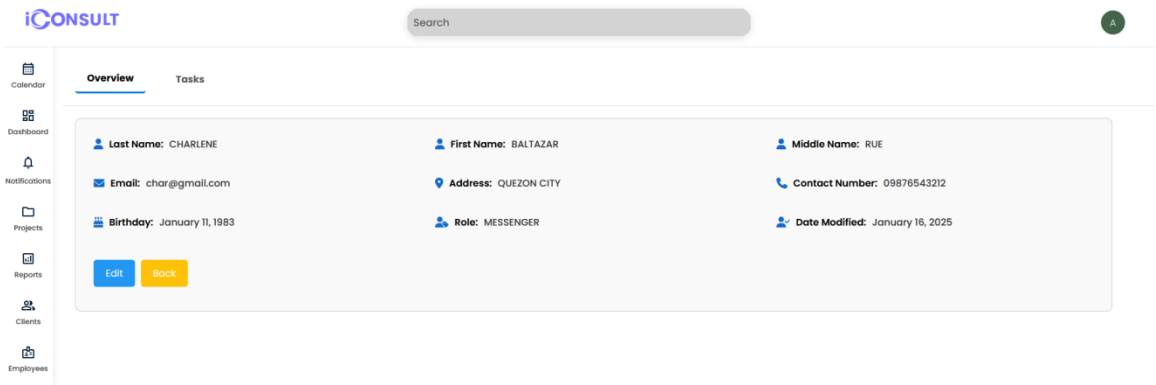


Figure 64 shows the Employee Overview section, displaying detailed information about a selected employee, including their personal details, contact information, role, and last modification date.

Figure 65

Consultant - Reports

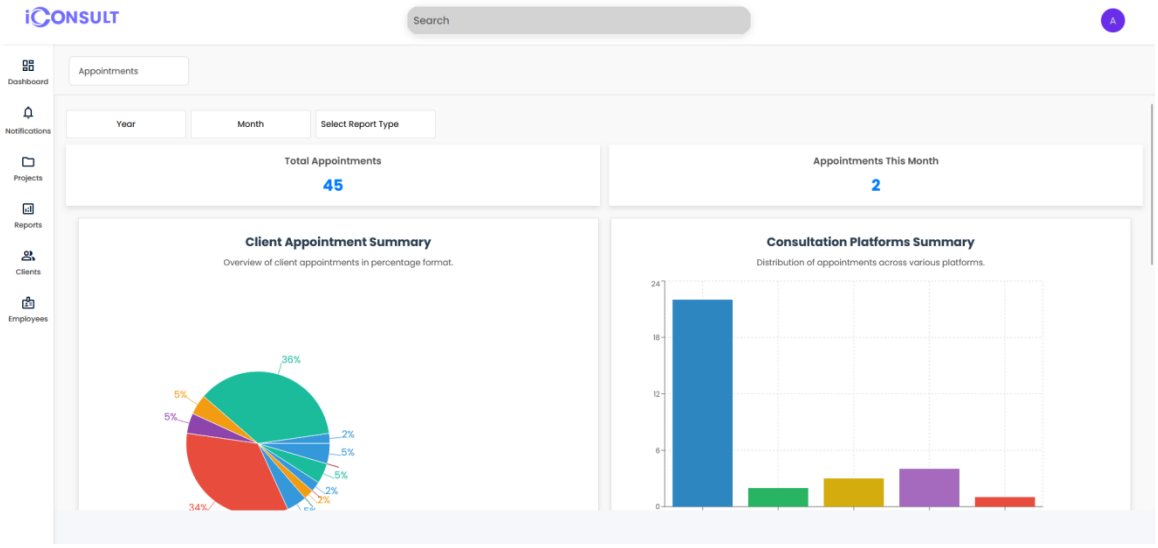


Figure 65 shows the Consultant Reports section, providing comprehensive insights into various business metrics, including appointment trends, client analysis, employee performance, project statuses, financial reports, and client feedback.

Figure 66

Consultant - Account Settings

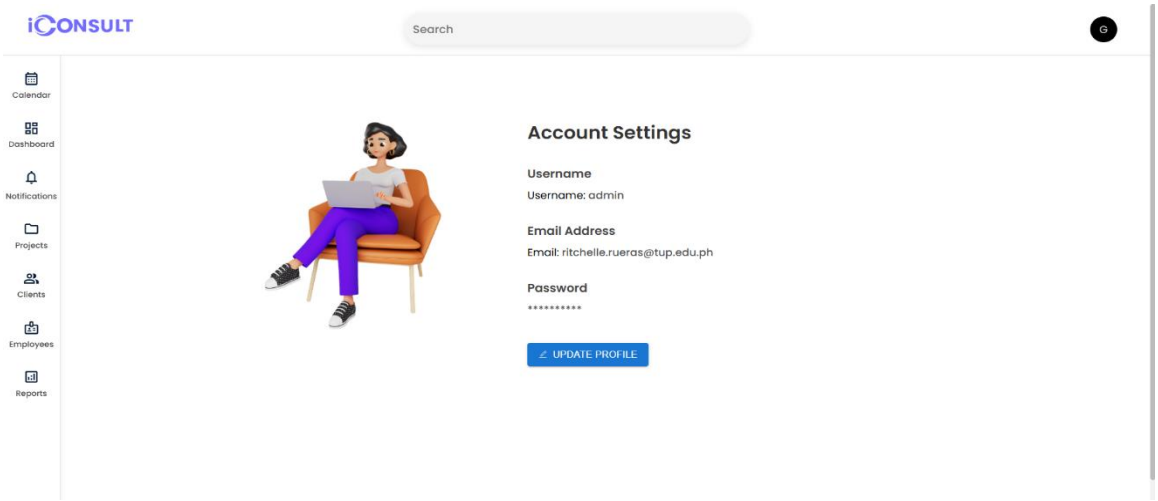


Figure 66 shows the Consultant Account Settings section, allowing users to manage their account details, including username, email address, and password.

Figure 67

Client - First Time Login

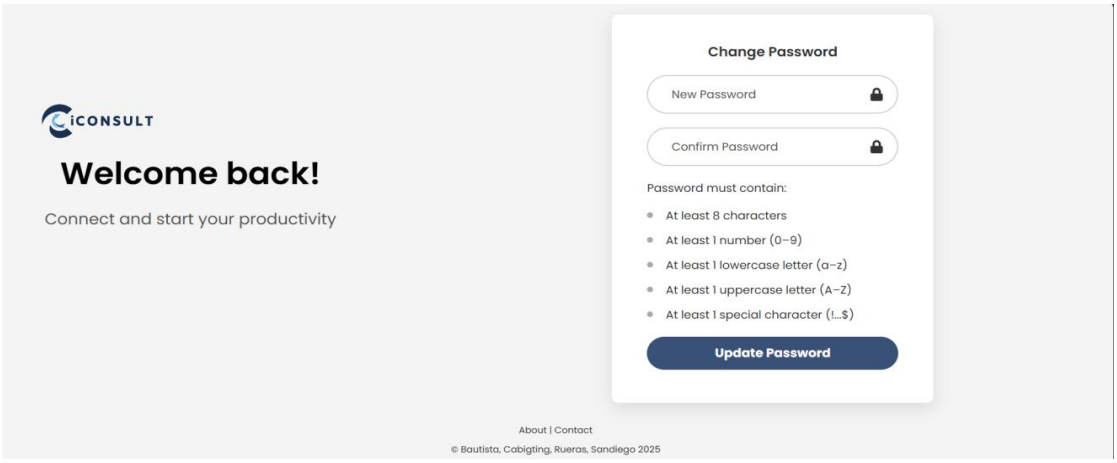


Figure 67 shows the change password. The client should type the new password and confirm the same password through the "confirm password" button. After the password is made, the client can now click the "Update Password".

Figure 68

Client - Calendar

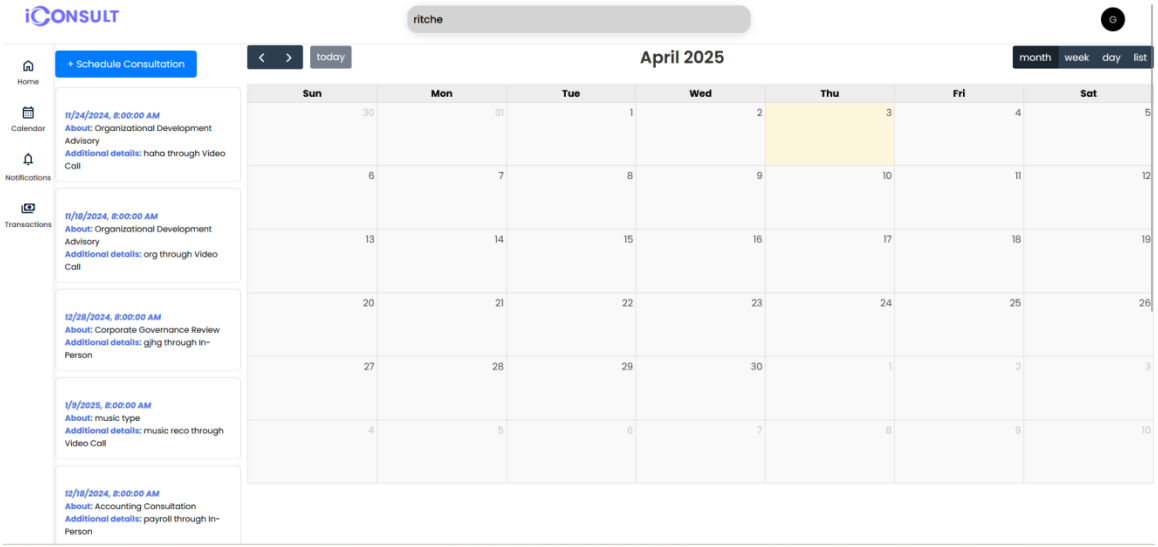


Figure 68 shows the consultation scheduling calendar. The calendar shows scheduled consultations for various services. These consultations are displayed in blue blocks on their respective dates. The client can use or check this calendar so as not to forget the consultation and be available by the time of the consultation.

Figure 69

Client Appointment Form - date and time

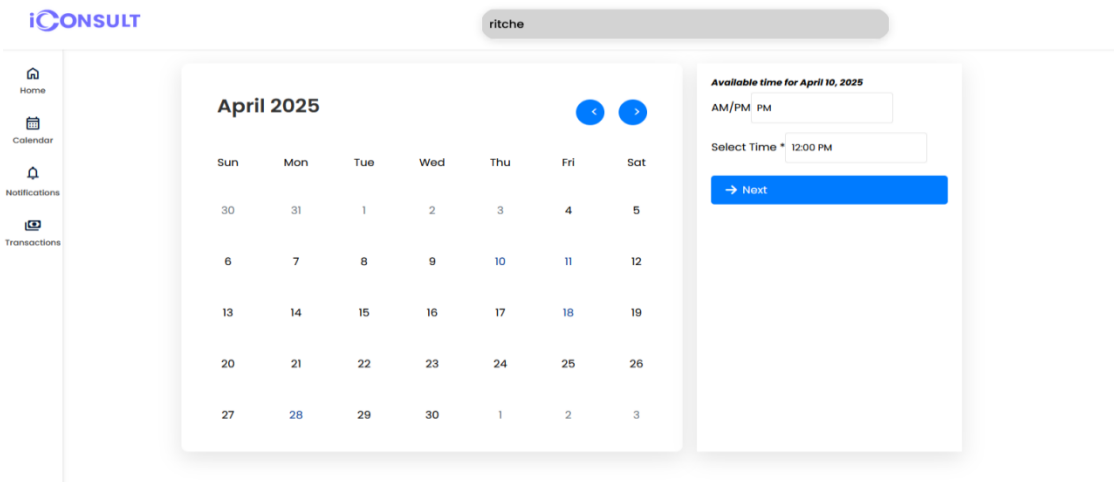


Figure 69 shows the Client Appointment Form, where clients can select a preferred date and time for their consultation.

Figure 70

Client Appointment Form

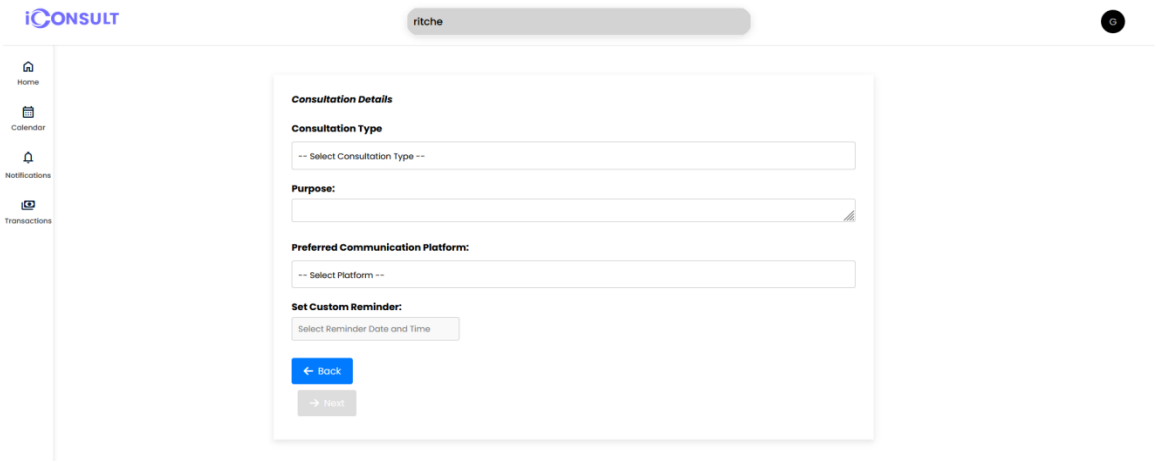


Figure 70 shows the Client Appointment Form, where clients can specify the type of consultation, state their purpose, choose a preferred communication platform, and set a custom reminder by selecting a date and time for notifications.

Figure 71

Client Appointment Form - Confirmation

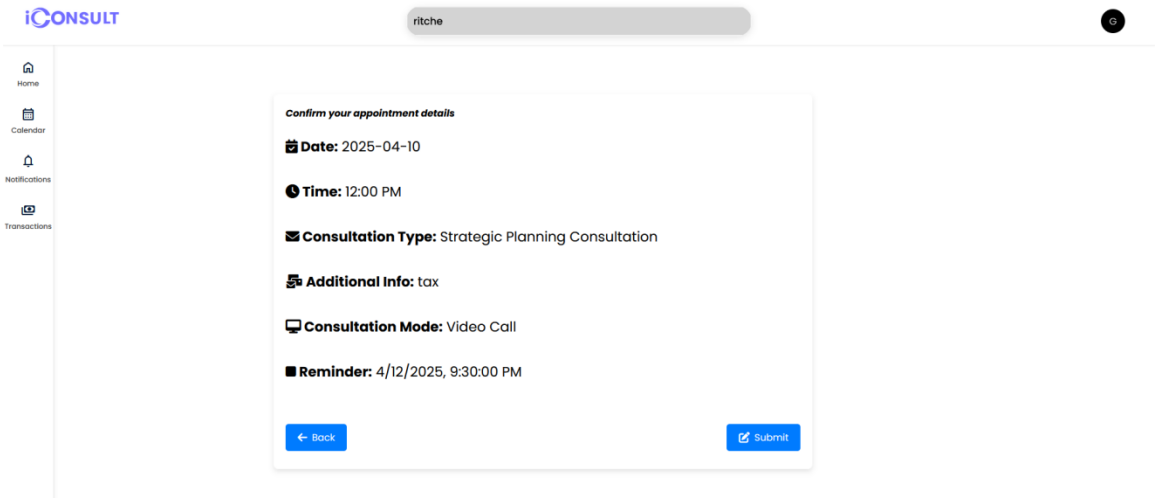


Figure 71 shows the Client Appointment Form - Confirmation section, where clients review their appointment details before finalizing the appointment.

Figure 72

Client - Notification tab

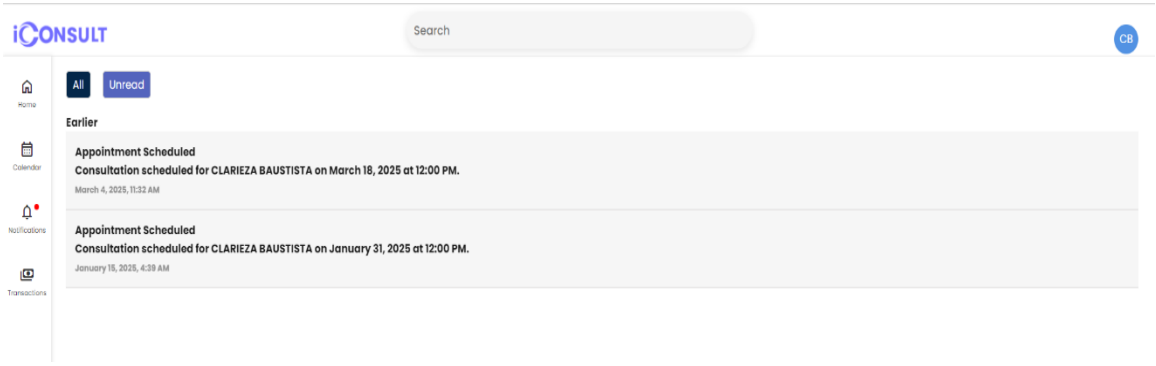


Figure 72 shows a client notification log. This displays a list of recent notifications.

Figure 73

Client - Project tab

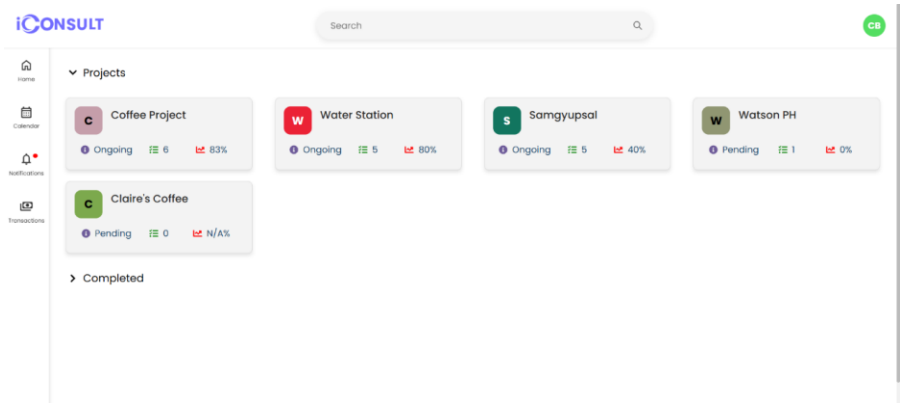


Figure 73 shows a display of the project, which shows the ongoing and completed projects. Each project has a status and progress that the client can check to monitor the projects.

Figure 74

Client - Project Overview

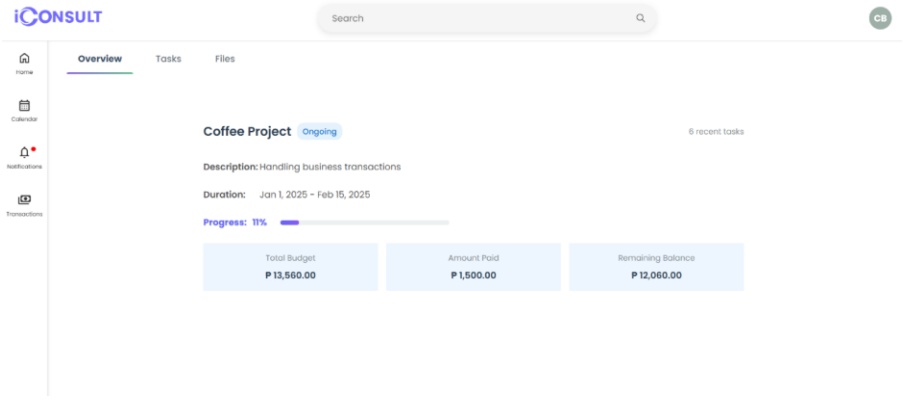


Figure 74 shows a detailed view of a specific project, providing essential information such as the project name, payment status, description, due date, and current status. It also includes a breakdown of the financial details, including the contract fee, total task cost, downpayment made, and remaining balance.

Figure 75

Client - Project Task Overview

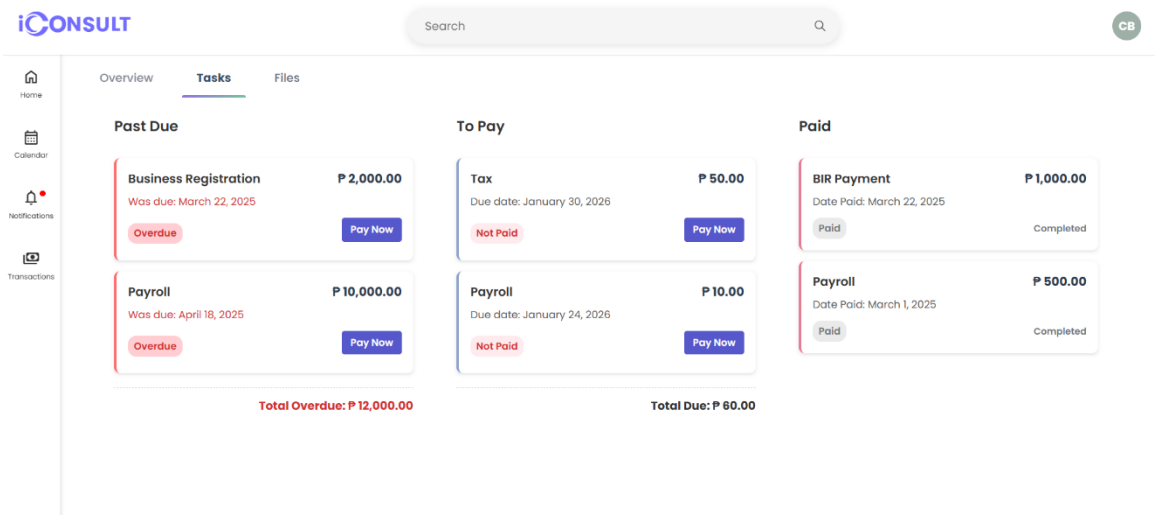


Figure 75 presents a detailed breakdown of tasks associated with a project. It includes task names, corresponding fees, due dates, statuses, and amounts.

Figure 76

Client - Project files upload and view

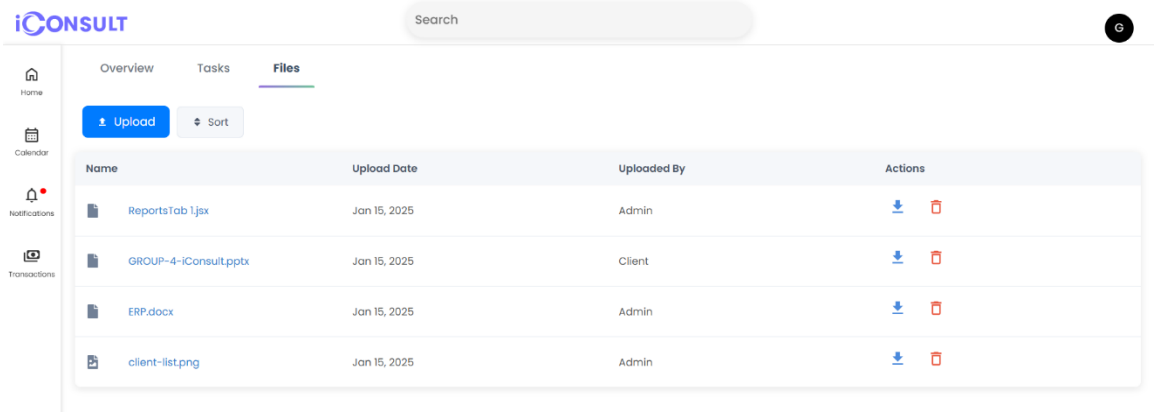


Figure 76 displays the project file management section, where clients can upload and view project-related files. It includes details such as file name, upload date, and the uploader's name.

Figure 77

Client - Payment

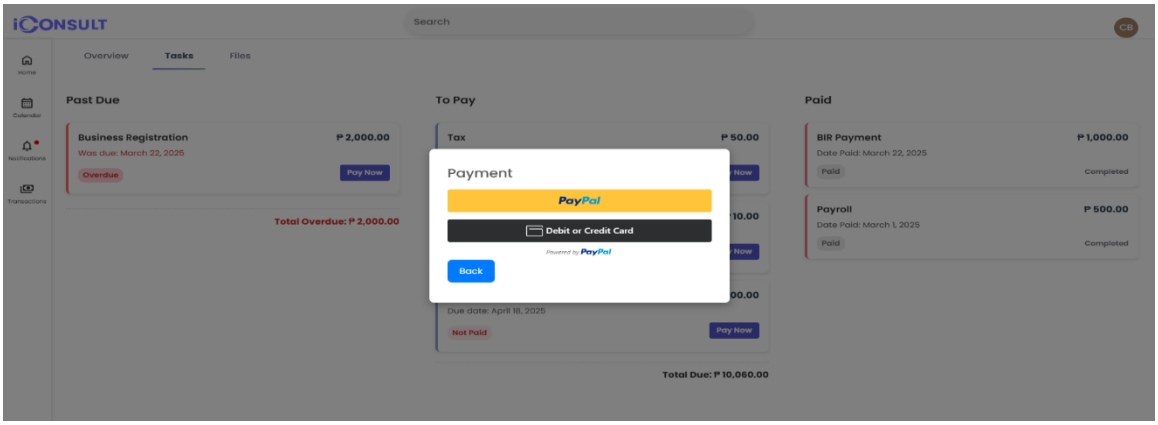


Figure 77 shows the payment gateway interface. The client is presented with payment options: PayPal, Debit, or Credit Card. The client can pay in different options.

Figure 78

Client – Review

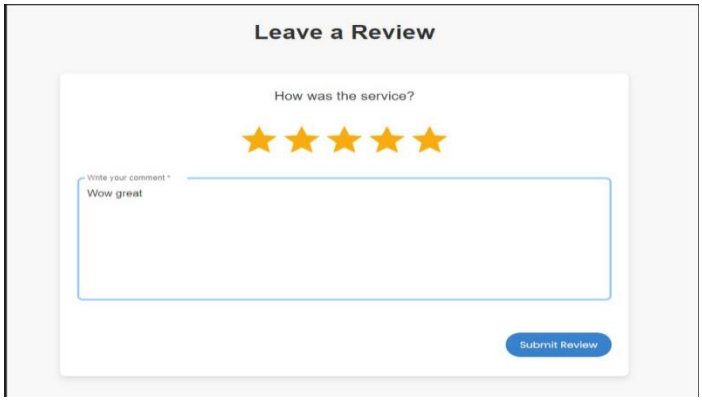


Figure 78 shows the client review section, where clients can provide feedback on completed projects.

Figure 79

Client - Transactions

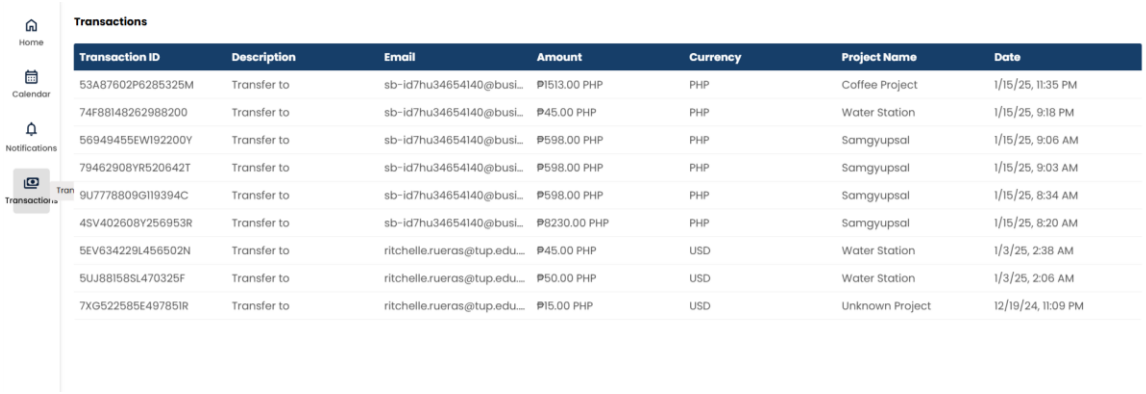


Figure 79 displays the client transactions section, where clients can view a detailed record of their financial transactions. Each entry includes the transaction ID, description, recipient email, amount, currency, associated project, and transaction date.

Figure 80

Client - Account Settings

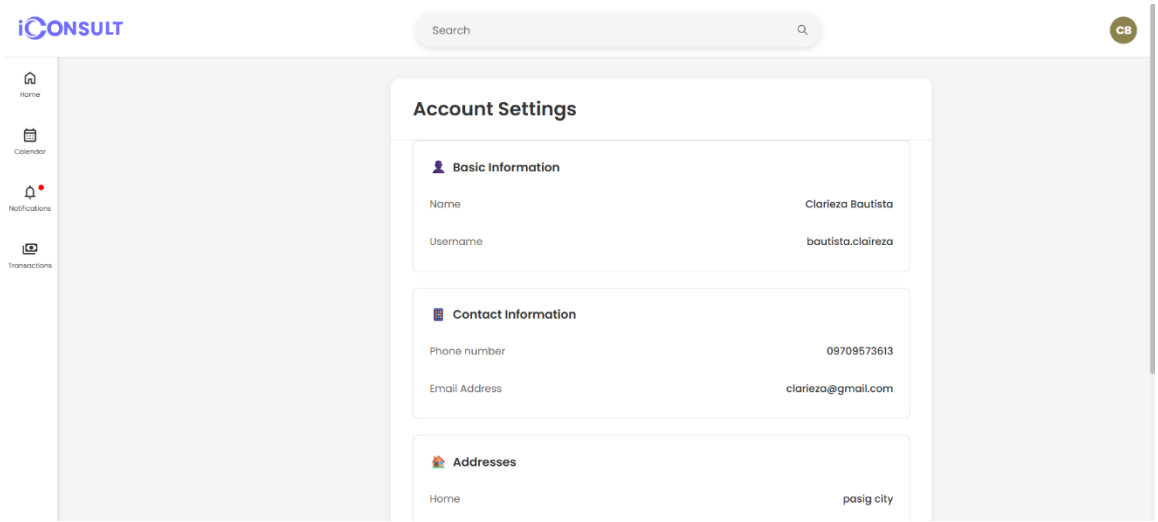


Figure 80 shows the client account settings page, where users can manage their personal information, contact details, addresses, and security settings.

Figure 81

Client - Feedbacks History

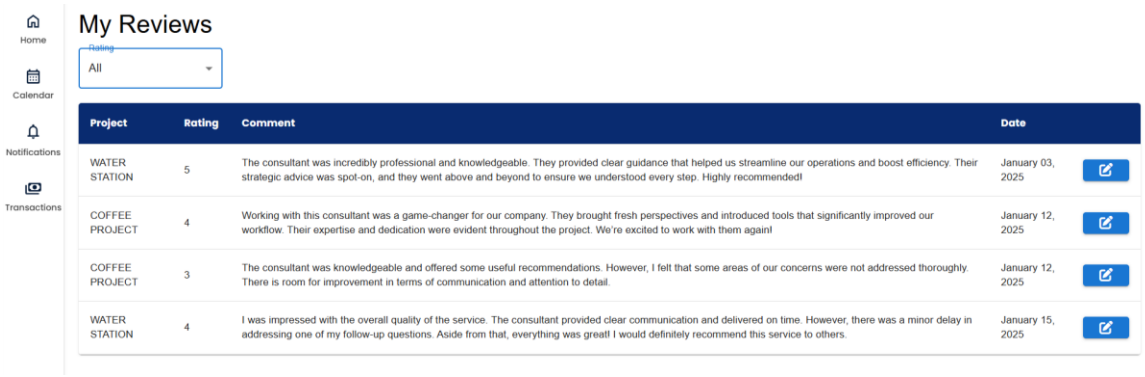


Figure 81 displays the client's feedback history, listing reviews given for various projects. Each review includes the project name, rating, comment, and submission date.

Project Test Results

This section outlines the data collected during the system's testing, focusing on its functionality and reliability. It also presents and analyzes the findings from the project evaluation, providing insights into the system's performance and effectiveness.

Functionality Test Results

The system was tested for functionality in different test case scenarios for the different actors of the system. Table 10 shows the overall summary of the test cases done for different actors of the system in terms of functionality.

Table 10

Overall Summary of Functionality Test Cases

Use Case	***No. of Test Cases***
Unregistered User	*1*
Verified User	*5*
Admin	*7*
Total	***13***

The testing process was conducted in two cycles, as summarized in Table 11, which presents the percentage of test cases that passed and failed in each cycle. In the first testing cycle, 95% of the executed test cases were marked as "passed," while the remaining 5% were marked as "failed," necessitating a second testing cycle. During the second cycle, all test cases executed received a "passed" remark, achieving a 100% success rate. Details of the functionality test cases are provided in Appendix D, while the failed test cases are documented in Appendix F.

Table 11

Functionality Testing Execution Summary

Text Execution	Expected Result	Active Result Cycle 1	Cycle 2
No. of Test Cases Results of Test Cases	100%	100%	100%
Passed	100%	95%	100%
Failed	0%	5%	0%
No. of Test Cases Not Executed	0%	0%	0%

The table provides a summary of test execution results across two cycles, presenting the percentage of test cases executed, their actual outcomes, and the pass/fail rates for each cycle. It indicates that all test cases were executed in both cycles, with Cycle 1 achieving a 95% pass rate and a 5% fail rate, while Cycle 2 recorded a 100% pass rate.

Reliability Test Results

Table 12

Overall Summary of Functionality Test Cases

Use Case	*No. of Test Cases*
WPC-REL-001	*To ensure that the admin can successfully register a client's account and that the system sends an email*
WPC-REL-002	*To ensure that the client can successfully monitor and track the progress of the project they are working on, allowing them to stay informed about project milestones, task completion, and overall status.*
WPC-REL-003	*To ensure that the admin can successfully select an available date and time slot from the calendar*
WPC-REL-004	*to successfully pay for a project using PayPal.*
WPC-REL-005	*To ensure that the system allows employees to efficiently assign tasks to enhance team collaboration.*
WPC-REL-006	*To ensure that the system will be able to manage appointment*
WPC-REL-007	*To ensure that the system will be able to manage project list*

The table outlines various use cases related to system functionality and management, each identified by a unique code. The "No. of Test Cases" column specifies the objectives or goals associated with each use case. The covered use cases include email verification for user security, project progress tracking, client management, user account administration, project listings, transaction handling, dashboard management, report generation, and

calendar management. This table serves as a reference to ensure that the system has undergone comprehensive testing and meets the specified requirements for each use case.

Project Evaluation

The developed web application was demonstrated to twelve (12) IT Students, eight (8) IT Professionals, six (6) Business Consultants, and four (4) IS Students. A total of thirty (30) respondents from this group of people evaluated the system in terms of functional suitability and reliability in accordance with ISO/IEC 25010, yielding an overall weighted mean rating of 3.575. The overall weighted mean rating falls within the scale range of 3.26 - 4.0, which is described as Highly Acceptable.

Table 13

Reliability Testing Execution Summary

Text Execution Summary	Expected Result	Active Result Cycle 1	Cycle 2
No. of Test Cases Results of Test Cases	100%	100%	100%
Passed	100%	100%	100%
Failed	0%	0%	0%
No. of Test Cases Not Executed	0%	0%	0%

Table 13 provides a summary of the reliability test execution, outlining the results of two testing cycles. It includes the number of executed test cases, their outcomes, and a comparison of the predicted versus actual results. All test cases in Cycles 1 and 2 were completed successfully with a 100% success rate. The "Results of Test Cases" section shows the percentage of test cases that passed and failed. Both Cycle 1 and Cycle 2

achieved a 100% success rate with no failures. All test cases were successfully executed, with no test cases left untested in either cycle.

Project Capabilities and Limitations

Capabilities:

1. **Project Management** - Create and manage projects with defined tasks and deadlines. Assign tasks to employees and track progress in real time. Provide clients with visibility into project milestones and progress.
2. **Client Management -** Allow clients to view project progress and access relevant documents. Each client has a dedicated dashboard providing a centralized overview of their projects, appointments, and relevant documents. This personalized view streamlines their interaction with the consultancy. Facilitate client feedback for improved service.
3. **Online Appointment Scheduling -** Enable clients to book appointments with consultants based on availability. Send real-time notifications to both consultants and clients.
4. **Integration with External APIs -** SendGrid for email notifications, Google Calendar for scheduling, and PayPal Sandbox for payment processing.

   Enhances system functionality with automated notifications and streamlined payment handling.

Limitations:

1. **No Real-Time Messaging:** Lacks an integrated messaging feature for direct communication between clients and consultants.
2. **Account Creation Dependency -** Clients cannot access the platform or book appointments until a consultant creates their account.
3. **Access Control -** The appointment setting and project creation is limited to the admin only. Therefore, employees are not able to manage the client's appointment and cannot create a project for the client.

Table 14

Functionality Suitability Evaluation Result

Criteria	Weighted Mean	Descriptive Rating
Functional Completeness	3.8	Highly Acceptable
Functional Correctness	3.8	Highly Acceptable
Functional Appropriateness	3.0	Very Acceptable
Overall Weighted Mean	**3.53**	**Highly Acceptable**

Table 14 shows that the system's functional completeness received a weighted mean of 3.80, earning a descriptive rating of "Highly Acceptable". This indicates that the developed web application effectively addressed all specified tasks and user goals. Similarly, the system scored 3.80 for functional correctness, also rated as "Highly Acceptable," reflecting its ability to produce precise and accurate outcomes, particularly when processing and utilizing user-provided data.

The system achieved a weighted mean of 3.00 for functional appropriateness with a

descriptive rating of "Highly Acceptable," demonstrating its capability to perform tasks efficiently for specific objectives. This is evident in features such as notifications and alert notices, which guide users in completing actions. Overall, the system's functional suitability was rated as "Very Acceptable" with an overall weighted mean of 3.53.

Table 15

Reliability Evaluation Result

Criteria	Weighted Mean	Descriptive Rating
Maturity	3.56	Highly Acceptable
Availability	3.56	Highly Acceptable
Fault Tolerance	3.63	Highly Acceptable
Recoverability	3.73	Highly Acceptable
Overall Weighted Mean	**3.62**	**Highly Acceptable**

Table 15 shows the evaluation results for reliability, where the system received a rating of “Highly Acceptable” with a weighted mean of 3.56 for maturity. This demonstrates that the system reliably meets its requirements under normal usage, as shown by its dependable performance without issues.

The system's availability was also scored as "Highly Acceptable," with a weighted mean of 3.56, meaning that even though it isn't perfect, it is still usable and available. With a fault tolerance score of 3.63, the system was deemed adequate to continue operating even in the event of hardware or software malfunctions. Lastly, the system received a weighted mean rating of 3.73 for recoverability, which indicates that it can restore data and resume optimal operation following disruptions or failures.

Table 16

Software Evaluation Results Summary

Criteria	Weighted Mean	Descriptive Rating
Functionality Suitability	3.53	Highly Acceptable
Reliability	3.62	Highly Acceptable
Overall Weighted Mean	**3.575**	**Highly Acceptable**

The evaluation findings are given in Table 16 according to two criteria. With an overall weighted mean grade of 3.575, the system was deemed "Highly Acceptable." This means that it complied with the dependability and functional appropriateness requirements specified in ISO/IEC 25010.

Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

This chapter provides the summary of findings, conclusions, and the corresponding recommendation drawn from the outcomes of the system's testing and evaluation.

Summary of Findings

The system was designed to transform business consultancy operations by seamlessly integrating project and client management. iConsult system features includes project tracking, online appointment scheduling, secure document storage, online payment, real-time notifications, and reports with data visualization. By analyzing the evaluation results of "iConsult: A Business Consultancy Integrated with Project Management System and Client Management System," indicates that this platform will likely improve management within business consultancy and improve client satisfaction through real-time updates, secure document access, and efficient task management.

Based on the actual test results, the system's reliability and functionality are satisfactory. To provide a more detailed understanding of the system's functionality, we assessed its functional completeness, correctness, and appropriateness, which was evaluated as "Highly Acceptable." This designation implies that the system is highly functional, where the system's features effectively achieve the intended outcomes under defined conditions and across various scenarios. Similarly, the system's reliability was also evaluated as "Highly Acceptable," indicating consistent operation over a specified period under stated conditions. This reliability assessment is informed by factors such as the system's recoverability, fault tolerance, availability, and maturity.

Conclusions

The following are the conclusions drawn from the findings of the study:

1. The development of “iConsult: A Business Consultancy Integrated with Project Management System and Client Management System” has been developed, delivering the following outcomes:

   a. A real-time online system that lets clients schedule consultations immediately.

   b. A project management feature that helps consultants organize their work, track tasks, and meet deadlines efficiently.

   c. A client management system that helps businesses manage client information, schedule meetings, track feedback, oversee projects, and handle payments all in one place.

   d. A document-sharing tool that lets users easily upload and access shared documents whenever needed.

   e. Payments are processed securely through integration with PayPal's sandbox testing system.

   f. Data visualization tools to monitor key metrics to track performance and identify areas to improve.

2. **Technologies Used**:

   a. The development process was conducted in Visual Studio Code.

   b. JavaScript served as the primary programming language, supported by a full-stack framework.

   c. For the user interface, React.js with Vite as a build tool is used, along with HTML and CSS for structure and styling.

d. Backend technologies: Express.js and Node.js.

e. MySQL as the database management system.

f. Version control managed via GitHub.

g. Figma used for designing the user interface.

h. Google Calendar API for scheduling, SendGrid Email API for email notifications, and PayPal Sandbox for payment processing.

i. Jaws DB MYSQL and Heroku for deployment.

3. **System Performance**: The web application was tested and demonstrated the system's functionality and reliability.
4. An evaluation conducted according to ISO/IEC 25010 standards graded the system's functionality and reliability as 'Highly Acceptable'.

Recommendations

To maximize the capabilities and potentials of this project as indicated by the survey that was conducted, the following suggestions are hereby put forward:

1. Enable document editing within the system to enhance accessibility for the admin.
2. Integrate live chat or chatbot support within the platform to provide real-time assistance for users experiencing issues or needing help scheduling.
3. Provide different payment methods other than Paypal Sandbox.
4. Add and data backup and recovery include an automated backup system and disaster recovery plan to protect data from accidental loss or system failures.
5. Allow integration with popular communication tools such as Zoom, Microsoft Teams, or Google Meet for video consultation scheduling and communication.

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APPENDICES

Appendix A

QUESTIONNAIRE FOR ISO/IEC25010: SOFTWARE QUALITY STANDARDS

Technological University of the Philippines- Manila
College of Science
Computer Studies Department

Name:_____
Occupation:_____
Date of Evaluation: _____ Signature of Evaluator:_____
(Signature Over Printed Name)

ISO 25010 EVALUATION FORM

Please rate the level by putting a check mark (✓) on the column that corresponds to your assessment using the four-point scale with its corresponding verbal descriptions (4 as the highest and 1 as the lowest). How does the developed system comply with the ISO/IEC 25010 software quality standards in terms of the following criteria. Use the following scale to rate each statement:

4 – Highly Acceptable 3 –Very Acceptable 2 – Fairly Acceptable 1 – Not Acceptable

A. Functional Stability

Evaluation Criteria	4	3	2	1
Functional Completeness *Does the system provide all the required features and capabilities?*				
Functional Accuracy *Are the features and functions delivering accurate and reliable results?*				
Functional Relevance *Are the system's features appropriately aligned with admin and user needs?*				

B. Reliability

Evaluation Criteria	4	3	2	1
System Stability *Is the system stable and reliable during extended use?*				
Operational Availability *Is the system consistently accessible without unexpected downtime?*				
Error Handling *Does the system effectively handle errors to minimize disruptions?*				
Recoverability *How well does the system recover data after an interruption or failure?*				

C. Performance Efficiency

Evaluation Criteria	4	3	2	1
Time Behavior *Does the system perform tasks in a timely manner?*				
Resource Utilization *Does the system use resource efficiently?*				
Capacity *Can the system handle large workloads effectively?*				

D. Usability

Evaluation Criteria	4	3	2	1
Appropriateness Recognizability *How well can you identify if the system fits your needs?*				
Learnability *How easy is it to learn how to use the system?*				

Operability *How easy is it to operate and control the system?*				
User Interface *Aesthetics How visually appealing is the user interface?*				
Accessibility *How accessible is the system for people with diverse abilities?*				

E. Security

Evaluation Criteria	4	3	2	1
Confidentiality *Is data accessible only to authorized users?*				
Integrity *Are data and programs protected from unauthorized changes?*				
Non-repudiation *Can actions or events be proven to have occurred?*				
Accountability *Can the actions of unauthorized users be traced?*				
Authenticity *Can the identity of users and resources be verified?*				

F. Maintainability

Evaluation Criteria	4	3	2	1
Modularity *Can part of the system be changed with minimal impact on others?*				
Reusability *Can components be used in other systems or projects?*				
Analyzability				

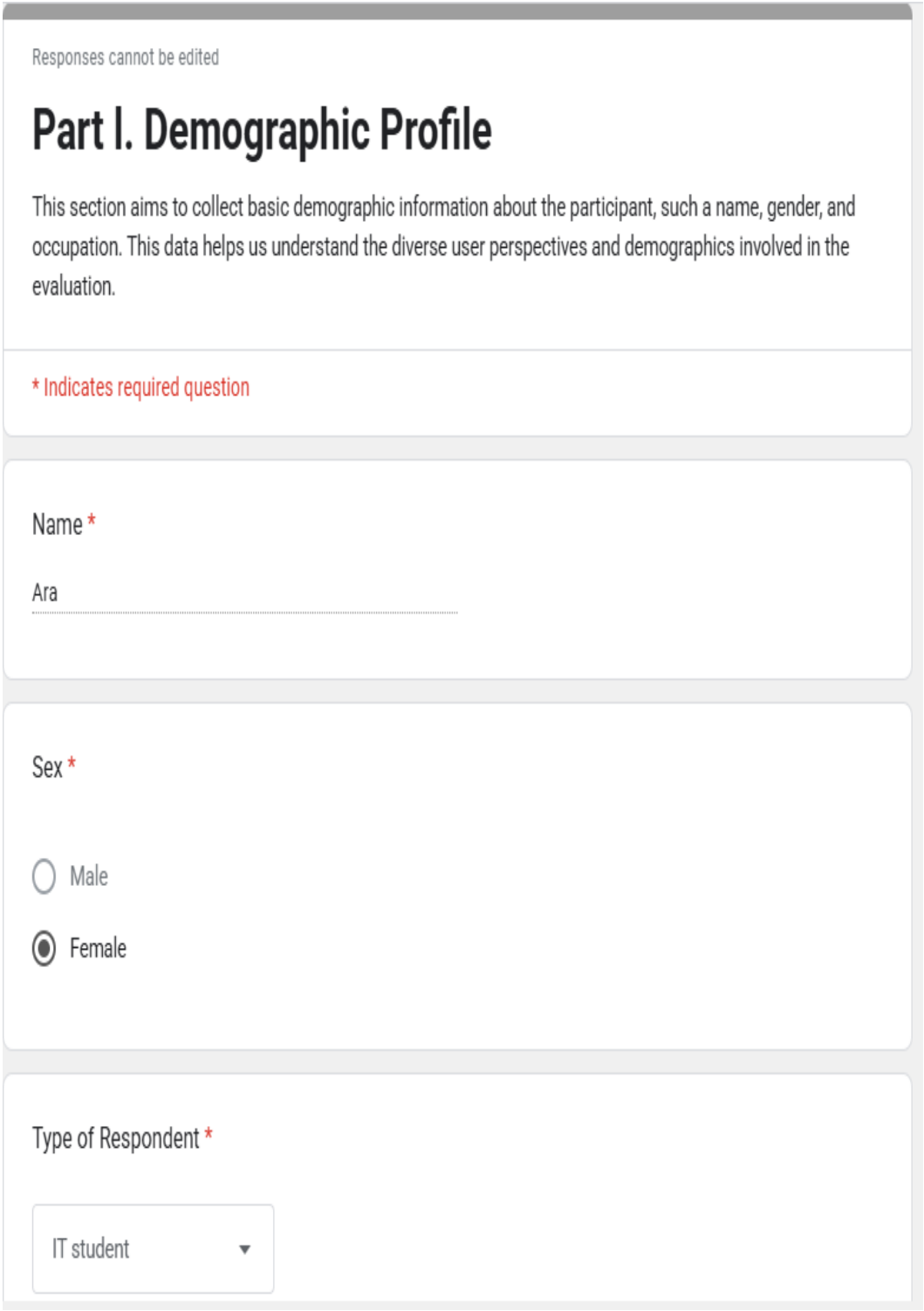
Can changes be assessed, and issues diagnosed effectively?				
Modifiability *Can changes be made without introducing new problems?*				
Testability *Can the system be tested effectively?*				

G. Portability

Evaluation Criteria	**4**	**3**	**2**	**1**
Adaptability *Does the web application work well across various browsers? Can the web application function on different devices such as smartphones, tablets, laptops?*				
Installability *Does the system load smoothly across different web browsers and/or devices without requiring a complicated setup process?*				
Replaceability *Is the system designed in a way that allows for easy integration of a new solution if necessary, without negatively impacting performance or user experience?*				

Appendix B

SAMPLE ANSWERED EVALUATION SHEET AND RESULTS SHEET



Responses cannot be edited

Part I. Demographic Profile

This section aims to collect basic demographic information about the participant, such a name, gender, and occupation. This data helps us understand the diverse user perspectives and demographics involved in the evaluation.

* Indicates required question

Name *

Ara

Sex *

Male

Female

Type of Respondent *

IT student

Part II. Functionality Suitability
In this section, we assess the functionality of our business consultancy system.

Please rate the level by putting a check mark (✓) on the column that corresponds to your assessment using the four-point scale with its corresponding verbal descriptions (4 as the highest and 1 as the lowest).

4 – Highly Acceptable
3 –Very Acceptable
2 – Fairly Acceptable
1 – Not Acceptable

Functional Completeness *
Does the system provide all the required features and capabilities

	1	2	3	4	
Not Acceptable	○	○	○	◉	Highly Acceptable

Functional Accuracy
Are the features and functions delivering accurate and reliable results?

	1	2	3	4	
Not Acceptable	○	○	○	◉	Highly Acceptable

Functional Relevance
Are the system's features appropriately aligned with admin and user needs?

	1	2	3	4	
Not Acceptable	○	○	○	◉	Highly Acceptable

Part III. Reliability
This section focuses on evaluating the reliability of our business consultancy system.

Directions: The following items describe the statement about the degree to which a system performs specified period of time. Please rate the level by putting a check mark (✓) on the column that corresponds to your assessment using the four-point scale

System Stability
Is the system stable and reliable during extended use?

	1	2	3	4	
Not Acceptable	○	○	◉	○	Highly Acceptable

Operational Availability
Is the system consistently accessible without unexpected downtime?

	1	2	3	4	
Not Acceptable	○	○	◉	○	Highly Acceptable

Error Handling
Does the system effectively handle errors to minimize disruptions?

	1	2	3	4	
Not Acceptable	○	○	○	◉	Highly Acceptable

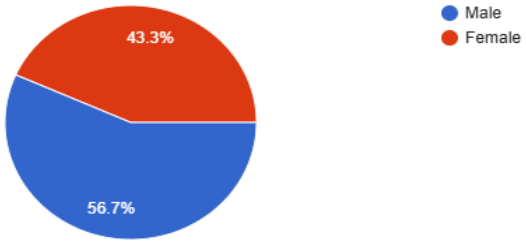
Recoverability
How well does the system recover data after an interruption or failure?

	1	2	3	4	
Not Acceptable	○	○	◉	○	Highly Acceptable

Sex

Copy chart

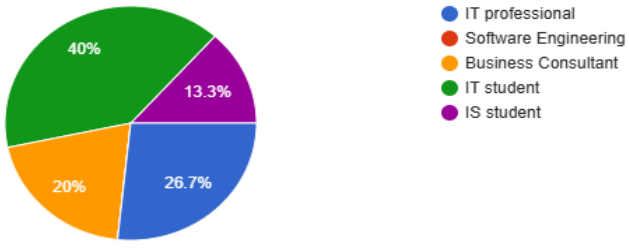
30 responses



Type of Respondent

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30 responses



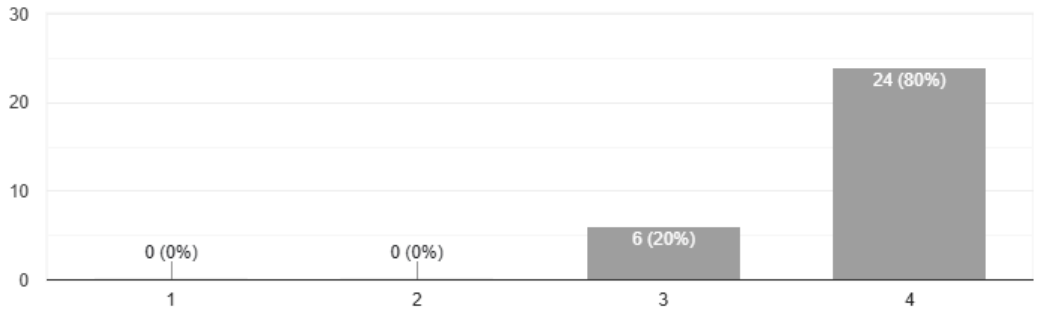
Part II. Functionality Suitability

Functional Completeness

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Does the system provide all the required features and capabilities

30 responses

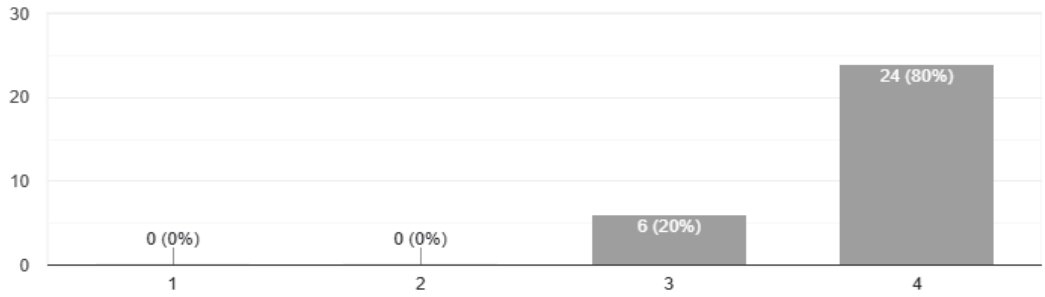


Functional Accuracy

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Are the features and functions delivering accurate and reliable results?

30 responses

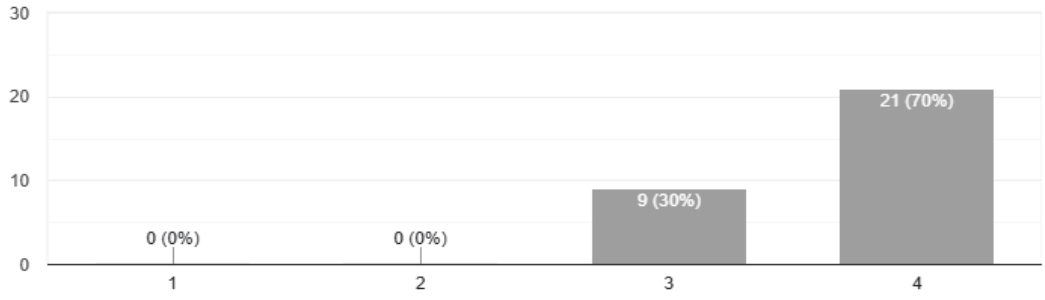


Functional Relevance

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Are the system's features appropriately aligned with admin and user needs?

30 responses



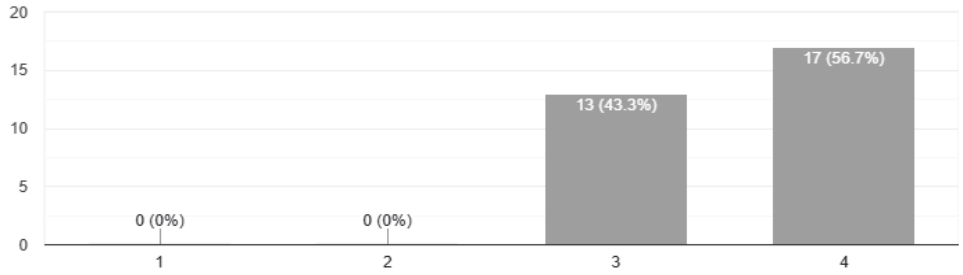
Part III. Reliability

System Stability

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Is the system stable and reliable during extended use?

30 responses

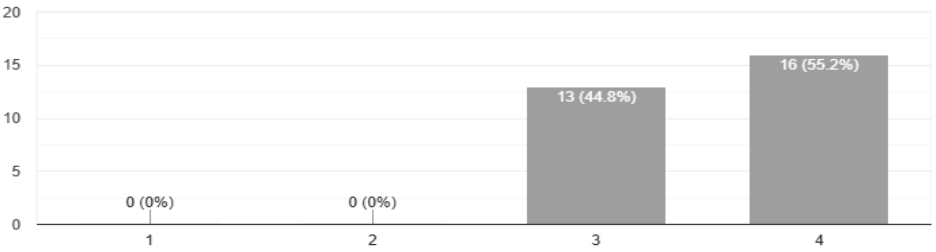


Operational Availability

Copy chart

Is the system consistently accessible without unexpected downtime?

29 responses

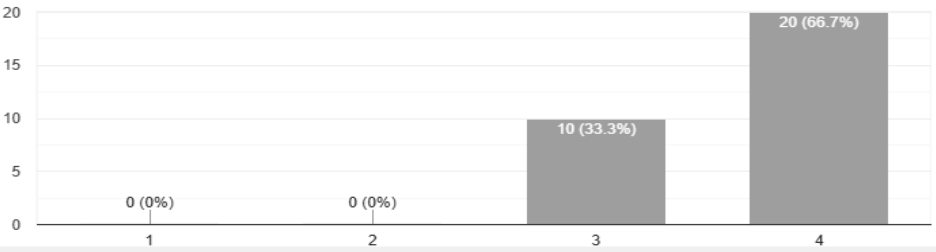


Error Handling

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Does the system effectively handle errors to minimize disruptions?

30 responses

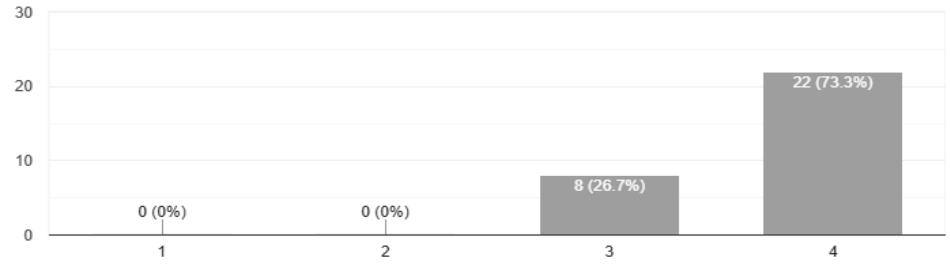


Recoverability

Copy chart

How well does the system recover data after an interruption or failure?

30 responses



Appendix C

SUMMARY OF RESPONDENT' EVALUATION

Respondents	Functional Completeness	Functional Correctness	Functional Appropriateness	Maturity	Availability	Fault Tolerance	Recoverability
1	3	3	3	3	3	4	4
2	4	4	4	3	3	4	3
3	4	3	3	4	4	4	4
4	4	4	4	4	4	4	4
5	4	4	4	4	4	4	4
6	4	4	4	4	4	4	4
7	4	4	4	3	3	3	3
8	4	4	4	4	3	3	3
9	4	4	4	4	4	3	4
10	4	4	4	3	4	4	4
11	4	4	4	3	3	3	3
12	4	4	4	4	4	3	4
13	4	4	4	4	3	3	3
14	4	4	4	3	3	3	3
15	4	4	4	4	4	4	4
16	4	4	3	3	3	3	4
17	4	4	4	3	3	4	4
18	3	4	4	3	3	4	4
19	3	3	3	4	4	3	3
20	4	4	3	3	4	4	3
21	4	4	4	4	4	4	4
22	4	4	4	3	4	4	4
23	4	4	4	4	4	4	4
24	3	3	3	3	3	4	4
25	3	3	3	4	4	4	4
26	3	4	4	4	3	3	4
27	4	4	3	4	4	3	4

28	4	3	4	4	3	4	4
29	4	4	3	4	4	4	4
30	4	4	4	3	4	4	4
Weighted Mean	**3.8**	**3.8**	**3.0**	**3.56**	**3.56**	**3.63**	**3.73**

Appendix D

TEST CASES FOR FUNCTIONALITY

Test Case ID		WPC-FUN-001		**UC Reference**	Admin
Objective		To allow the admin to securely sign into their account or recover access if the password is forgotten.			
Assumptions/ Preconditions		The admin has access to a valid account			
Actions		**Expected Result**		**Actual Result**	
• The admin signs into the admin account. • Enter Username and password • If you forget your password, "Click" Forget password.		• They are successfully logged into the admin account and redirected to the admin dashboard		• The admin successfully logged.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-002		**UC Reference**	Verified User
Objective		Verify that the system enforces password change requirements for clients logging in for the first time.			
Assumptions/ Preconditions		The client is logged into the system.			
Actions		**Expected Result**		**Actual Result**	
• Navigate to the "Change Password" form. • Enter a valid new password that meets all the criteria. • Re-enter the same password in the "Confirm Password" field. • Click the "Update Password" button.		• The system displays a success message indicating the password has been successfully updated.		• The system validated the new password, successfully updated it.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-003		**UC Reference**	Admin, Verified User
Objective		To verify that the system appropriately handles invalid login attempts by displaying an error message when incorrect credentials are entered,			
Assumptions/ Preconditions		A registered user exists in the system.			
Actions		**Expected Result**		**Actual Result**	
• Click on the "Sign In" option in the navigation bar. • Enter invalid credentials (e.g., incorrect email or password). • Click on the "Sign In" button.		• A validation error message should be displayed appropriately and prominently, indicating that the log-in credentials are invalid.		• The validation error message is displayed accordingly and prominently, indicating that the log-in credentials are invalid.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-005		**UC Reference**	Admin
Objective		Sign in with valid credentials as an admin.			
Assumptions/ Preconditions		Must have registered admin credentials.			
Actions		**Expected Result**		**Actual Result**	
• Click on the "Sign In" option in the navigation bar. • Enter "Username and Password". • Click the "Sign In" button.		• The admin should be successfully logged in and redirected to the dashboard page.		• The admin was successfully logged in and redirected to the dasboard page.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-006		**UC Reference**	Unregistered User
Objective		To ensure that an unregistered user cannot successfully Log in			
Assumptions/ Preconditions		Must be an unregistered user attempting to register.			
Actions		**Expected Result**		**Actual Result**	
• Click on the "Sign In" option in the navigation bar. • Enter a invalid "Username and Password. • Click the "Log In" button.		• Form validation error message(s) should be displayed accordingly and in the correct position, indicating the invalid credentials		• Form validation error message(s) were displayed accordingly and in the correct position, indicating the invalid credentials	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-007		**UC Reference**	Verified User
Objective		Send email to user's email for resetting password.			
Assumptions/ Preconditions		Must be a registered user who wants to reset their password.			
Actions		**Expected Result**		**Actual Result**	
• Click on the "Forgot Password" option. • Enter the registered email address.		• The user should be able to enter their email address and receive an email providing a link to reset their password.		• The user was able to receive an email containing a link to reset their forgotten password.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-008		**UC Reference**	Admin
Objective		To ensure that an employee is added successfully using valid data.			
Assumptions/ Preconditions		The admin is logged in as a valid admin account.			
Actions		**Expected Result**		**Actual Result**	
• Fill in all other required fields. • Click "Add Employee".		• The new employee's information should be viewable in the employee list.		• The admin successfully adds a new employee, the new employee's information is displayed in the employee list.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-009		**UC Reference**	Admin
Objective		To verify the successful editing of an existing availability slot.			
Assumptions/ Preconditions		The user attempting to edit the availability slot has the necessary permissions or access rights to do so			
Actions		**Expected Result**		**Actual Result**	
• Select a date from the calendar • Click the "EDIT" button for an existing availability slot. • Click the "DEL" button for an existing availability slot. • Click the "SAVE" button.		• The new availability slot is added to the "Availability" section below.		• The new availability slot is added to the "Availability" page.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-010		**UC Reference**	Verified User
Objective		To allow the client to successfully select a date and time for an appointment or meeting.			
Assumptions/ Preconditions		The client has the necessary access or permissions to make an appointment or schedule a meeting.			
Actions		**Expected Result**		**Actual Result**	
• Select a date from the calendar • Choose a time slot from the "Available Times for • Click the "Next" button to continue with the appointment booking process.		• The user successfully selects a date from the calendar.		• The user successfully booked appointment from the calendar tab	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-FUN-011		**UC Reference**	Admin
Objective		To ensure that the system will be able to manage projects.			
Assumptions/ Preconditions		• The system has an available interface for creating tasks.			
Actions		**Expected Result**		**Actual Result**	
• Enter a valid last name, first name, and middle initial. • Enter a valid email address. • Select "Active" for the status. • Click the "Add Employee" button.		A new project is successfully created and displayed in the project list.		• A new project was successfully created and displayed in the project list.	
Status	PASSED	**Severity**		**Priority**	

Appendix E

TEST CASES FOR RELIABILITY

Test Case ID		WPC-REL-001		**UC Reference**	
Objective		To ensure that the admin can successfully register a client's account and that the system sends an email			
Assumptions/ Preconditions		The admin has the necessary permissions and access to register new client accounts within the system			
Actions		**Expected Result**		**Actual Result**	
• The admin registers client's account. • -The system emails the client registered email address a link.		• The admin successfully registers the client's account using the provided details.		• The admin was successfully registers the client's account using the provided details.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID		WPC-REL-002		**UC Reference**	
Objective		To ensure that the client can successfully monitor and track the progress of the project they are working on, allowing them to stay informed about project milestones, task completion, and overall status.			
Assumptions/ Preconditions		The client has the necessary permissions to view and monitor the project's progress in the dashboard page			
Actions		**Expected Result**		**Actual Result**	
• The client logs into the system using their credentials to access the client management • The client navigates to the project or task dashboard where the project's progress is displayed		• The client is able to log into the system and navigate to the dasboard page		• The client is able to access the system and navigate to the dasboard successfully.	
Status	PASSED	**Severity**		**Priority**	

Test Case ID	WPC-REL-003	**UC Reference**	
Objective	To ensure that the admin can successfully select an available date and time slot from the calendar		
Assumptions/ Preconditions	The admin has the necessary access rights to view and modify the calendar to select available date and time slots.		
Actions	**Expected Result**	**Actual Result**	
• The admin can select the available date and time slots from the calendar, choosing a suitable time based on the availability.	• The admin successfully logs in and has access to the calendar interface with the necessary permissions to select available time slots.	• The admin was successfully changed available time slots.	

Test Case ID		WPC-REL-004		UC Reference	
Objective					
Assumptions/ Preconditions		The user is logged in and has a valid account.			
Actions		Expected Result		Actual Result	
• The user signs in to their account. • The user "click" on "Project" from the dashboard. • The user click the project and pay using paypal		• The system successfully paid using paypal account		• The system was successfully paid using paypal account and automatic "paid" to payment status in the project managment	
Status	PASSED	Severity		Priority	

Test Case ID		WPC-REL-006		UC Reference	
Objective		To ensure that the system allows employees to efficiently assign tasks to enhancing team collaboration.			
Assumptions/ Preconditions		The system has the functionality to assign tasks.			
Actions		Expected Result		Actual Result	
• The system enables the employee to assign tasks to employees		• The admin successfully assigns the task to a specific employee		• The admin was successfully assigns the task to a specific employee.	
Status	PASSED	Severity		Priority	

Test Case ID		WPC-REL-007		UC Reference	
Objective		To ensure that the system will be able to manage appointment			
Assumptions/ Preconditions		The user is logged in as a valid admin account.			
Actions		Expected Result		Actual Result	
• "Click" the Calendar page • "Click" the "Delete" button to delete booked appointment. • Confirm Delation, "Click" Delete button if yes		• The system succefully deleted booked appoinment		• The system was succefully deleted booked appoinment	
Status	PASSED	Severity		Priority	

Test Case ID		WPC-REL-008		**UC Reference**	
Objective		To ensure that the system will be able to manage project list			
Assumptions/ Preconditions		The admin is logged in as a valid admin account.			
Actions		**Expected Result**		**Actual Result**	
• The admin signs into the admin account. • The admin clicks on "Project" from the dashboard • The admin "Click" the Ongoing project		• The system enables the admin to .		• The system successfully allows the admin to manage shopping carts.	
Status	PASSED	**Severity**		**Priority**	

Appendix F

TEST INCIDENT LOGS

Test Case ID		WPC-FUN-001		**UC Reference**	Admin
Objective		To allow the admin to securely sign into their account or recover access if the password is forgotten.			
Assumptions/ Preconditions		The admin has access to a valid account			
Actions		**Expected Result**		**Actual Result**	
• The admin signs into the admin account. • Enter Username and password • If you forget your password, "Click" Forget password.		• They are successfully logged into the admin account and redirected to the admin dashboard		• The admin successfully logged.	
Status	FAILED	**Severity**	MINOR	**Priority**	MEDUIM

Test Case ID		WPC-FUN-004		**UC Reference**	Verified User, Admin
Objective		Allowing users to reset their password by providing a valid email address			
Assumptions/ Preconditions		A registered user forgot his/her password.			
Actions		**Expected Result**		**Actual Result**	
• Click the forgot password in Sign-in. • Enter an existing email to send the link for password reset. • Reset your password.		• The user receives a password reset email in their inbox, containing a secure link to reset their password.		• An email containing a link is sent to the user that allows them to reset their password.	
Status	FAILED	**Severity**	MINOR	**Priority**	MEDUIM

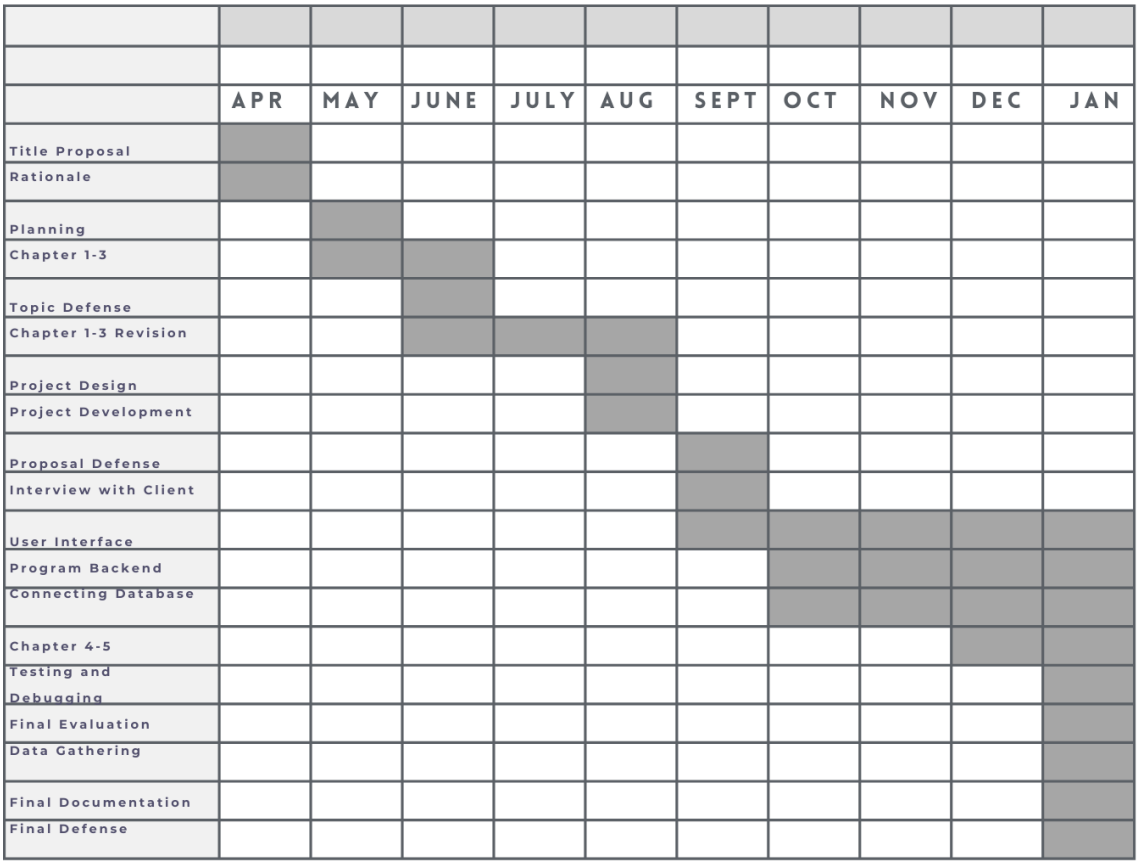
Test Case ID		WPC-FUN-010		**UC Reference**	Verified User
Objective		To allow the client to successfully select a date and time for an appointment or meeting.			
Assumptions/ Preconditions		The client has the necessary access or permissions to make an appointment or schedule a meeting.			
Actions		**Expected Result**		**Actual Result**	
• Select a date from the calendar • Choose a time slot from the "Available Times for • Click the "Next" button to continue with the appointment booking process.		• The user successfully selects a date from the calendar.		• The user successfully booked appointment from the calendar tab	
Status	FAILED	**Severity**	MAJOR	**Priority**	HIGH

Test Case ID		WPC-FUN-008		**UC Reference**	Admin
Objective		To ensure that an employee is added successfully using valid data.			
Assumptions/ Preconditions		The admin is logged in as a valid admin account.			
Actions		**Expected Result**		**Actual Result**	
• Fill in all other required fields. • Click "Add Employee".		• The new employee's information should be viewable in the employee list.		• The admin successfully adds a new employee, the new employee's information is displayed in the employee list.	
Status	FAILED	**Severity**	MAJOR	**Priority**	HIGH

Test Case ID		WPC-FUN-011		**UC Reference**	Admin
Objective		To ensure that the system will be able to manage projects.			
Assumptions/ Preconditions		• The system has an available interface for creating tasks.			
Actions		**Expected Result**		**Actual Result**	
• Enter a valid last name, first name, and middle initial. • Enter a valid email address. • Select "Active" for the status. • Click the "Add Employee" button.		A new project is successfully created and displayed in the project list.		• A new project was successfully created and displayed in the project list.	
Status	FAILED	**Severity**	MAJOR	**Priority**	HIGH

Appendix G

GANTT CHART



Appendix H

USER MANUAL

I. Consultant User Guide:

A. Account Login:

- Open Browser & Navigate: Open your web browser and go to the iConsult website
- Login Screen: You should see a login screen with fields for "Username" and "Password".
- Enter Credentials: Enter your assigned username and password.
- Click Login: Click the "Login" button. If your credentials are correct, you'll be taken to your dashboard. If not, you'll receive an error message. Contact support if you need help with your login.

B. Project Management:

- Dashboard Overview: Your dashboard provides a quick summary of your projects, including progress percentages and upcoming appointments.

1. Calendar (Scheduling):

- Click on the "Calendar" tab or icon.
- Select the date and time you are available for consultations.
- Save your availability. The system will likely highlight your already scheduled appointments.

2. Projects Tab:

- Click the "Projects" tab.
- This page lists all your current projects. You can view details for each project by clicking on its name. Information displayed should include:
- Project Name
- Client Name
- Timeline (Start and End Dates)
- Project Status (e.g., In Progress, Completed)
- Down Payment Received
- Total Project Cost
- Payment Status (e.g., Paid, Partially Paid, Unpaid)

3. **Client Management:**

- Click the "Clients" tab.
- View existing clients or add new clients by clicking an "Add Client" button. You'll likely need to provide client details such as name, contact information, etc.

4. **Employee Management:**

- Click the "Employees" tab.
- Manage your employees. Add new employees using an "Add Employee" button. You'll need to provide employee details.

5. **Transaction History:**

- Click the "Transactions" tab.
- View a complete history of all transactions, including transaction IDs, client IDs, and payment amounts.

II. **Client User Guide:**

A**. Account Setup (First-Time Login):**

1. Access Website: Go to the iConsult website: [website address].
2. Initial Login: You may need to use a temporary password or a link sent to your email.
3. Password Creation: You will be prompted to create a new password and confirm it. Follow the on-screen instructions.

B**. Using the Client Dashboard:**

1. Dashboard Overview: The dashboard shows your project's progress percentage.
2. Calendar (Scheduling):
3. Click on the "Calendar" or "Schedule Consultation" tab.
4. Select a date and time for your consultation. Red dates indicate the consultant is unavailable; blue dates show existing appointments.

3. **Consultation Details:**

- Select the type of consultation you need.
- Describe the purpose of your consultation in the provided text box.
- Choose your preferred communication method (online or onsite).

4**. Notifications:** Check the "Notifications" tab for updates on your appointments and payments.

5. **Payment:**

- Click the "Pay" button or similar.
- Choose your payment method (PayPal or Credit/Debit Card).
- PayPal: You will be redirected to the PayPal website to complete your payment.
- Credit/Debit Card: Fill in the required credit card information (card number, expiry date, CSC, billing address). Click "Pay" to submit your payment.

Congratulations! You now have a comprehensive understanding of the iConsult platform. This system enhances business consultancy operations by integrating project management and client relationship management, ensuring a streamlined and user-friendly experience for all stakeholders. For further assistance, please contact the system administrator or refer to this user manual.

Appendix I

THESIS GRAMMARIAN CERTIFICATE

	TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES Ayala Blvd., Ermita, Manila, 1000, Philippines Tel No. +632-5301-3001 local 608| Fax No. +632-8521-4063 Email: cos@tup.edu.ph | Website: www.tup.edu.ph	Index No.	REF-COS-3.5-INT-TGC
		Revision No.	00
		Effectivity Date	06132022
VAA-COS	THESIS GRAMMARIAN CERTIFICATION	Page	1 / 1

THESIS GRAMMARIAN CERTIFICATION

This is to certify that the thesis entitled,

ICONSULT: A BUSINESS CONSULTANCY INTEGRATED WITH PROJECT MANAGEMENT AND CLIENT MANAGEMENT SYSTEM

authored by

RITCHELLE T. RUERAS
CLARIEZA H. BAUTISTA
TRISHIA I. SAN DIEGO
MONIQUE KYLE CABIGTING

has undergone editing and proofreading by the undersigned.

This Certification is being issued upon the request
for whatever purposes it may serve them.

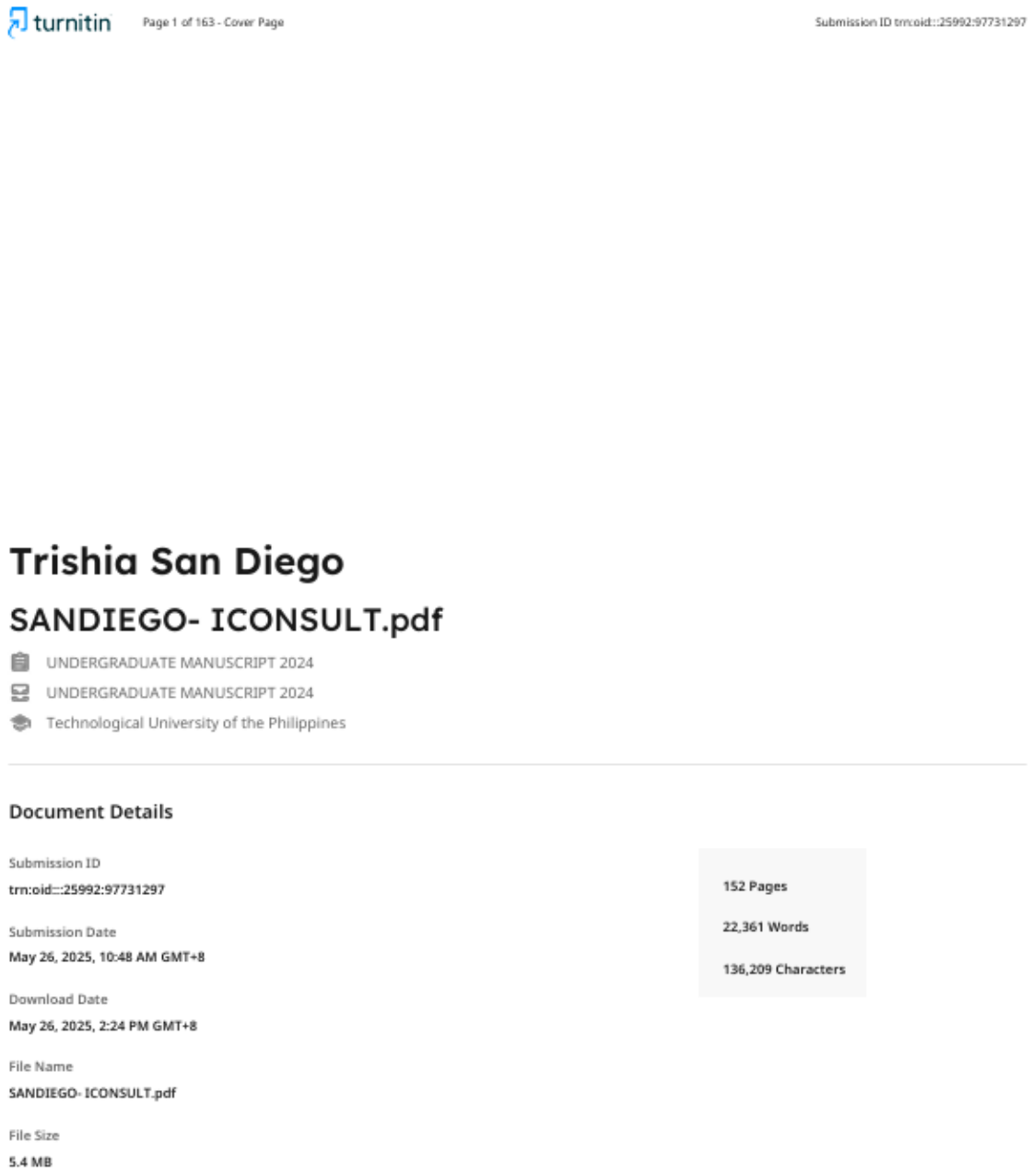
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Grammarian

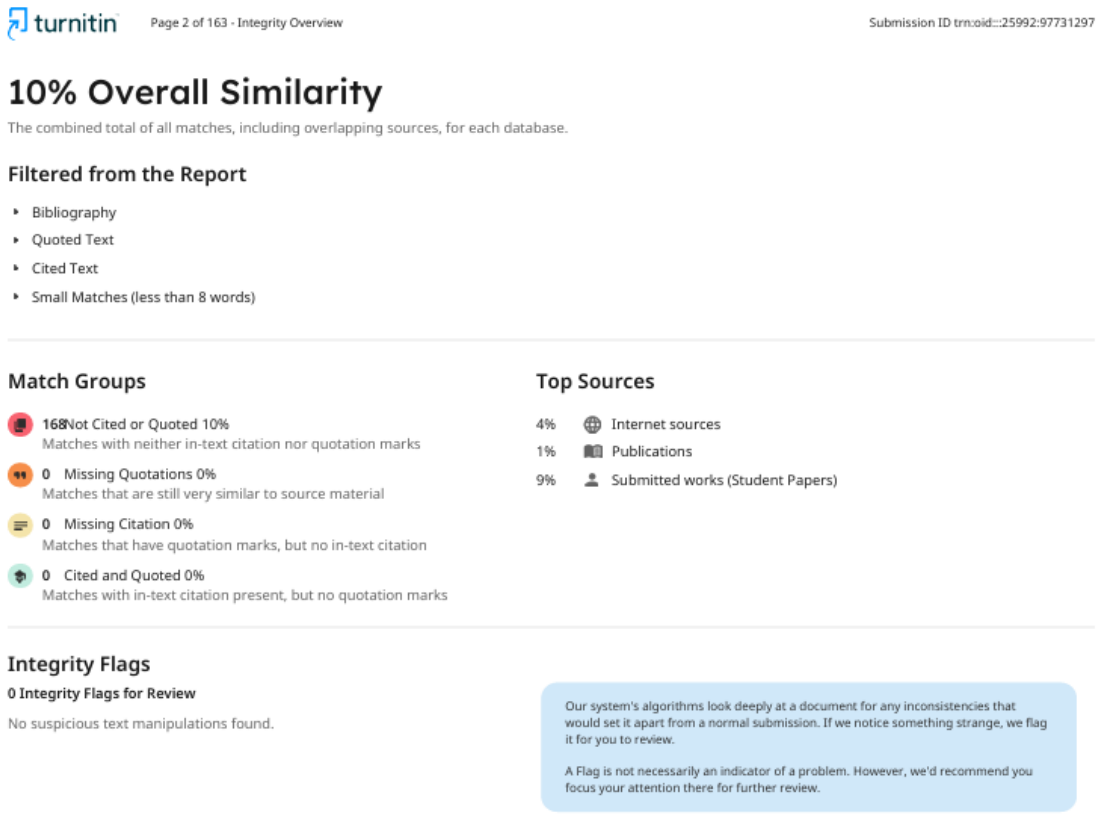
Technological University of the Philippines

April 24, 2025

Appendix J

CERTIFICATION OF SIMILARITY INDEX USING TURNITIN





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Appendix K

CERTIFICATION OF SIMILARITY INDEX USING TURNITIN FROM URDS

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		Revision No.	01
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VRE-URD	**CERTIFICATE OF SIMILARITY INDEX USING TURNITIN**	Page	1 / 1
		QAC No.	CC-04132021

This is to certify that the manuscript entitled,

ICONSULT: A BUSINESS CONSULTANCY INTEGRATED WITH PROJECT MANAGEMENT SYSTEM AND CLIENT MANAGEMENT SYSTEM

authored by

RITCHELLE T. RUERAS
TRISHIA I. SAN DIEGO
MONIQUE KYLE CABIGTING
CLARIEZA H. BAUTISTA

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RESEARCHER'S PROFILE



CLARIEZA H. BAUTISTA

CONTACT

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clariezabautista@gmail.com

Rosario, Pasig City

SKILLS

- Programming: Python and JavaScript.
- Web Development: HTML, CSS, Ant Design, ReactJS, NuxtJS
- Database Management: PostgreSQL, MySQL
- Design & Prototyping: Figma
- Development Tools: Visual Studio
- Graphic Design: Canva, Illustrator, Adobe

LANGUAGES

- Filipino
- English

INTERESTS

- Watching programming videos
- Reading and Writing
- Playing online games

OBJECTIVES

A passionate and detail-oriented designer with skills in programming, web development and UI/UX design. Experienced in prototyping, and graphic design with a keen interest in technology and creative problem-solving. Seeking opportunities to apply my technical and design expertise in a dynamic environment while continuously learning and growing.

WORK EXPERIENCE

Senior High School (SHS) Work Immersion *Oct 2019 - Jan 2020*
Human Resource Management Office (HRMO)
Municipality of Pililla Rizal

- Answering citizen questions about HR policies and processes.
- Sorting employee records, contracts, and other HR-related files.
- Recording requests or inquiries from citizens (e.g., certificates, IDs, or HR-related services).

PROJECTS

iCONSULT: A Business Consultancy Integrated with Project Management System and Client Management System *2024 - Present*

- A client-based capstone project to develop a platform for managing projects, tasks, clients, and employees.

WanderPets: Pet Adoption and Rehoming Mobile Application *2024*

- A mobile app for processing pet adoptions and browsing available pets, connecting users with shelters or owners.

EDUCATION

Bachelor of Science in Information System *2021 - Present*
Technological University of the Philippines

- Academic Achiever

Mobile App and Web Development (MAWD) Senior High School *2017 - 2019*
STI College Ortigas-Cainta Extension

- With Honors



TRISHIA I. SAN DIEGO

CONTACT ME

0991-461-7184

trishiasandiego24@gmail.com

1949 Matibay St. Kahilom II
Pandacan manila

SKILLS

- Programming Language: C, C++, Python Node.js
- Web Development: HTML, CSS, PHP
- Database Management: MySQL
- Design and Prototyping: Figma
- Operating System: Windows and Linux
- Development Tools: Github, Visual Studio Code

EDUCATION

2008 - 2013
Bagong Diwa Elementary School
Primary Education

2013 - 2021
Mariano Marcos Memorial High School
Secondary Education

2021 - to present
Technological University of the Philippines
Tertiary Education

ACADEMIC ACHIEVEMENTS

- With Honor
  Mariano Marcos Memorial High School
  S.Y 2019 - 2020
- 2nd Place Mathematics Competition
  S.Y 2019 -2020
- With High Honor
  Mariano Marcos Memorial High School
  S.Y 2020-2021
- Dean's Lister
  S.Y 2022 - 2023
  Technological University of the Philippines
- Dean's Lister
  S.Y 2023 - 2024
  Technological University of the Philippines



RITCHELLE T. RUERAS

09709573613
ritchellerueras@gmail.com
1151-I I. delos Reyes St., Sampaloc, Manila

OBJECTIVE

To be able to work and apply my analytical, organizational, and technical skills in a dynamic work environment that fosters professional growth, while contributing to process efficiency and quality improvement.

EDUCATION

- **2013 - 2014** — **SURIGAO EDUCATION CENTER**
  ELEMENTARY
- **2017 - 2018** — **UNIVERSITY OF SAN CARLOS**
  JUNIOR HIGH SCHOOL
- **2019 - 2020** — **SURIGAO DEL NORTE NATIONAL HIGH SCHOOL**
  SENIOR HIGH SCHOOL
- **2021-2025** — **TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES**
  COLLEGE- present — BACHELOR OF SCIENCE IN INFORMATION SYSTEM

SKILLS

- Knowledgeable in Microsoft Office
- Basic knowledge in programming languages such as C, C++, Python, and JavaScript
- Leadership and Teamwork
- Time Management

PROJECTS

- **WANDERPETS: PET ADOPTION AND REHOMING MOBILE APP**
  Developed a React Native mobile app that facilitates responsible pet adoption and rehoming
- **VISITOR MANAGEMENT SYSTEM**
  Developed a system to streamline visitor appointments and monitoring for organizations

Cabigting, Monique Kyle

3500 D. Buenos Aires St. Sta Mesa Manila

09086579046 moniquecabigting01@gmail.com



OBJECTIVE

Brings diverse experience in customer service, having worked in a milk tea shop, sizzling restaurant, and vape shop. Skilled in handling high-volume orders, inventory management, and creating positive customer experiences in fast-paced environments. Adaptable, reliable, and a quick learner, committed to quality service and operational efficiency. Ready to contribute strong work ethic and enthusiasm to a dynamic team.

BASIC SKILLS

- Customer Service
- Communication Skills
- Time Management
- Teamwork
- Multitasking
- Adaptability
- Computer Skills

TECHNICAL SKILLS

- Programming Languages: JavaScript, C, C++
- Database Management: phpMyAdmin
- Web Development: React.js, HTML, CSS

WORK EXPERIENCE

Service Crew
- Milk Tea Shop | 2021 – 2022

Kitchen Crew / Server
- Sizzling Restaurant | 2022

Sales Associate
- Vape Shop | 2023 – 2024

On-the-Job Training (OJT)
- Mariana Academy Maritime Studies Inc.
- Department: Under of Quality Management Officer

PERSONAL DETAILS

Name: Monique Kyle Cabigting
Age: 22 yrs. old
Course: Bachelor of Science in Information Systems
Birth Date: Dec 1, 2002
Birth Place: Manila
Sex: Female

EDUCATION

ELEMENTARY
PADRE BURGOS ELEM. SCHOOL
2009-2015

JUNIOR HIGH SCHOOL
VICTORINO MAPA HIGH SCHOOL
2015-2019

SENIOR HIGH SCHOOL
AMA UNIVERSITY
2019-2021

COLLEGE
TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES

Bachelor of Science in Information Systems

2021-2025

REFERENCES

Larry Joaquin
Owner of sizzling
09467981259

Raymond Geolagon
Owner of Vape Shop
09637177069